Observers Are All You Need

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Abstract

Observer-Patch Holography starts from a finite holographic screen and one requirement: neighboring observer patches must agree on their overlap. The outer description supplies the finite screen. The inner universe is the effective world reconstructed by observers who live on that screen. The central quantitative object is the pixel ratio $P = a_{\mathrm{cell}}/\ell_P^2$, the size of one screen cell measured in Planck areas. The framework finds the fine-structure constant by asking for the nonzero detuning of a holographic screen cell such that the cell's outer geometric displacement from perfect self-similar equilibrium equals the electromagnetic observation scale emitted by the universe living on that same screen. The fixed-point calculation gives a numerical closure witness, with an interval-wide uniqueness certificate outside the theorem core. The framework also admits an inverse observational use case in which observers infer P and total screen capacity from inside the world while the outer screen settings remain externally fixed screen data. Independently of that quantitative closure, the framework recovers the Lorentz/Einstein branch and the realized Standard Model quotient with exact hypercharge and counting data. On the gravity lane the local Einstein equation is fixed only up to a metric term, and the external screen-capacity input closes that same branch globally by fixing the cosmological constant and the de Sitter static-patch package. The electroweak sector, the Higgs/top surface, the quark and neutrino families, the local gravity-facing readout, and the controlled large-N_{edge} world-sheet effective-description lane descend from the same P; the cosmological-capacity closure uses the separate external input N_{scr}. The direct theorem-grade zero-momentum electromagnetic certificate, charged-lepton absolute masses, and hadrons are work in progress.

A reader who follows the mathematical proofs presented should conclude that, contrary to popular belief, space, time, and matter are emergent appearances, while information and computation form the fundamental substrate of reality.

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1 Introduction

This synthesis paper assembles the current paper corpus on one unified synthesis surface. It brings together the recovered core from Ref. [1], the finite patch-net consensus spine from Ref. [2], the screen-microphysics engineering lane from Ref. [3], and the downstream particle-spectrum lane from Ref. [4]. The guiding question is whether overlap consistency for local observer descriptions on a finite-capacity screen is strong enough to recover the structural features of low-energy physics, and if so, how much of that recovery is theorem-level and how much depends on explicit scaling-limit, categorical, or continuation premises.

Canonical premise key. The six labeled technical items used across the paper corpus are introduced up front here so the overview can use them without forcing the reader to hunt for a later table. **T1** is loop coherence on the ordinary compact-gauge route. **T2** is the controlled refinement/scaling-limit scope clause for the Lorentz/null-modular/Einstein branch. **T3** is internalized on the compact surface as the derived fixed-cap generalized-entropy stationarity theorem for admissible fixed-cap MaxEnt variations on the realized cap-label-preserving MaxEnt family used in the D5 Einstein branch. **T4**–**T6** are the bosonic gauge-reconstruction premises: symmetric braiding, a faithful bosonic fiber functor or explicit super-Tannakian fork, and the directed colimit of transportable edge sectors with objectwise finite-dimensional fibers. The formal premise ledger appears in Section 3.2.6.

On this synthesis surface, recovered-core theorem status and claim tier inherit from the compact recovered-core ledger in Ref. [1], while lane-specific claim tier remains with the corresponding companion papers, including Refs. [4, 2, 3]; the present paper organizes those outputs across lanes but does not upgrade them by restating them.

The answer is sharply tiered. Fixed-cutoff collar structure, higher-gauge defect transport, the consensus/fixed-point formulation together with observable-level quotient confluence on the declared physical algebras, the record-algebra / Born-Lüders package with its quantitative stability surface, and the checkpoint/restoration package, including checkpoint data, restoration maps, future-law stability under restoration error, and an operational observer-identity criterion imported from Ref. [3], are theorem-bearing on their stated finite surfaces. The Lorentz/null-modular/Einstein chain remains a conditional scaling-limit branch, starting from finite type-I regulators but targeting an extracted scaling-limit observer algebra that may leave that class, and compact gauge reconstruction remains a conditional categorical branch. The particle lane is the most tiered part of the corpus: the structural carrier skeleton and realized Standard Model branch are fixed, the electroweak stage remains a forward-emitted non-core quantitative-closure branch whose internal transmutation data are fixed from P, with target-free public W/Z rows and a compare-only frozen validation pair, the Higgs/top stage is an exact source-only split theorem on the declared running, matching, and threshold surface with a compare-only exact inverse slice beneath it, the neutrino lane closes on its weighted-cycle theorem branch with diagnostic sidecars beneath it, the charged-lepton lane is fixed only up to exact centered readback plus a closed common-shift no-go, the quark lane closes on the public physical quark frame class selected by P without claiming a global classification of all quark frame classes, and the hadron lane remains unfinished.

The paper is organized by lane. Section 2 gives the synthesis-level map. Section 3 carries the recovered-core trunk: axioms, premise ledger, collar tools, gluing, the Lorentz/gravity branch, and the compact-gauge/Standard-Model branch. Section 4 records the consensus lane. Section 5 summarizes the particle lane. Section 6 records the screen-microphysics and observer lane. Section 7 records the controlled worldsheet effective-description lane. Section 8 gives the cross-lane theorem

boundary and open directions. The appendices collect supplementary architecture and synthesis-level material that is useful in this paper but not part of the recovered core itself.

2 Overview of Results

The current paper corpus supports the following lane-level split.

1. **At fixed cutoff, local structure and repair are sharp.** Technically, the collar theorem package is explicit and finite-range, the genuinely noncentral higher-gauge branch is closed at fixed cutoff by the crossed-module defect package, and the consensus lane gives a unique schedule-independent quotient normal form together with schedule-independent physical observable values on the declared fixed-cutoff physical algebras once the declared recovery-derived repair moves are committed by the touched-overlap local-fit contract that yields strict decrease of the inconsistency potential Φ, disjoint accepted repairs commute by support locality, the fixed-cutoff center-sector / higher-gauge gluing package makes overlapping repairs parenthesization-independent on the physical quotient, and repair completeness holds on the stated finite patch-net branch. The remaining open item on that genuinely noncentral lane is the refinement-limit transportable-sector lift used by the later compact-gauge branch. Microscopic representative lifts may still differ by gauge or sector relabelings inside one declared quotient-local glued state without changing the resulting observable values on those algebras.

2. **Relativity appears as a controlled continuum branch.** Technically, the Lorentz/null-modular/Einstein chain is a refinement/scaling-limit branch. It depends on the controlled refinement/scaling limit, the theorem-local realized transported geometric cap-local system whose carried-collar schedule sits above the transported fixed-local-collar Markov/faithfulness datum, and ordered cut-pair rigidity on the extracted prime geometric subnet; on that branch the finite cap regulators are type-I, but the scaling-limit observer algebra may be non-type-I, so the theorem is automorphism-level there and $K_C = 2\pi B_C$ is only the special type-I limit form; stronger BW / canonical-cap phrasing remains theorem-external unless the extraction-and-rigidity package is supplied. The null bridge also uses the theorem-local null-cut center transfer, the extra inherited left/right strip-split condition needed for the spatial-collar-type tensor decomposition, the exact-or-controlled Markov hypotheses on one fixed inherited strip model, and the geometric scaling action on the null half-line blow-up net that derives the half-sided modular pair, while the same-half-line local modular-Hamiltonian input and the separate bounded-interval projective branch enter only later; on that surface the exact-or-controlled strip additivity, the endpoint-Lipschitz renormalized half-line family, the weak tail generator, the derived half-sided modular inclusion, and the positive null-translation generator on its Stone domain with the affine half-line modular relation are internal to the bridge itself. The Einstein step adds the derived fixed-cap generalized-entropy stationarity theorem for admissible fixed-cap MaxEnt variations on the realized cap-label-preserving MaxEnt family, the internal small-ball bridge from the geometric cap generator together with the separate bounded-interval projective branch, the locally Lorentzian $d = 4$ scaling regime, the small-ball constancy and D4 carried-remainder hypotheses, and the all-directions/all-reference-states tensor-upgrade premise. On that same gravity lane, the local Einstein equation is fixed only modulo Λg_{ab}, and the additional screen-capacity input N_{scr} closes the same branch globally by $\Lambda = 3\pi/(GN_{\mathrm{scr}})$, thereby fixing the de Sitter static-patch package $(S_{\mathrm{dS}}, r_{\mathrm{dS}}, t_\Lambda)$; the observed cosmic age remains a downstream FLRW benchmark rather than an additional D6 theorem output.

3. **The Standard Model gauge skeleton appears as a constrained consistency output.** Technically, compact gauge reconstruction and the realized $\mathrm{SU}(3) \times \mathrm{SU}(2) \times \mathrm{U}(1)/\mathbb{Z}_6$ branch follow under the stated transportability and categorical premises, with the exact hypercharge lattice and the realized counting chain $N_g = 3$ then $N_c = 3$.

4. **The particle lane has real outputs and a nonuniform closure profile.** Technically, the structural carriers are exact, the electroweak stage remains on a non-core Phase-II quantitative-closure branch with target-free public W/Z rows, a Ward-projected electromagnetic transport family whose low-energy endpoint is tracked by the same forward closure program, with a numerical fixed-point witness and a separate certificate boundary, and a compare-only frozen validation pair. Its internal transmutation data are fixed forward from P; measured couplings enter as comparisons. The Higgs/top stage closes on a source-only split theorem surface, the charged-lepton lane carries exact centered readback plus a closed common-shift no-go, the neutrino lane closes on its weighted-cycle theorem branch with diagnostic sidecars below it, the quark lane closes on the public physical quark frame class selected by P without claiming a global classification of all quark frame classes, and the hadron lane remains an execution-contract surface without a mass-derivation theorem. At the local numerical level the same declared input P organizes the bosonic mass trunk and the gravity-side shared edge-entropy identity beneath the local G readout.

5. **The screen-microphysics lane is explicit and concrete.** Technically, it contains an explicit regulated reference architecture, sharpened patch-net and edge-sector branches, a closed fixed-cutoff record-algebra / Born-Lüders package with quantitative stability bounds, and a closed checkpoint/restoration package.

6. **The UV picture is sharper at fixed cutoff than in the continuum.** Technically, OPH fixes the physical UV branch only modulo implementation hiding and inert ancillary refinement; the UV burden is the realized transported geometric cap-local system with its derived carried-collar schedule above the transported fixed-local-collar Markov/faithfulness datum, then ordered cut-pair rigidity on the emitted scaling-limit geometric cap pair, and separately the refinement-limit transportable-sector lift.

7. **The consensus lane is exact but not overpromoted.** Technically, on each fixed finite patch net the normal form is a finite-state decidable object, approximate stability is only carried by the local splice and record error controls, and no universality theorem for a uniform family of growing patch nets and repair laws is part of the current five-paper corpus.

8. **The string/worldsheet lane is controlled but non-core.** Technically, it is carried as a theorem-level effective description on a distinct large-N_{edge} branch above the heat-kernel edge-sector theorem, with fixed-$\tau = tN_{\mathrm{edge}}$ window and uniform genus-remainder control explicit in the premises, but it is not part of the recovered core.

Companion Papers

The detailed depth surfaces used throughout this synthesis are:

1. *Recovering Relativity and the Standard Model from the OPH Package Rooted in Observer Consistency* [1], which carries the premise ledger, the recovered-core Lorentz/gravity route, and the compact-gauge/Standard-Model branch.

2. *Reality as a Consensus Protocol: The Fixed-Point Computation That Implements Physics* [2], which carries the finite patch-net fixed-point, defect, quotient, and operator-record consensus package.

3. *Screen Microphysics, Patches, and Observer Synchronization in OPH* [3], which carries the regulated reference architecture, the fixed-cutoff edge heat-kernel / Casimir theorem, and the measurement and observer checkpoint/restoration packages.

4. *Deriving the Particle Zoo from Observer Consistency* [4], which carries the structural-to-family particle route and the per-sector claim-tier ledger.

3 Recovered Core: Foundations and Structural Branches

Observer-Patch Holography is a reconstruction program for fundamental physics. Physical data live on a horizon screen S^2, and reality is encoded by the mutual consistency of overlapping patch descriptions. This paper works with five OPH axioms, one explicit external input used only in the cosmological branch, one Phase-II pixel variable fixed on a later outer/inner closure branch, and theorem-local technical premises stated where needed for the scaling, gauge, and transport arguments below.

Canonical premise key. The six labeled technical items used across the paper corpus are introduced up front here so the overview can use them without forcing the reader to hunt for a later table. **T1** is loop coherence on the ordinary compact-gauge route. **T2** is the controlled refinement/scaling-limit scope clause for the Lorentz/null-modular/Einstein branch. **T3** is internalized on the compact surface as the derived fixed-cap generalized-entropy stationarity theorem for admissible fixed-cap MaxEnt variations on the realized cap-label-preserving MaxEnt family used in the D5 Einstein branch. **T4**–**T6** are the bosonic gauge-reconstruction premises: symmetric braiding, a faithful bosonic fiber functor or explicit super-Tannakian fork, and the directed colimit of transportable edge sectors with objectwise finite-dimensional fibers. Section 3.2.4 states the same list in formal form.

3.1 Overview

The recovered core of the paper is the D1–D5 and D7–D9 chain on the declared fixed-cutoff and recovered-core branches: overlap-consistency normal form, collar and edge-center structure, the conditional Lorentz and Einstein branches, compact gauge reconstruction, and the realized Standard Model branch with exact hypercharge structure and the counting chain $N_g = 3$ then $N_c = 3$. Here and throughout the synthesis, labels such as D1–D12 are documentary node labels for the dependency ledger; the local key appears in Section 3.2.6, the SM/GR derivation paper *Recovering Relativity and the Standard Model from the OPH Package Rooted in Observer Consistency* gives the same ledger in shorter form [1], and the public paper ledger has no separate D11 row. D6 is not part of the recovered core because it needs total screen capacity as an external input, but it is the input-dependent local/global theorem stack for the same Einstein branch: the local null-data route leaves the metric term undetermined, the external screen-capacity identification closes that same branch, and the same D6 stack emits the de Sitter static-patch parameter package. D12 collects phenomenological continuations kept distinct from the recovered core. Within D1, the consensus paper classifies the declared fixed-cutoff status sharply: exact normal-form computation is finite-state and decidable, approximate stability is only collar-local through the splice

and record estimates, no global approximate-consensus theorem for long asynchronous noisy repair is claimed, and no theorem-grade universality claim is made for a uniform family of growing patch nets and repair laws. The imported D1 package is the fixed-cutoff consensus theorem stack: asynchronous confluence for the declared recovery-derived repair law, the cycle-obstruction and higher-gauge defect hierarchy, gauge-quotient descent, observable-level confluence on the declared physical observable algebras, and the record-algebra theorem on the observer-accessible surface. The imported remaining D1 burdens are repair completeness and the stated Petz support/CPTP clause where that branch is used. If the finite-candidate law-selection model is mentioned, its status stays exactly as in the consensus paper: a monotonicity theorem on a finite candidate space, not a universality theorem and not a literal cosmological dynamics claim. The scaling-limit bridge from that fixed-cutoff patch-net package to the later Lorentz, Einstein, and compact-gauge branches stays separate.

The Lorentz/null-modular/Einstein chain is a scaling-limit statement on the declared recovered-core branch. It depends on the controlled refinement/scaling limit, the theorem-local realized transported geometric cap-local system with its derived carried-collar schedule above the transported fixed-local-collar Markov/faithfulness datum, and the later ordered cut-pair rigidity step. On that branch the scaling-limit observer algebra may leave the finite type-I regulator class, so the theorem is automorphism-level there and $K_C = 2\pi B_C$ is only the special type-I limit form. The null bridge also uses the theorem-local null-cut center transfer, the extra inherited left/right strip-split condition needed to inherit the spatial-collar-type tensor decomposition, the exact-or-controlled Markov hypotheses, and the geometric scaling action on the null half-line blow-up net that derives the half-sided modular pair and the explicit positive null-translation generator on its Stone domain with the affine half-line modular relation, while the same-half-line local modular-Hamiltonian input enters only in the later generator/charge identification. The Einstein step then adds the derived fixed-cap generalized-entropy stationarity theorem for admissible fixed-cap MaxEnt variations on the realized cap-label-preserving MaxEnt family, the internal small-ball bridge from the geometric cap generator together with the separate bounded-interval projective branch, the locally Lorentzian $d = 4$ scaling regime, the small-ball constancy and D4 carried-remainder hypotheses, and the all-directions/all-reference-states tensor-upgrade premise. The gauge chain stays on the compact paper's recovered Standard Model surface: compact gauge reconstruction needs explicit loop coherence, symmetric braiding, and the transportable refinement-directed sector colimit with a faithful bosonic fiber functor; the realized one-generation plus one-Higgs branch then yields the realized Standard Model quotient, exact hypercharge lattice, and the counting chain $N_g = 3$ then $N_c = 3$. The controlled large-N_{edge} worldsheet effective-description branch, together with flavor, neutrino, hadron, dark-sector, and spectroscopy branches, is kept separate from the recovered core.

One external continuous input and one Phase-II pixel variable appear only where declared. The pixel area $P \equiv a_{\text{cell}}/\ell_P^2$ is fixed on the outer/inner closure branch summarized in Section 5, and the screen capacity $N_{\text{scr}} \equiv \log \dim \mathcal{H}_{\text{tot}}$ enters the cosmological-capacity branch. They are not additional axioms.

3.2 Model and Axioms

3.2.1 Observers and access model

An observer O is a tuple $(P_O, \mathcal{A}(P_O), \rho_O, R_O)$ where:

- $P_O \subset S^2$ is a connected screen patch (the observer's access region).
- $\mathcal{A}(P_O)$ is the von Neumann algebra associated to P_O.
- ρ_O is the local state, obtained by restricting the global state to $\mathcal{A}(P_O)$.

- R_O is a finite record algebra generated by stable internal record projectors within P_O.

Observers are internal patterns in the global state. Different observers correspond to different patches and their compatible marginals.

3.2.2 Screen, patches, and algebra net

We work in a single static patch with a horizon screen S^2. Each connected subregion $P \subset S^2$ is assigned a von Neumann algebra $\mathcal{A}(P)$. The net satisfies isotony:

$$P \subset Q \implies \mathcal{A}(P) \subset \mathcal{A}(Q).$$

A global state ω is a positive linear functional on the inductive-limit algebra. Overlap consistency is imposed algebraically: for overlaps $P_1 \cap P_2$, ω restricted to $\mathcal{A}(P_1 \cap P_2)$ is the same from either side.

3.2.3 Five OPH axioms

This paper uses five OPH axioms. The only labeled technical premises used below are the canonical premises **T1**–**T6** introduced above and stated formally in Section 1.4. Other theorem-local hypotheses are written out in prose where needed rather than assigned a separate label family.

1. Screen Net. A horizon screen S^2 carries a net of algebras $P \mapsto \mathcal{A}(P)$.

2. Overlap Consistency. Local states agree on shared observables for any overlap.

3. Local MaxEnt and Refinement Stability. At the UV scale the realized state maximizes entropy subject to a finite family of gauge-invariant local constraints, and the resulting family of states is stable under coarse-graining so symmetry-allowed relevant operators are not held at zero by unexplained fine tuning.

4. Recoverable Generalized Entropy. A generalized entropy functional exists on caps, obeys quantum focusing on null generators, and comes with the recoverability structure used throughout the manuscript: collar tripartitions have small conditional mutual information with controlled recovery maps, with stronger collar/null-strip hypotheses stated explicitly where needed.

5. Minimal Admissible Realization (MAR). On the admissible low-energy branch, the realized sector package is the lexicographically minimal one under the complexity vector $C(\mathfrak{S})$. MAR is an explicit structural-economy axiom on admissible realized branches, not a theorem derived from the preceding four axioms.

3.2.4 External inputs and theorem-local technical premises

The quantitative branches discussed here use one Phase-II pixel variable together with one external continuous input:

$$P \equiv a_{\text{cell}}/\ell_P^2, \qquad N_{\text{scr}} \equiv \log \dim \mathcal{H}_{\text{tot}} \sim 10^{122}.$$

Here P is the local UV-area ratio fixed on the outer/inner closure branch summarized in Section 5, while N_{scr} remains the external screen-capacity input used by the cosmological-capacity branch. Any retained capacity-level neutrino side estimate on that input is legacy bookkeeping rather than the live weighted-cycle theorem lane. These are quantitative-branch quantities, not extra axioms.

The list below introduces the only labeled technical premises used later, namely **T1**–**T6**. Fixed-cutoff regulator bookkeeping, quasi-local propagation, endpoint-Lipschitz control, collar decomposition, and related branch-internal statements are cited directly from axioms and earlier theorems rather than through a second premise-label system.

Local MaxEnt at the regulator scale. At the regulator scale ℓ_{UV}, the global state ω maximizes von Neumann entropy subject to:

1. A finite set $\{O_a\}$ of gauge-invariant local operators, each supported on a ball of radius $\leq r_0 = O(\ell_{\mathrm{UV}})$.
2. Constraint equations $\langle O_a(x)\rangle = c_a$ for each cell x in the UV lattice.
3. Optionally, a finite number of global constraints (total energy, charge).

This is the minimal specification needed to derive the local Gibbs form of Lemma 2.6 from Axiom 3.

Clarification (MaxEnt $\neq$ thermal equilibrium). MaxEnt here is **local state selection** given constraints, not "the universe is in thermal equilibrium." The Lagrange multipliers (inverse temperatures) may vary slowly in space and time. Non-equilibrium physics appears as gradients in these multipliers and as controlled violations of exact Markov additivity under the explicit mixing hypotheses stated later in Section 2.3. Equilibrium is an approximation regime with explicit error terms.

Rotationally invariant constraint branch. Constraint sets are SO(3)-invariant on S^2.

Gauge as overlap redundancy at fixed cutoff. Overlap identifications are not unique; the freedom that leaves overlap observables invariant forms a local groupoid.

Central-defect subbranch. On triple overlaps, when the failure of strict coherence is central, one has

$$\varphi_{ij}\varphi_{jk}\varphi_{ki} = \mathrm{Ad}(z_{ijk}), \qquad z_{ijk} \in Z(\mathcal{A}_{ijk}).$$

Collar double-scaling hypothesis. There exists a UV length ℓ_{UV} such that for any cap C and collar width δ, in the refinement limit $\delta \to 0$ and $\ell_{\mathrm{UV}} \to 0$ with $\delta/\ell_{\mathrm{UV}} \to \infty$, the Markov error satisfies

$$I(A_\delta : D_\delta \mid B_\delta)_\omega \leq \varepsilon(\delta/\ell_{\mathrm{UV}}), \qquad \varepsilon(x) \to 0 \text{ as } x \to \infty.$$

See Section 2.3 for the collar tripartition definitions and the regulated EC proof that yields this limit from the fixed-cutoff realized presentation.

Geometric-subnet BW lift. The continuum Lorentz statement is asked only on the extracted prime geometric subnet for caps, together with the controlled collar/refinement limit that carries the Markov and recovery remainders explicitly. The finite cap regulators are type-I algebras, but the scaling-limit observer algebra may leave that class, so the theorem is automorphism-level on the emitted geometric cap pair and $K_C = 2\pi B_C$ is only the special type-I limit form. In the canonical numbered list below, the only global item here is the scaling-limit scope clause **T2**; the theorem-local UV scaffold is a realized transported geometric cap-local system whose carried-collar schedule is derived from the transported fixed-local-collar Markov/faithfulness datum, followed by ordered cut-pair rigidity on the emitted scaling-limit geometric cap pair.

Derived quasi-local propagation and modular-locality control. At scale ℓ_{UV}, the automorphism group generated by the MaxEnt generator of Lemma 2.6, or more generally by any branch generator lying in the same bounded-support algebraic closure, has a finite Lieb–Robinson velocity v_{LR}: for local operators A, B supported on regions separated by distance d,

$$\|[A(t), B]\| \leq c\|A\|\|B\| \min(|R_A|, |R_B|)e^{-(d-v_{\mathrm{LR}}|t|)/\xi}$$

for $d > v_{\mathrm{LR}}|t|$, where $\xi = O(\ell_{\mathrm{UV}})$.

This is the branch-internal control statement that turns the quasi-local structure of the local-Gibbs branch into explicit support control for time-evolved operators. It is not an extra premise beyond the third axiom.

Approximate modular covariance on the fixed-cutoff realized branch. Under the fixed-cutoff realized presentation, Lemma 2.6 (local Gibbs), the derived quasi-local propagation bound above, and the collar double-scaling plus mixing hypotheses of Section 2.3, the modular flow $\sigma_t^{\omega,C}$ maps $\mathcal{A}(R)$ into a slightly thickened region algebra:

$$\sigma_t^{\omega,C}(\mathcal{A}(R)) \subseteq \mathcal{A}(R^{+v_{\mathrm{mod}}|t|}) \quad \text{up to error } \eta(d - v_{\mathrm{mod}}|t|),$$

where R^{+s} denotes the s-neighborhood thickening, v_{mod} is a "modular propagation velocity" controlled by v_{LR} and local norm bounds, and d is the distance from R to ∂C.

Heuristic motivation. The modular Hamiltonian $K_C = -\log \rho_C$ is quasi-local by Lemma 4.1a-b (modular additivity localizes it to the collar). The derived Lieb-Robinson control then bounds support spreading under $e^{iK_C t}$. In the double-scaling collar limit ($\delta/\ell_{\mathrm{UV}} \to \infty$), the thickening vanishes in macroscopic units. This is heuristic support for the tangent-limit package used later.

Continuum-limit heuristic. Define the induced region flow

$$f_t^C(R) := \lim_{\ell_{\mathrm{UV}} \to 0} R^{+v_{\mathrm{mod}}|t|}.$$

Then $\sigma_t^{\omega,C}(\mathcal{A}(R)) = \mathcal{A}(f_t^C(R))$ becomes exact in the continuum limit, with error controlled by

$$\eta(\delta) \lesssim 2\sqrt{\ln 2 \cdot c \cdot |\partial C|_{\mathrm{UV}}}\, e^{-\delta/(2\xi)}.$$

The discussion above is heuristic support for the tangent-limit package used in Theorem 4.2.

Refinement-stable MaxEnt branch. Choose any family of coarse-graining channels $\Phi_{\ell\to L}$ between UV scale ℓ and IR scale L that is compatible with the third OPH axiom. Then on the realized branch one may write

$$\Phi_{\ell\to L}(\omega_\ell(\lambda)) = \omega_L(R_{\ell\to L}(\lambda)),$$

for an induced map $R_{\ell\to L}$ on one common finite-dimensional multiplier family. Because the same finite constraint family is preserved, the regulator states lie in that shared multiplier family rather than in unrelated state spaces at different cutoffs, and the realized branch is the persistent trajectory or invariant subset selected inside it under the induced refinement maps. This is a state-side persistence statement; it does not by itself upgrade fixed-cutoff edge labels to a transportable refinement-persistent sector colimit.

Fixed-cutoff realized presentation. At a UV scale ℓ_{UV}, local patch algebras are type-I with finite-dimensional Hilbert spaces, and gauge-as-gluing is realized as a boundary action on a lifted finite-dimensional presentation. Section 2.3 proves the resulting fixed-cutoff collar package directly

from overlap consistency on this realized presentation, while distinguishing the lifted fixed-point algebra from the sector-preserving collar algebra used in the EC theorem.

External mathematical inputs: SSA and recovery theorems (Petz 1986, 1988; Fawzi and Renner 2015), the fixed-volume area-variation identity for small geodesic balls used in the D5 rest-frame relation, and the local quadratic-polarization argument used in the D5 tensor upgrade. The null modular bridge and the small-ball kernel are carried internally on the compact surface rather than imported from a separate EFT small-ball law. For SM contact we also use the Doplicher-Roberts reconstruction (Doplicher and Roberts 1989, 1990) once localized transportable sectors are assumed in the small-region limit. Full citations appear in the References.

Canonical technical premises. The only labeled technical premises used in this manuscript are:

- **T1**: the loop-coherence branch criterion kept explicit whenever the ordinary compact-group route is invoked, written $[z] = 0$ on the central-defect branch; on the genuinely noncentral fixed-cutoff branch the parallel criterion is $q_\Sigma = 0$, but that branch is handled by crossed-module data rather than by the ordinary DR field-algebra route.
- **T2**: all Lorentz/null-modular/Einstein statements are claims about a controlled refinement/scaling limit of the regulator net, not literal exactness at fixed finite cutoff.
- **T3**: the derived fixed-cap generalized-entropy stationarity theorem for admissible fixed-cap MaxEnt variations on the realized cap-label-preserving MaxEnt family used in the D5 Einstein branch.
- **T4**: symmetric braiding in the $3+1$D EFT branch.
- **T5**: a bosonic Tannakian fiber-functor premise, or else an explicit super-Tannakian fork.
- **T6**: whenever compact gauge reconstruction is invoked, the sector category means the directed colimit of transportable edge sectors with objectwise finite-dimensional fibers.

3.2.5 Notation

- ρ_C: reduced state on cap C.
- $K_C := -\log \rho_C^\omega$: modular Hamiltonian of the reference state.
- B_C: geometric generator of the cap-preserving conformal dilation.
- $S_{\mathrm{gen}}(C)$: generalized entropy on a cap.
- ℓ_{UV}: UV length scale of the refined screen net.
- δ: collar width around a cap boundary.

3.2.6 Summary: premise ledger and dependency DAG

For reference, the paper uses the five OPH axioms above, the closure-selected local pixel ratio P, the external capacity input N_{scr}, and the theorem-local technical premises declared in Section 1.4.

Item	Meaning, typical use, and status
Axiom 1	Screen Net; use: global net structure; status: axiom.
Axiom 2	Overlap Consistency; use: compatibility on overlaps; status: axiom.
Axiom 3	Local MaxEnt and Refinement Stability; use: state selection and RG stability; status: axiom.
Axiom 4	Recoverable Generalized Entropy; use: focusing plus recoverability structure; status: axiom.
Axiom 5	Minimal Admissible Realization (MAR); use: selection on admissible gauge branches; status: axiom.
P	Pixel area $a_{\mathrm{cell}}/\ell_P^2$; use: closure-selected local quantitative scale on the particle/gravity branch; status: fixed on the outer/inner closure branch.
N_{scr}	Total screen capacity; use: cosmological-capacity branch; status: external input.
T1	Loop-coherence branch criterion for the gauge route; use: central ordinary compact-group reconstruction, with the higher-gauge q_Σ analogue tracked separately at fixed cutoff; status: canonical technical premise when invoked.
T2	Controlled refinement/scaling-limit scope clause; use: Lorentz/null-modular/Einstein branch; status: canonical technical premise.

Item	Meaning, typical use, and status
T3	Derived fixed-cap generalized-entropy stationarity for admissible fixed-cap MaxEnt variations on the realized cap-label-preserving MaxEnt family; use: D5 Einstein branch; status: internal theorem-level criterion on the compact surface.
T4	Symmetric braiding in the $3+1$D EFT branch; use: compact gauge reconstruction; status: canonical technical premise.
T5	Bosonic Tannakian fiber functor or explicit super-Tannakian fork; use: compact gauge reconstruction; status: canonical technical premise.
T6	Directed colimit of transportable edge sectors with objectwise finite-dimensional fibers; use: compact gauge reconstruction; status: canonical technical premise.

Master dependency map. The completion program is organized into the following nodes. The point of the table is to make explicit which results use the OPH axioms alone, which import standard mathematics, and which require extra physical premises or external inputs.

Node	Main-manuscript output	Immediate OPH ingredients	Standard mathematics used	Extra physical premises / external inputs	Claim tier
D1	Theorem 3.1: schedule-independent normal-form statement for the declared recovery-derived repair relation	overlap-consistency setup on finite patch nets together with the declared fixed-cutoff collar recovery step, its touched-overlap local-fit acceptance contract, support-local disjoint commutation, and restriction-compatible union-collar gluing on the physical quotient	local-to-global Lyapunov descent on the inconsistency potential; local-diamond completion from overlap-associative center-sector / higher-gauge union-collar gluing; Newman's lemma	repair completeness; stated Petz support/CPTP control where that branch is used	structural theorem
D2	Section 2.3 and §5.4: EC/Markov collar and generalized-entropy split	Axioms 1–4 plus the derived fixed-cutoff regulator/collar package when invoked	HJPW; Petz; Fawzi–Renner	exact-Markov or idealized recoverability hypotheses when exact identities are used; coarse-grained $A/(4G)$ dictionary	structural lemma/ proposition layer

Node	Main-manuscript output	Immediate OPH ingredients	Standard mathematics used	Extra physical premises / external inputs	Claim tier
D3	Theorems 4.2–4.3: geometric modular flow and Lorentz branch	Axioms 1–4 together with SO(3)-invariant constraint data	Bisognano–Wichmann modular geometry; conformal classification on S^2	T2 together with the theorem-local realized transported geometric cap-local system whose carried-collar schedule is derived from the transported fixed-local-collar Markov/faithfulness datum, and ordered cut-pair rigidity on the emitted scaling-limit geometric cap pair; the scaling-limit observer algebra may leave the finite type-I regulator class, so the theorem is automorphism-level there and $K_C = 2\pi B_C$ is only the special type-I limit form	conditional scaling-limit theorem
D4	Section 5.2: null modular bridge with exact-or-controlled strip additivity, endpoint-Lipschitz renormalized half-line families, the weak tail generator, the derived half-sided modular inclusion, and the explicit positive half-line null-translation generator on its Stone domain with affine half-line modular relations	D2+D3	Borchers–Wiesbrock positive-generator theorem for standard half-sided inclusions; Stone's theorem; endpoint differentiation	the theorem-local null-cut center transfer, the extra inherited left/right strip-split condition needed for the spatial-collar-type tensor decomposition, and the exact-or-controlled Markov hypotheses of Section 5.2, plus T2; quasi-local propagation and endpoint-Lipschitz control are supplied internally by the local finite-constraint MaxEnt branch, the geometric scaling action on the null half-line blow-up net derives the half-sided modular pair, the same-half-line local modular-Hamiltonian formula is imported only for the later generator/charge identification, and bounded-interval formulas still require the separate interval-preserving projective branch	conditional bridge theorem

Node	Main-manuscript output	Immediate OPH ingredients	Standard mathematics used	Extra physical premises / external inputs	Claim tier
D5	Theorem 5.1: Einstein branch	D3+D4	fixed-volume area-variation identity for small geodesic balls; local quadratic-polarization argument for the tensor upgrade	the derived fixed-cap generalized-entropy stationarity theorem for admissible fixed-cap MaxEnt variations on the realized cap-label-preserving MaxEnt family, together with the internal small-ball bridge from the geometric cap generator, the separate interval-preserving projective branch used to transport the D4 half-line identification to bounded intervals, the locally Lorentzian $d = 4$ scaling regime, small-ball constancy assumptions, D4 carried remainder negligible at order ℓ^4, and the all-directions/all-reference-states premise used in the tensor upgrade	conditional scaling-limit theorem
D6	local/global theorem stack for Λ: null-invisible metric ambiguity, screen-capacity closure of the same Einstein branch, and the derived de Sitter static-patch parameter package, plus a legacy capacity-level neutrino side estimate	D5 local Einstein recovery leaves $+\Lambda g_{ab}$ undetermined	de Sitter entropy relation, static-patch radius/time formulas, and dimensional analysis	external input N_{scr} and the screen-capacity identification	input-dependent global closure / estimate
D7	Theorem 6.1: compact gauge reconstruction	Axioms 1–4	Doplicher–Roberts / Tannaka reconstruction	T1 and T4–T6 on the ordinary or central-defect bosonic branch; the genuinely noncentral fixed-cutoff branch is handled separately by crossed-module data	conditional structural theorem
D8	Section 6.2: product gauge structure up to finite quotient	D7 + Axiom 5 (MAR)	compact Lie representation classification; Schur's lemma	the same ordinary or central-defect bosonic branch as D7, together with a connected admissible class, one connected abelian factor, and faithful action on the minimal coupled carrier	realized-branch theorem

Node	Main-manuscript output	Immediate OPH ingredients	Standard mathematics used	Extra physical premises / external inputs	Claim tier
D9	Theorem 6.13, Proposition 6.9, Theorem 6.14, Proposition 6.6: hypercharge lattice, the counting chain $N_g = 3$ then $N_c = 3$, and the $\mathbb{Z}_6$ quotient	D8	anomaly algebra; Witten anomaly; CKM CP counting	realized one-generation/one-Higgs package and MAR admissibility premises	realized-branch theorem/corollary chain
D12	downstream flavor, dark-sector, baryogenesis, spectroscopy, and string continuations	various subsets of D6 and D9	branch-specific EFT and phenomenological manipulations	explicit additional ansätze	phenomenological continuation

In particular, D3–D5 are scaling-limit branches rather than fixed-cutoff matrix-algebra theorems; D7–D9 are realized-branch structural outputs in the bosonic internal-gauge sector; D6 is the input-dependent global closure of the D5 Einstein branch together with its derived de Sitter static-patch parameter package; and D12 remains phenomenological.

3.3 Information-Theoretic Tools

3.3.1 Strong subadditivity and Markov states

For any tripartite state ρ_{ABC},

$$I(A : C \mid B) := S(AB) + S(BC) - S(B) - S(ABC) \geq 0.$$

Exact Markov states satisfy $I(A : C \mid B) = 0$ and admit a recovery map:

$$\rho_{ABC} = (\mathrm{id}_A \otimes \mathcal{R}_{B\to BC})(\rho_{AB}).$$

3.3.2 Approximate recovery

If $I(A : C \mid B) \leq \varepsilon$ (bits), there exists a CPTP recovery map $\mathcal{R}$ with

$$\|\rho_{ABC} - (\mathrm{id}_A \otimes \mathcal{R})(\rho_{AB})\|_1 \leq 2\sqrt{\ln 2\, \varepsilon}.$$

3.3.3 Collar refinement and sufficient mechanisms

Fix a cap $C \subset S^2$ with boundary circle. For collar width δ define

$$B_\delta := \{x \in S^2 : \mathrm{dist}(x, \partial C) \leq \delta\}, \qquad A_\delta := C \setminus B_\delta, \qquad D_\delta := (S^2 \setminus C) \setminus B_\delta.$$

Then $S^2 = A_\delta \cup B_\delta \cup D_\delta$ with A_δ and D_δ interacting only through B_δ. The collar double-scaling hypothesis is the requirement that $I(A_\delta : D_\delta \mid B_\delta) \to 0$ in the refinement limit. At fixed regulator scale, the same finite-dimensional realized presentation supplies the patch-net and overlap-gluing data used below. We therefore isolate that realized presentation first, then separate the exact-Markov and quantitative routes.

Derived regulator realization. Choose a finite UV cellulation at scale ℓ_{UV} and let R be a finite union of cells. Hilbertize each cell's finite local data to a finite-dimensional space $\tilde{\mathcal{H}}_i \cong \mathbb{C}^{n_i}$. The extended algebra before quotienting by overlap redundancy is

$$\tilde{\mathcal{A}}(R) = \mathcal{B}(\tilde{\mathcal{H}}_R), \qquad \tilde{\mathcal{H}}_R = \bigotimes_{i \subset R} \tilde{\mathcal{H}}_i.$$

Overlap-preserving changes of local trivialization act only on cut data. On each finite-dimensional chart, their unitary image has compact closure, so one may represent the boundary gluing redundancy by a compact group $G_{\partial R} \subset U(\tilde{\mathcal{H}}_R)$. The lifted boundary-invariant endomorphism algebra is

$$A_{\mathrm{inv}}(R) = \tilde{\mathcal{A}}(R)^{G_{\partial R}} = \mathcal{B}(\tilde{\mathcal{H}}_R)^{G_{\partial R}}.$$

This is the fixed-cutoff realized presentation used throughout the collar analysis. The later EC theorem works on the invariant-state realization and uses the sector-preserving induced collar algebra there rather than the entire fixed-point algebra on the unreduced tensor product. When the consensus paper later speaks of quotient repair, the physical repair law is defined on this fixed-point / overlap-invariant data and any representative-level map is only a lift of that quotient update. Descent to the gauge quotient is therefore built into the physical-algebra formulation, not added later as a separate covariance axiom on hidden representatives. On the declared fixed-cutoff collar branch, the local repair step itself is read from exact Markov splice or a declared Petz/Fawzi–Renner recovery channel. What remains above that step on the declared fixed-cutoff branch is repair completeness, together with control on the declared Petz domain where that branch is used. The declared touched-overlap acceptance contract makes Φ the finite-patch Lyapunov functional for accepted moves, and the support-local disjoint-commutation plus restriction-compatible union-collar glue that turn overlap associativity into the local diamond are part of the declared repair law itself. On the same fixed-cutoff quantum lift, the terminal expectation functional on each declared physical observable algebra is therefore unique even when microscopic representative lifts differ by gauge labels globally or by sector labels inside one quotient-local glued state. Stability of that compatibility package under refinement or branch change is a separate question. The same consensus surface is classified explicitly as a finite-state exact theorem package: decidable normal-form computation with the Lyapunov step bound, collar-local approximate stability only through the splice and record controls, and no universality theorem for a uniform family of growing patch nets and repair laws.

Theorem 2.3 (EC from regulated overlap gluing). For a collar B_δ around a cap boundary Σ, there is a canonical decomposition

$$H_{B_\delta} = (\tilde{\mathcal{H}}_{B_L} \otimes \tilde{\mathcal{H}}_{B_R})^{G_\Sigma} = \bigoplus_\alpha (H_{b_L^\alpha} \otimes H_{b_R^\alpha}),$$

and the sector-preserving collar algebra

$$A_{\mathrm{EC}}(B_\delta) := \bigoplus_\alpha \big(B(H_{b_L^\alpha}) \otimes B(H_{b_R^\alpha})\big)$$

with

$$Z(A_{\mathrm{EC}}(B_\delta)) = \bigoplus_\alpha \mathbb{C}\,\mathbf{1}_\alpha,$$

such that $\mathcal{A}(A_\delta B_\delta)$ acts only on $H_{b_L^\alpha}$ and $\mathcal{A}(B_\delta D_\delta)$ acts only on $H_{b_R^\alpha}$ within each block.

Proof. Split the collar into half-collars B_L and B_R meeting on $\Sigma = \partial C$. By the realized regulator presentation above, the physical collar Hilbert space is the diagonal invariant subspace $(\tilde{\mathcal{H}}_{B_L} \otimes \tilde{\mathcal{H}}_{B_R})^{G_\Sigma}$. Decompose each side into irreps:

$$\tilde{\mathcal{H}}_{B_L} = \bigoplus_\alpha (V_\alpha \otimes H_{b_L^\alpha}), \qquad \tilde{\mathcal{H}}_{B_R} = \bigoplus_\beta (V_\beta^* \otimes H_{b_R^\beta}).$$

Then

$$\tilde{\mathcal{H}}_{B_L} \otimes \tilde{\mathcal{H}}_{B_R} = \bigoplus_{\alpha,\beta} (V_\alpha \otimes V_\beta^*) \otimes (H_{b_L^\alpha} \otimes H_{b_R^\beta}).$$

By Schur's lemma,

$$(V_\alpha \otimes V_\beta^*)^{G_\Sigma} \cong \begin{cases} \mathbb{C}, & \alpha = \beta, \\ 0, & \alpha \neq \beta. \end{cases}$$

Therefore the invariant subspace is

$$H_{B_\delta} = \bigoplus_\alpha (H_{b_L^\alpha} \otimes H_{b_R^\alpha}),$$

as claimed. The induced sector-preserving collar algebra is

$$A_{\mathrm{EC}}(B_\delta) = \bigoplus_\alpha (B(H_{b_L^\alpha}) \otimes B(H_{b_R^\alpha})),$$

so the center is generated by the block projectors. Adjacent region algebras act on the left or right factor only because the gluing action is supported on Σ. QED.

Remark. On the central-defect subbranch, replace G_Σ by its central extension. The sector label α then ranges over irreps of the extension; the decomposition is unchanged.

We refer to the decomposition in Theorem 2.3 as **edge-center completion (EC)**.

Exact Markov route. If the reference state on $A_\delta B_\delta D_\delta$ is exact Markov, or in an explicitly stated idealized limit that reduces to exact Markovity, then

$$\rho_{A_\delta B_\delta D_\delta} = \bigoplus_\alpha p_\alpha \big(\rho_{A_\delta b_L^\alpha} \otimes \rho_{b_R^\alpha D_\delta}\big), \qquad I_\omega(A_\delta : D_\delta \mid B_\delta) = 0.$$

This is the HJPW normal form applied to the EC blocks, and it is the exact identity used for literal Markov-modular equalities.

Interpreting collar refinement as the inductive limit of these regulators with $\delta/\ell_{\mathrm{UV}} \to \infty$, Theorem 2.3 supplies the kinematic edge-center decomposition. Exact Markovity is one idealized route. The following lemma and axiom provide the quantitative decay route used when the manuscript keeps the approximation explicit.

Lemma 2.6 (MaxEnt with local constraints implies local Gibbs form). Under the local MaxEnt branch of Axiom 3 and the finite-dimensional regulator realization above, the MaxEnt state has the Gibbs form

$$\omega = \frac{e^{-H_{\mathrm{eff}}}}{\mathrm{Tr}\, e^{-H_{\mathrm{eff}}}}, \qquad H_{\mathrm{eff}} = \sum_x \sum_a \lambda_a O_a(x) + (\text{global terms}),$$

where the sum runs over UV cells x and constraint operators O_a. The effective Hamiltonian H_{eff} is quasi-local with range $O(\ell_{\mathrm{UV}})$.

Proof. On a finite-dimensional algebra, the unique state maximizing $S(\rho) = -\mathrm{Tr}(\rho \log \rho)$ subject to linear constraints $\mathrm{Tr}(\rho O_i) = c_i$ is given by Lagrange multipliers:

$$\rho = \frac{e^{-\sum_i \lambda_i O_i}}{\mathrm{Tr}\, e^{-\sum_i \lambda_i O_i}}.$$

Strict concavity of von Neumann entropy ensures uniqueness. When the constraints are "translated local" (the same O_a at each cell x), the exponent is a sum of local terms. QED.

Exponential mixing hypothesis. There exist constants c and correlation length $\xi = O(\ell_{\mathrm{UV}})$ such that

$$I_\omega(A_\delta : D_\delta \mid B_\delta) \le c\, |\partial C|_{\mathrm{UV}}\, e^{-\delta/\xi}, \qquad |\partial C|_{\mathrm{UV}} \sim \frac{\mathrm{length}(\partial C)}{\ell_{\mathrm{UV}}}.$$

This is the standard fixed-cutoff clustering/mixing condition used for local Gibbs states. One sufficient realization on a chosen fixed-cutoff collar family is a Dobrushin-type uniqueness estimate for the corresponding local Gibbs specifications, or a spectral-gap estimate uniform across that family. Used here, it is a collar-local recoverability condition on the selected branch; it does not imply infinite-volume uniqueness and it does not decide whether a separately supplied refinement-limit gauge-sector colimit is trivial or nontrivial.

Theorem 2.5 (Local Gibbs + mixing implies collar refinement). Under Lemma 2.6 and the exponential mixing hypothesis above, the collar double-scaling hypothesis holds in the limit $\delta \to 0$, $\ell_{\mathrm{UV}} \to 0$ with $\delta/\ell_{\mathrm{UV}} \to \infty$.

Proof. The mixing bound above has polynomial growth in $|\partial C|_{\mathrm{UV}}$ and exponential decay in $\delta/\ell_{\mathrm{UV}}$. In the double-scaling limit the exponential dominates, so $I_\omega(A_\delta : D_\delta \mid B_\delta) \to 0$. QED.

This bound is the quantitative hinge for constructive gluing.

3.3.4 Concrete UV realization: quantum link models

Section 2.3 internalizes the regulator package: the fixed-cutoff type-I algebra and boundary fixed-point structure are the realized form of the screen net plus overlap gluing. Quantum link models are included here only as an explicit microscopic example of that structure, not as a separate axiom layer.

UV regulator. Triangulate S^2 at scale ℓ_{UV}, giving vertices v, oriented links ℓ, and plaquettes p. Refinement corresponds to $\ell_{\mathrm{UV}} \to 0$ with increasing lattice size.

Degrees of freedom. Attach to every oriented link ℓ a **finite-dimensional** Hilbert space $\mathcal{H}_\ell$. In ordinary Wilson lattice gauge theory, the continuum/refinement-limit edge description is modeled by $L^2(G)$ (infinite-dimensional for continuous G), but the microscopic OPH regulator is instead a **quantum link model** with finite-dimensional link Hilbert spaces that preserve gauge symmetry in operator form [32]. Optionally attach matter Hilbert spaces $\mathcal{H}_v$ at vertices. Then:

$$\tilde{\mathcal{H}}_{\mathrm{total}} = \bigotimes_\ell \mathcal{H}_\ell \otimes \bigotimes_v \mathcal{H}_v,$$

finite-dimensional on any finite lattice. This is a concrete realization of the extended type-I presentation used in Section 2.3.

Boundary gluing as Gauss-law invariants. Define a local gauge transformation group G_v at each vertex v acting on incident links (and matter at v). Physical states satisfy:

$$|\psi\rangle \in \mathcal{H}_{\text{phys}} \quad \Longleftrightarrow \quad U(g_v)|\psi\rangle = |\psi\rangle \;\; \forall\, v,\, g_v \in G_v.$$

Equivalently: $\mathcal{H}_{\text{phys}} = \tilde{\mathcal{H}}_{\text{total}}^{\prod_v G_v}$.

Region algebras. For any region $R \subset S^2$, define an extended Hilbert space $\tilde{\mathcal{H}}_R$ from the links/vertices in R. The **boundary gauge group** $G_{\partial R}$ acts on the cut degrees of freedom (the "half-links" ending on ∂R). Define:

$$\mathcal{A}_{\text{inv}}(R) = \mathcal{B}(\tilde{\mathcal{H}}_R)^{G_{\partial R}}.$$

This is the concrete quantum-link instance of the lifted fixed-point presentation used in Section 2.3. The same gauge constraints then induce the sector-preserving EC algebra on collars.

Why EC follows immediately. Take a cap C and a collar B_δ around ∂C. Because the *only* coupling between inside and outside is through the boundary gauge constraint, the collar Hilbert space decomposes into superselection blocks labeled by boundary irreps:

$$\mathcal{H}_{B_\delta} \cong \bigoplus_\alpha (H_{b_L^\alpha} \otimes H_{b_R^\alpha}),$$

with center generated by the projectors P_α. This is precisely the Schur-lemma mechanism of Theorem 2.3. The labels α are the familiar edge-mode / electric-flux labels appearing whenever one factorizes gauge theories across an entangling cut [33]. Exact Markovity depends on the state and is not forced by the decomposition alone.

Dynamics and MaxEnt. The natural Hamiltonian is a 2+1D lattice gauge Hamiltonian on the screen worldvolume: plaquette ("magnetic") terms, electric terms on links, vertex Gauss terms as constraints, plus local matter couplings. In quantum link form this remains finite-dimensional per link while behaving like gauge theory in the continuum limit. Then the MaxEnt assumption becomes concrete: the MaxEnt state is a Gibbs state $\rho \propto e^{-\sum_i \lambda_i O_i}$ with quasi-local O_i, precisely the local-Gibbs regime.

Geometry and G. This microphysics naturally supplies the emergent geometric objects:

- **Edge entropy / area operator:** $L_C = \sum_\alpha (\log d_\alpha) P_\alpha$ becomes "log of boundary irrep dimension" in the gauge link model.
- **Newton constant G:** the conversion factor between edge entropy density per boundary UV cell and macroscopic geometric area.

Thus area is an operator living in the center of the boundary algebra, because in gauge systems the center is where the cut labels live.

What this example does and does not close. The quantum-link realization makes the fixed-cutoff matrix/fixed-point bookkeeping concrete. What it does **not** automatically guarantee is that modular flow on caps becomes geometric conformal dilation with the 2π KMS normalization. The remaining continuum burden is the **T2** scaling-limit package together with the theorem-local geometric-subnet extraction and ordered cut-pair rigidity objects stated later. Viable architectures for this include holographic quantum error-correcting codes [34] and quantum double / string-net Hamiltonians [35].

3.3.5 Conformal-modular fixed-point microphysics

After Section 2.3, the remaining gap is narrower. On the local finite-constraint MaxEnt branch of the third axiom, the logarithm of the selected state is a quasi-local UV generator, and the refinement-stable branch lies inside one common finite-dimensional multiplier family. The unresolved question is which algebraic phase the refinement limit realizes. At each regulator stage the patch and cap algebras are type-I matrix algebras, but the finite regulator class is not refinement-closed, so the scaling-limit observer algebra may leave that class. The fixed-cutoff analysis therefore proves a local-Gibbs/refinement-stable branch with propagation and endpoint-Lipschitz control, but it does *not* prove that every such branch emits the prime geometric cap pair with geometric modular flow. The conditional boundary is the **T2** scaling-limit clause together with the theorem-local geometric-subnet extraction and ordered cut-pair rigidity inputs: the target algebra is the extracted geometric subnet, while the internal extraction of its realized scaling-limit cap pair depends on the derived carried-collar schedule above the transported fixed-local-collar Markov/faithfulness datum and the later weak-$*$ extraction plus GNS gluing step.

Local finite-constraint MaxEnt branch. The constraint family $\mathcal{C}$ is generated by finitely many gauge-invariant local densities $\{O_a(x)\}$ of UV range $O(\ell_{\mathrm{UV}})$, with the same finite label set retained under refinement. This is not an extra postulate beyond the third OPH axiom; it is that axiom unpacked at regulator level.

Theorem 2.6 (Local constraints imply a local-Gibbs form). If the MaxEnt constraints are expectations of finitely many quasi-local operators $\{O_a\}$ with bounded support size at scale ℓ_{UV}, then the entropy maximizer is

$$\omega \propto \exp\left(-\sum_a \lambda_a O_a\right),$$

so the MaxEnt generator $H_{\mathrm{MaxEnt}} = -\log\omega$ is a UV-range quasi-local sum. This is exactly the local-Gibbs form used later.

Proof. Standard exponential-family result: maximum entropy subject to linear constraints $\langle O_a\rangle = c_a$ yields the Gibbs state with Lagrange multipliers λ_a. QED.

Derived propagation control on the same branch. Because H_{MaxEnt} is a finite-range or quasi-local sum on a finite type-I regulator net, standard Lieb–Robinson estimates [6] apply to the automorphism group it generates, or to any branch generator lying in the same bounded-support algebraic closure. Thus there is a finite propagation velocity v_{LR} and constants C, ξ such that for local observables A_X, B_Y,

$$\big\|[\tau_t(A_X), B_Y]\big\| \le C\,\|A_X\|\,\|B_Y\|\,\min(|X|,|Y|)\,e^{-(d(X,Y)-v_{\mathrm{LR}}|t|)/\xi}.$$

Because changing a bounded interval endpoint only adds or removes an $O(|\Delta v|)$ collar of local terms once the central endpoint piece is separated off, the same local branch also gives bounded-interval endpoint-Lipschitz matrix elements,

$$\big|\langle\psi, (K[I'] - K[I])\phi\rangle\big| \le C_{\psi,\phi,I_{\max}}\,|I'\Delta I|,$$

used later in the null-modular bridge. So quasi-local propagation and endpoint-Lipschitz control are not external regularity selectors; they are the local-constraint MaxEnt branch written in dynamical form.

Refinement-stable multiplier branch. Because the same finite constraint family is preserved under coarse-graining, the regulator states lie in one common finite-dimensional multiplier family rather than in unrelated state spaces at different cutoffs. The "refinement-stable" language

used later therefore means persistence along the stable or fixed branch of this multiplier family. This state-side notion is enough to compare realized states across cutoffs. It does *not* by itself upgrade fixed-cutoff edge labels to transportable refinement-persistent sectors, and it does not show that the directed colimit of such sectors is nontrivial. Accordingly, whenever the gauge derivation speaks of a refinement-stable directed colimit of sectors, the entire transportable directed-system/fiber clause is the explicit content of **T6**; the third OPH axiom supplies only the realized state branch along which one asks whether such sector transport persists.

Scaling limit and algebraic type. The regulator presentation gives a family of finite type-I algebras

$$\mathcal{A}_\ell(C) \cong \mathcal{B}(\mathcal{H}_{C,\ell}).$$

A scaling limit of this family need not remain type I. In the continuum-QFT case of interest one expects the local limit algebra $\mathcal{A}_\infty(C)$ to be non-type-I, typically type III. Accordingly the fundamental modular datum in the limit is the automorphism group of the pair $(\mathcal{A}_\infty(C), \omega_\infty^C)$, not a density matrix inside $\mathcal{A}_\infty(C)$.

Proposition 2.6 (Geometric modular action on caps on the extracted prime geometric subnet). Under the **T2** scaling-limit clause and the theorem-local geometric cap-pair extraction and ordered cut-pair rigidity hypotheses used later in Theorem 4.2, for any extracted scaling-limit geometric cap pair $(\mathcal{A}_\infty^{\mathrm{geo}}(C), \omega_\infty^{\mathrm{geo},C})$, let $\alpha_{\lambda_C(s)}$ be the automorphism induced by the standard cap-preserving conformal subgroup $\lambda_C(s)$. Then

$$\sigma_t^{\omega_\infty^{\mathrm{geo},C}} = \alpha_{\lambda_C(2\pi t)}.$$

If the limit cap algebra happens to remain type I, this may be written as $K_C = 2\pi B_C$. In the generic continuum case, the same statement is an outer modular action on a non-type-I algebra.

Proof. Ordered cut-pair rigidity on the extracted prime geometric cap pair identifies the scaling-limit cap modular group with the standard geometric cap subgroup up to normalization, and the modular KMS condition fixes the normalization to 2π. If the limit algebra is type I, the automorphism statement may be represented by a modular Hamiltonian K_C; otherwise the automorphism statement is the full content. QED.

Alternative derivation via net regularity. The same modular-covariance property can also be read off from a scaling-limit support map when the net satisfies the outer-regularity / minimal-support condition used later. This does *not* complete the BW lift by itself; it clarifies how the geometric labeling of the extracted limit net is read once the geometric cap pair has been emitted.

(NR) Outer regularity / minimal support. For any operator O, the intersection of all connected regions P with $O \in \mathcal{A}(P)$ is again a connected region, denoted $\mathrm{supp}(O)$.

Proposition 2.7 (Modular covariance from net regularity). Under (NR), define for any region $R \subset C$

$$f_t^C(R) := \bigcup_{O \in \mathcal{A}(R)} \mathrm{supp}\Big(\sigma_t^{\omega,C}(O)\Big).$$

Then $\sigma_t^{\omega,C}(\mathcal{A}(R)) = \mathcal{A}(f_t^C(R))$, which is exactly the desired modular-covariance property.

Proof. Since $\sigma_t^{\omega,C}$ is an automorphism of $\mathcal{A}(C)$, and (NR) allows one to read support from the net labeling, the map $R \mapsto f_t^C(R)$ is well-defined and consistent. QED.

Null-surface modular structure. On the same **T2** scaling branch and extracted geometric-subnet branch, the null-surface modular machinery narrows as follows:

- **Fixed-cutoff null-strip bridge.** The null-strip package proves transferred cut-center data, the theorem-local inherited left/right strip-split condition needed for the spatial-collar-type

tensor decomposition, exact-or-controlled four-term strip additivity on one inherited strip model, renormalized endpoint-Lipschitz control up to the weak tail generator, and the derived half-sided modular pair whose Borchers–Wiesbrock consequence is an explicit positive null-translation generator on its Stone domain with the affine half-line modular relation.

- **Derived half-sided modular inclusion.** Nested null half-line algebras satisfy half-sided modular inclusion on the declared geometric scaling branch by Corollary 5.2e; Borchers–Wiesbrock then identifies the positive null-translation generator on its Stone domain together with the affine half-line modular relation [10].

- **Weak continuity and finite variation.** The bounded-interval and half-line endpoint-Lipschitz control follow from the local MaxEnt branch and define the weak tail generator at fixed cutoff. The **T2** scaling branch together with the extracted geometric cap pair supplies the continuum null-generator setting in which that weak-tail data can be matched to the geometric null modular action of the relativistic phase.

Constraint set specification. On the local finite-constraint branch, the "correct fixed-cap constraint set" becomes explicit: constraints are the local conserved charges of the symmetries used in the derivation:

1. **Edge/cap label constraints:** fix the distribution of collar-sector labels, equivalently $\langle L_C \rangle$ for each cap size, giving the area term.

2. **Gauge charges:** fix boundary flux or charge operators.

3. **Geometric (conformal) charges:** fix the expectation of the conformal Killing charges that preserve the cap, i.e. the generator B_C or its microscopic lattice approximation.

MaxEnt therefore selects the unique finite-stage invariant state compatible with those conserved charges. What it does *not* prove by itself is that the scaling-limit branch lands in the BW / canonical cap phase with standard geometric modular action. The stronger statement is the **T2** scaling-limit clause together with the theorem-local extraction of a realized geometric cap pair and the later ordered cut-pair rigidity step. The corresponding shorthand in this manuscript means only the case in which such a geometric cap pair has been emitted; if the limit algebra is non-type-I, the geometric modular action on that branch is outer.

QNEC internalization. QNEC has rigorous QFT proofs in broad settings [14]. On the **T2** scaling branch with the extracted geometric cap pair in place, the Recoverable Generalized Entropy axiom can be supported internally by:

- EC + MaxEnt derive $S_{\text{gen}} = S_{\text{bulk}} + \langle L_C \rangle$ (Section 5.4).

- Once the downstream null-stress bridge and Einstein branch are in place, focusing becomes a semiclassical consequence in the same scaling regime.

Summary. The CMFP package therefore separates the dependency structure into two layers:

- **Internal branch consequences:** the local-Gibbs form, quasi-local propagation, bounded-interval endpoint-Lipschitz control, and the realized state-side refinement branch along which later transport questions are asked, all from the local finite-constraint MaxEnt branch.

- **Theorem-external branch statement:** that the scaling limit emits the prime geometric cap pair and that ordered cut-pair rigidity holds on it. On that branch the limit algebra may be non-type-I, the modular action may be outer, and the 2π normalization and later null half-sided-inclusion bridge follow.

The internal extension route is a two-step chain, and the pressure point is the first object. First, one needs a realized scaling-limit observer cap-pair extraction on the geometric subnet from transported marginals. The theorem contract visible in the manuscript is that the realized transported cap-local system packages the geometric cap-local test family, the projectively compatible transported marginal family, and the asymptotic transport-equivalence certificate. The remaining emitted witness is a derived vanishing carried-collar schedule on fixed local collars, explicitly decomposed into the constructive-recovery remainder and the faithful modular-defect term, namely convergence of the carried quantity

$$r_{\mathrm{FR}}(\varepsilon_{n,m,\delta}) + 4\lambda_*^{-1}\,\delta^{\mathrm{M}}_{A_{n,m,\delta}:B_{n,m,\delta}:D_{n,m,\delta}}(\varepsilon_{n,m,\delta})$$

to zero after transport to each fixed local collar model; only then can local weak-$*$ extraction and GNS gluing emit the realized scaling-limit geometric cap pair. Beneath that derived schedule, the sharper raw datum is the transported fixed-local-collar Markov/faithfulness package, and the single missing clause inside that lower datum is the eventual common floor on the finite modular-transport family feeding the faithful modular-defect term. No separate exact-Markov-reference faithfulness input is missing there: on one fixed collar model, the exact-Markov comparison marginals inherit the same eventual floor once the exact-Markov modulus tends to zero. The manuscript does not derive a refinement-uniform common floor on the finite modular-transport marginals; this single clause remains external to the emitted theorem chain. Second, one needs ordered cut-pair rigidity on that realized limit. That stage remains symbolic until the realized scaling-limit geometric cap pair exists. The weak-$*$ extraction plus rigidity route is therefore an explicit internal extension route, not part of the proved theorem surface.

3.4 Overlap Consistency and Gluing

The constructive part of overlap consistency is the tree-gluing theorem below. The structural part is the origin of the gluing redundancy itself. On the ordinary or central-defect branch, gauge-as-gluing is the finite-regulator overlap redundancy of local chart presentations: once the overlap algebras are realized in finite-dimensional charts, overlap-consistent rechartings of a cut form a compact unitary transition system. The collar theorem of Section 2.3 should therefore be read with its boundary group G_Σ understood as shorthand for this derived compact boundary action, not as a model-specific add-on. On the genuinely noncentral branch, the same weak gluing data are encoded by a compact crossed-module change system, so the fixed-cutoff collar theorem upgrades to a higher-gauge statement rather than failing. The fixed-cutoff topological package is therefore closed on all three branches. A second structural point is UV underdetermination. If each patch Hilbert space is tensored with an inert finite ancillary factor and the observable patch algebras are embedded as $\mathcal{A}(P) \otimes \mathbf{1}$, then the physical observables, collar conditional mutual information, carried Markov errors, and quotient normal forms are unchanged. So the physical UV branch is fixed only modulo gauge or implementation hiding together with such ancillary stabilization, not a unique microscopic presentation; the unique theorem-grade object is the induced family of terminal expectation functionals on the declared physical observable algebras, not a preferred microscopic representative.

3.4.1 Constructive gluing on tree covers

Theorem 3.1 (tree gluing). Let a rooted tree of patches satisfy a tree-ordered overlap structure and a tripartite factorization (A_k, B_k, C_k) at step k. If a target state ρ^* obeys $I(A_k : C_k \mid B_k) \le \varepsilon_k$, then there exist recovery maps $\mathcal{R}_k$ such that

$$\|\rho^*_{A_kB_kC_k} - (\mathrm{id}_{A_k} \otimes \mathcal{R}_k)(\rho^*_{A_kB_k})\|_1 \le \delta_k,$$

with

$$\delta_k = 2\sqrt{\ln 2\, \varepsilon_k}.$$

The iteratively glued state $\hat{\rho}$ satisfies

$$\|\hat{\rho} - \rho^*\|_1 \le \min\left(2, \sum_{k=2}^{n} \delta_k\right).$$

Proof. Induct on k. The recovery error contracts under CPTP maps, so the errors add. QED.

3.4.2 Gauge-as-gluing and loops

At finite regulator scale, the fixed-cutoff gauge-as-gluing package identifies the overlap-consistency redundancy of local chart presentations. Choose finite-dimensional local presentations of the overlap algebras on a connected cut Σ. Because the overlap algebras are matrix algebras, any overlap-consistent change of chart is inner and is implemented by a unitary on the cut Hilbert space. Fixing a reference chart, the compact closure of the subgroup generated by all such recharting unitaries is the boundary gluing group K_Σ; before fixing the reference chart, the same data form a compact unitary groupoid.

Proposition 3.2a (Derived gauge-as-gluing at finite regulator). Let a finite regulator chart be chosen for the patches meeting along a connected interface Σ. Then overlap consistency determines a compact unitary transition system on the cut data. On the ordinary or central-defect branch, this transition system reduces to a compact boundary group K_Σ, and when the triple-overlap defect is central its projective composition law lifts to a genuine action of a compact central extension $\widehat{K_\Sigma}$. Gauge is therefore the overlap redundancy itself, not an extra primitive.

Proof. On a finite-dimensional matrix algebra every *-automorphism is inner, so each overlap-consistent recharting is conjugation by a unitary on the cut Hilbert space. The subgroup generated by those unitaries has compact closure inside a finite-dimensional unitary group. If triple-overlap defects are central, the resulting projective composition law lifts to a central extension. QED.

Lemma 3.2b (trees vs loops). If the patch adjacency graph is a tree, one can choose local charts h_i so that the overlap labels satisfy $g_{ij} = h_i^{-1} h_j$ on all edges. If loops exist, the loop holonomy

$$H(\gamma) = g_{i_1 i_2} g_{i_2 i_3} \cdots g_{i_n i_1}$$

is invariant under local frame changes. Nontrivial holonomy is the obstruction to global trivialization. QED.

Corollary 3.2c (Collar consequence on the ordinary or central-defect branch). Let $B_\delta = B_L \cup B_R$ be a collar around a cap boundary Σ, and set $\widehat{K_\Sigma} = K_\Sigma$ on the ordinary branch. Then the EC theorem of Section 2.3 has the derived interpretation

$$H_{B_\delta} = (\tilde{\mathcal{H}}_{B_L} \otimes \tilde{\mathcal{H}}_{B_R})^{\widehat{K_\Sigma}} \cong \bigoplus_\alpha (H_{b_L^\alpha} \otimes H_{b_R^\alpha}),$$

with

$$Z(A_{\mathrm{EC}}(B_\delta)) = \bigoplus_\alpha \mathbb{C}\, \mathbf{1}_\alpha.$$

The right half-collar carries the contragredient action because it sees the inverse transport across the same cut. Exact Markov normal forms used later require the additional state hypothesis $I_\omega(A_\delta : D_\delta \mid B_\delta) = 0$ or the explicitly stated idealized recoverability limit; the block decomposition itself is kinematic and follows from the derived boundary action.

Proof sketch. Decompose the left boundary data into irreps $(V_\alpha \otimes H_{b_L^\alpha})$ of $\widehat{K}_\Sigma$ and the right boundary data into the dual modules $(V_\beta^* \otimes H_{b_R^\beta})$. Then Schur's lemma leaves a singlet only when $\alpha = \beta$, producing the displayed direct sum. QED.

3.4.3 Loop obstruction class (central defect)

On the central-defect subbranch, define central defects z_{ijk} by

$$\varphi_{ij}\varphi_{jk}\varphi_{ki} = \mathrm{Ad}(z_{ijk}) \quad \text{on } \mathcal{A}_{ijk}.$$

Then $\{z_{ijk}\}$ is a Čech 2-cocycle, and its cohomology class $[z]$ is gauge invariant. Central defects do not obstruct the collar block decomposition: they only replace K_Σ by its central extension $\widehat{K}_\Sigma$. Ordinary loop-coherent gluing exists iff $[z] = 0$. (A full proof appears in Section 6.4 below, in the algebra-net language.)

3.4.4 Non-central obstruction (2-group cocycle)

When defects are not central, the natural coefficient data is a crossed module $(H \to G)$ with an action of G on H by conjugation. Here G is the reconstructed gauge group, and H is the unitary group acting on edge multiplicity spaces, with boundary map $\partial : H \to G$.

A crossed module is a homomorphism $\partial : H \to G$ together with an action of G on H such that

$$\partial(g \triangleright h) = g\,\partial(h)\,g^{-1}, \qquad \partial(h) \triangleright h' = hh'h^{-1}.$$

On a good cover $\{P_i\}$, a weakly coherent gluing is encoded by:

$$g_{ij} : P_{ij} \to G, \qquad h_{ijk} : P_{ijk} \to H,$$

obeying the 2-cocycle conditions

$$g_{ij}g_{jk} = \partial(h_{ijk})g_{ik},$$

and on quadruple overlaps,

$$h_{jkl}h_{ijl} = (g_{ij} \triangleright h_{ikl})h_{ijk}.$$

Gauge changes act by 1- and 2-cochains in the standard way for crossed-module cohomology.

Theorem 3.4 (non-central obstruction). Loop-coherent gluing exists iff the 2-cocycle (g_{ij}, h_{ijk}) is equivalent to the trivial cocycle in nonabelian Čech H^2 with values in the crossed module $(H \to G)$.

Proof sketch. Strict gluing corresponds to $h_{ijk} = 1$ and $g_{ij}g_{jk} = g_{ik}$. Gauge changes are exactly the crossed-module coboundaries, so strictification exists iff the 2-class is trivial. QED.

The central-defect case is the abelian truncation with H central and trivial action, which reduces to Section 3.3. A genuinely noncentral class is the point where the fixed-cutoff collar theorem upgrades from the ordinary-group package to the higher-gauge one below.

Corollary 3.4a (Fixed-cutoff higher-gauge EC and transportability). For a finite regulator chart on a connected cut Σ, the genuinely noncentral branch admits a compact crossed-module change system

$$\mathcal{T}_\Sigma = C^1(N_\Sigma, H_\Sigma) \rtimes C^0(N_\Sigma, G_\Sigma).$$

The physical higher-gauge collar is

$$\mathcal{H}_B^{2g} = (\widetilde{\mathcal{H}}_{B_L} \otimes \widetilde{\mathcal{H}}_{B_R})^{\mathcal{T}_\Sigma} \cong \bigoplus_\lambda (\mathcal{H}_{b_L^\lambda} \otimes \mathcal{H}_{b_R^\lambda}),$$

with

$$Z(\mathcal{A}_{2g}(B)) = \bigoplus_\lambda \mathbb{C}\,\mathbf{1}_\lambda,$$

and the defect class

$$q_\Sigma = [(g,h)] \in \check{H}^2(N_\Sigma, H_\Sigma \to G_\Sigma)$$

is invariant under local rechartings, classifies fixed-cutoff genuinely noncentral sectors, and vanishes iff the defect is removable.

Proof sketch. Pair-overlap rechartings are inner, while triple-overlap associators strictify to compact crossed-module data. Finite-dimensional unitary $\mathcal{T}_\Sigma$-modules decompose semisimply, and Schur matching leaves only diagonal left/right blocks. The transport statement is the crossed-module Čech analogue of the central-defect case. QED.

3.5 Modular Flow and Lorentz Kinematics

3.5.1 Modular additivity in the Markov collar limit

Consider a collar tripartition $A : B : D$ around a cap boundary, with the EC decomposition of Section 2.3 understood. Define, for a faithful reference state ω,

$$\Delta K(\omega) := K_{ABD}(\omega) - K_{AB}(\omega) - K_{BD}(\omega) + K_B(\omega).$$

Exact modular additivity is not a consequence of EC alone. It is available only on the exact Markov set, or along a controlled fixed-cutoff family that approaches that set on one fixed collar model. Three quantities must therefore be kept separate:

1. the raw collar conditional mutual information $I(A : D \mid B)$;
2. the constructive Fawzi–Renner comparison error

$$r_{\mathrm{FR}}(\varepsilon) := 2\sqrt{1 - e^{-\varepsilon}} \le 2\sqrt{\varepsilon};$$

3. the fixed-collar exact-Markov replacement modulus

$$\delta^{\mathrm{M}}_{A:B:D}(\varepsilon) := \sup\left\{ \inf_{\sigma \in \mathfrak{M}_{A:B:D}} \|\rho - \sigma\|_1 : I(A : D \mid B)_\rho \le \varepsilon \right\},$$

where

$$\mathfrak{M}_{A:B:D} := \{\sigma_{ABD} : I(A : D \mid B)_\sigma = 0\}.$$

Lemma 4.1a (Exact Markov implies exact additivity up to a central term). If $I(A:D\mid B)_\omega = 0$, then the EC decomposition of Section 2.3 puts the state in HJPW block form

$$\omega_{ABD} = \bigoplus_\alpha p_\alpha\, \omega^{(\alpha)}_{Ab^\alpha_L} \otimes \omega^{(\alpha)}_{b^\alpha_R D},$$

and $\Delta K(\omega)$ is central. On the canonical HJPW block model one may take

$$\Delta K(\omega) = 0.$$

Equivalently, there exists a central operator $K_{\partial,ABD}(\omega) \in Z(\mathcal{A}(B))$ such that

$$K_{ABD}(\omega) = K_{AB}(\omega) + K_{BD}(\omega) - K_B(\omega) + K_{\partial,ABD}(\omega).$$

Proof. On each HJPW block the modular Hamiltonians of ABD, AB, BD, and B are the logarithms of tensor-product states with the same classical block weight p_α. The blockwise logarithms therefore cancel exactly in the combination $K_{ABD} - K_{AB} - K_{BD} + K_B$. If one keeps the blockwise endpoint-label bookkeeping explicit rather than absorbing it into the canonical block identification, the remainder is a direct sum of block constants, hence central. QED.

Proposition 4.1b (Controlled exact-Markov replacement on a fixed collar). Because the state space is compact and conditional mutual information is continuous in finite dimension,

$$\delta^{\mathrm{M}}_{A:B:D}(\varepsilon) \longrightarrow 0 \qquad (\varepsilon \downarrow 0).$$

Thus small collar CMI converges to the exact HJPW normal form only in this controlled fixed-cutoff sense; the manuscript does not claim a dimension-free one-shot trace-norm bound from small $I(A:D\mid B)$ directly to an exact Markov state.

Corollary 4.1c (Carried collar defect operator). Let ω_ε satisfy $I(A:D\mid B)_{\omega_\varepsilon} \le \varepsilon$, and choose $\sigma_\varepsilon \in \mathfrak{M}_{A:B:D}$ with

$$\|\omega_\varepsilon - \sigma_\varepsilon\|_1 \le \delta^{\mathrm{M}}_{A:B:D}(\varepsilon).$$

Define the carried defect operator

$$\mathfrak{D}_{A:B:D}(\omega_\varepsilon, \sigma_\varepsilon) := \Delta K(\omega_\varepsilon) - \Delta K(\sigma_\varepsilon).$$

Then

$$K_{ABD}(\omega_\varepsilon) = K_{AB}(\omega_\varepsilon) + K_{BD}(\omega_\varepsilon) - K_B(\omega_\varepsilon) + K_{\partial,ABD}(\sigma_\varepsilon) + \mathfrak{D}_{A:B:D}(\omega_\varepsilon, \sigma_\varepsilon),$$

where $K_{\partial,ABD}(\sigma_\varepsilon) = \Delta K(\sigma_\varepsilon)$ is central. For every bounded observable X supported on $A \cup B$,

$$|\mathrm{Tr}[X(\omega_\varepsilon - \sigma_\varepsilon)]| \le \|X\|_\infty\, \delta^{\mathrm{M}}_{A:B:D}(\varepsilon).$$

If the relevant marginals of ω_ε and σ_ε are uniformly faithful with lower spectral bound $\lambda_* > 0$, then

$$\|\mathfrak{D}_{A:B:D}(\omega_\varepsilon, \sigma_\varepsilon)\|_\infty \le 4\lambda_*^{-1}\, \delta^{\mathrm{M}}_{A:B:D}(\varepsilon).$$

Independently, Fawzi–Renner recovery supplies a constructive comparison state $\omega^{\mathrm{rec}}_\varepsilon$ with

$$\|\omega_\varepsilon - \omega^{\mathrm{rec}}_\varepsilon\|_1 \le r_{\mathrm{FR}}(\varepsilon), \qquad r_{\mathrm{FR}}(\varepsilon) := 2\sqrt{1 - e^{-\varepsilon}} \le 2\sqrt{\varepsilon}.$$

This $O(\varepsilon^{1/2})$ term is the finite-stage observable error. The modulus $\delta^{\mathrm{M}}_{A:B:D}(\varepsilon)$ is instead the fixed-collar quantity that justifies replacing the physical state by an exact Markov normal form in the later geometric modular arguments.

Accordingly, the manuscript's later exact collar formulas take one of two forms:

1. literal exactness when the reference state is exact Markov on the relevant collar; or
2. a controlled collar family for which $\delta^{\mathrm{M}}_{A:B:D}(\varepsilon_\delta) \to 0$, with the constructive Fawzi–Renner remainder and the carried operator defect $\mathfrak{D}_{A:B:D}$ kept explicit at finite stage.

3.5.2 Theorem: BW_{S^2} on the extracted prime geometric subnet

The collar analysis proves a fixed-cutoff statement on the finite type-I regulator net: the reduced cap state has a literal density matrix, its modular Hamiltonian exists, and its nonadditive part is confined to a shrinking collar up to carried errors. The Lorentz claim is therefore not a fixed-cutoff matrix-algebra identity. Its target is the refinement-limit geometric cap pair $(\mathcal{A}_\infty^{\mathrm{geo}}(C), \omega_\infty^{\mathrm{geo},C})$ singled out by the controlled scaling clause **T2** together with the theorem-local geometric cap-pair extraction. Nothing in the local MaxEnt branch by itself proves that the scaling-limit cap algebra remains type I, and nothing in the fixed-cutoff MaxEnt/collar package by itself emits that prime geometric cap pair. Ordered cut-pair rigidity is then what turns the extracted pair into the BW statement.

Fix a cap $C \subset S^2$ and a shrinking collar family $(A_\delta, B_\delta, D_\delta)$ around ∂C. Write

$$\varepsilon_\delta := I(A_\delta : D_\delta \mid B_\delta)_\omega, \qquad r_{\mathrm{FR}}(\varepsilon_\delta) := 2\sqrt{1 - e^{-\varepsilon_\delta}} \le 2\sqrt{\varepsilon_\delta},$$

and, on each fixed faithful collar model,

$$\eta_\delta^{\mathrm{M}} := 4\lambda_*^{-1}\, \delta^{\mathrm{M}}_{A_\delta:B_\delta:D_\delta}(\varepsilon_\delta),$$

where $\lambda_* > 0$ is the lower spectral bound used to compare modular Hamiltonians. All exact collar formulas in this subsection are therefore to be read either literally at exact Markovity or along a controlled collar family satisfying

$$\delta^{\mathrm{M}}_{A_\delta:B_\delta:D_\delta}(\varepsilon_\delta) \to 0 \qquad (\delta \downarrow 0),$$

with $r_{\mathrm{FR}}(\varepsilon_\delta)$ and η_δ^{M} carried explicitly.

For each cap C, let $\lambda_C(s) \subset \mathrm{Conf}^+(S^2)$ denote the standard cap-preserving conformal one-parameter subgroup, normalized so that the null blow-up near a smooth cut acts by $v \mapsto e^{-s}v$, and let $\alpha_{\lambda_C(s)}$ denote the induced automorphism of the scaling-limit cap net.

Theorem 4.2 (BW_{S^2} on the extracted prime geometric subnet). Let $C \subset S^2$ be a round cap. Assume the OPH axioms, the derived fixed-cutoff regulator/collar package established earlier, the controlled scaling-limit scope clause **T2**, and the theorem-local UV inputs that:

1. the transported cap marginals on the geometric subnet admit a realized scaling-limit geometric cap pair $(\mathcal{A}_\infty^{\mathrm{geo}}(C), \omega_\infty^{\mathrm{geo},C})$ through the carried-collar local weak-$*$ / GNS extraction package; and

2. ordered cut-pair rigidity holds on that extracted prime geometric pair.

Let $\lambda_C(s)$ denote the standard cap-preserving conformal one-parameter subgroup, normalized so that the null blow-up near a smooth cut acts by $v \mapsto e^{-s}v$. Then the scaling-limit modular automorphism group is

$$\sigma_t^{\omega_\infty^{\mathrm{geo},C}} = \alpha_{\lambda_C(2\pi t)}.$$

At finite regulator stage the nonadditive part of

$$K_C^{(\delta)} := -\log \rho_C^{(\delta)}$$

is confined to the shrinking collar up to the carried errors $r_{\mathrm{FR}}(\varepsilon_\delta)$ and η_δ^{M}. No separate cap-isotropy/SO(2)-equivariance selector or Euclidean-regularity premise is used. If the scaling-limit cap algebra remains type I, the same automorphism identity may be represented by

$$K_C = 2\pi B_C;$$

otherwise the automorphism identity is the full statement and the action is outer. Consequently, for every controlled collar family with $\delta^{\mathrm{M}}_{A_\delta:B_\delta:D_\delta}(\varepsilon_\delta) \to 0$ and surviving faithfulness bound, replacing the finite-stage modular action by the geometric cap-dilation action incurs only the carried errors $r_{\mathrm{FR}}(\varepsilon_\delta)$ and η^{M}_δ, and these vanish in the refinement limit.

Proof. Markov locality localizes the modular defect to the shrinking collar, with carried finite-stage errors $r_{\mathrm{FR}}(\varepsilon_\delta)$ and η^{M}_δ. By **T2**, the theorem concerns only scaling-limit cap pairs. By the theorem-local extraction hypothesis, the relevant limit object is the prime geometric cap pair. Ordered cut-pair rigidity identifies the scaling-limit modular automorphism group with the standard cap-preserving conformal flow up to normalization:

$$\sigma_t^{\omega_\infty^{\mathrm{geo},C}} = \alpha_{\lambda_C(\kappa_C t)}$$

for some $\kappa_C > 0$. Because s is normalized by the null blow-up $v \mapsto e^{-s}v$, the modular KMS condition fixes $\kappa_C = 2\pi$. Thus

$$\sigma_t^{\omega_\infty^{\mathrm{geo},C}} = \alpha_{\lambda_C(2\pi t)}.$$

If the limit algebra remains type I, this automorphism identity is represented by $K_C = 2\pi B_C$; otherwise the automorphism identity is the full statement and the action is outer. QED.

Scope note. The theorem fixes the 2π normalization and the carried refinement limit on the extracted geometric cap pair. It does *not* prove that MaxEnt alone emits that pair. The remaining UV-side scaffold is exactly a realized transported geometric cap-local system whose carried-collar schedule

$$r_{\mathrm{FR}}(\varepsilon_{n,m,\delta}) + 4\bar{\lambda}^{-1}_{m,\delta}\delta^{\mathrm{M}}_{A_{n,m,\delta}:B_{n,m,\delta}:D_{n,m,\delta}}(\varepsilon_{n,m,\delta})$$

is derived from transported CMI together with an eventual fixed-local-collar modular-transport common floor, followed by ordered cut-pair rigidity on the emitted scaling-limit geometric cap pair. No separate cap-isotropy or Euclidean-regularity premise is used, and no second comparison-state faithfulness clause is missing.

3.5.3 Theorem: BW_{S^2} implies Lorentz kinematics

Theorem 4.3 (Lorentz kinematics on the screen). Under the hypotheses of Theorem 3.5.2,

$$\mathrm{Conf}^+(S^2) \cong \mathrm{PSL}(2,\mathbb{C}) \cong \mathrm{SO}^+(3,1).$$

The cap modular flows are therefore the standard one-parameter Lorentz boost/dilation subgroups in the celestial-sphere realization, and the induced local kinematic group is the connected Lorentz group.

Proof. Orientation-preserving conformal maps of S^2 are exactly the Möbius transformations, so

$$\mathrm{Conf}^+(S^2) \cong \mathrm{PSL}(2,\mathbb{C}) \cong \mathrm{SO}^+(3,1).$$

By Theorem 3.5.2, on the extracted prime geometric subnet the realized scaling-limit cap modular automorphism groups are the standard cap-preserving conformal dilations with the 2π normalization fixed internally. Hence the local kinematics induced by modular flow is the connected Lorentz group. QED.

3.6 Gravity from Fixed-Cap Generalized-Entropy Stationarity

3.6.1 Cap first law

Section 4.2 fixes the cap modular statement first at the automorphism level. On the extracted prime geometric subnet, the scaling-limit cap modular flow is geometric with the standard 2π normalization,

$$\sigma_t^{\omega_\infty^{\mathrm{geo},C}} = \alpha_{\lambda_C(2\pi t)}.$$

If the realized scaling-limit cap algebra happens to remain type I, one may choose a density matrix $\rho_C^{\omega_\infty^{\mathrm{geo},C}}$ and modular Hamiltonian

$$K_C := -\log \rho_C^{\omega_\infty^{\mathrm{geo},C}},$$

and then for a perturbation $\rho(\varepsilon)$ with $\rho(0) = \omega_\infty^{\mathrm{geo},C}$ the first-law identity reads

$$\delta S_C = \delta\langle K_C\rangle = 2\pi\,\delta\langle B_C\rangle.$$

In the generic continuum/non-type-I case, Section 4.2 does not supply an inner operator identity $K_C = 2\pi B_C$; the theorem surface there is only the geometric modular automorphism statement, so the cap first-law formula is conditional on the special type-I realization.

3.6.2 Null-surface modular bridge

This subsection extends the fixed-cutoff weak-tail-generator boundary to the derived positive null-translation stage and the exact half-line generator/charge identification. At fixed cutoff one first transfers cut-center data to regulated null strips, then imposes the extra inherited split condition needed for the spatial-collar-type tensor decomposition, obtains exact or controlled strip additivity on one inherited strip model, and derives a weak tail generator for the renormalized half-line family. On the scaling-limit geometric-cap branch of Theorem 4.2, the null half-line blow-up net then inherits geometric dilation and therefore half-sided modular inclusion, so Borchers–Wiesbrock supplies an explicit positive null-translation generator on its Stone domain. The remaining downstream boundary is narrower: bounded-interval formulas still require the separate interval-preserving projective branch, and the later tensor upgrade still carries the null-invisible metric ambiguity. Accordingly the c.12-local bridge consists of:

- null-cut center transfer yielding the central cut-label decomposition, together with the extra inherited split condition needed for the spatial-collar-type tensor decomposition;
- exact or controlled four-term strip additivity on one fixed inherited strip model;
- endpoint-Lipschitz control for the renormalized half-line family and the resulting weak tail generator;
- derived half-sided modular inclusion on the null half-line blow-up net, and the resulting Borchers positive translation generator.

The later density formula and T_{kk} bridge are kept below only as explicit downstream templates rather than as fixed-cutoff theorems proved in this subsection. Lemma 5.2f makes the positive null-translation generator itself explicit: it records the Borchers unitary group, positivity and self-adjointness on the Stone domain, the corrected affine commutator relation, and the half-line modular-Hamiltonian identity $K_a = K_0 - 2\pi a P_\Omega$. Bounded-interval formulas remain downstream.

Proposition 5.2a (Null-cut center transfer and inherited split). Fix a regulated null tripartition

$$I_- = (v_1, v_2), \qquad J = (v_2, v_3), \qquad I_+ = (v_3, v_4),$$

with cuts $\Gamma_- := \{v = v_2\}$ and $\Gamma_+ := \{v = v_3\}$. Under the fixed-cutoff realized presentation, assume that the two null cuts inherit the ordinary or central-defect boundary-redundancy data used in the spatial collar branch, so that in a compatible type-I regulator presentation one has

$$\tilde{\mathcal{H}}_{I_-} \cong \bigoplus_{\alpha_-} W_{\alpha_-} \otimes \mathcal{H}_{i_-^{\alpha_-}},$$

$$\tilde{\mathcal{H}}_J \cong \bigoplus_{\alpha_-,\alpha_+} W^*_{\alpha_-} \otimes \mathcal{H}_{j^{\alpha_-,\alpha_+}} \otimes W_{\alpha_+},$$

$$\tilde{\mathcal{H}}_{I_+} \cong \bigoplus_{\alpha_+} W^*_{\alpha_+} \otimes \mathcal{H}_{i_+^{\alpha_+}},$$

where opposite sides of each cut carry inverse transport. If the strip algebra is the commutant of the transported cut actions, then

$$\mathcal{A}(J) \cong \bigoplus_{\alpha_-,\alpha_+} \mathcal{B}(\mathcal{H}_{j^{\alpha_-,\alpha_+}}), \qquad Z(\mathcal{A}(J)) = \bigoplus_{\alpha_-,\alpha_+} \mathbb{C}\, P_{\alpha_-,\alpha_+}.$$

If, in addition, each multiplicity space factors as

$$\mathcal{H}_{j^{\alpha_-,\alpha_+}} \cong \mathcal{H}_{j_L^{\alpha_-,\alpha_+}} \otimes \mathcal{H}_{j_R^{\alpha_-,\alpha_+}},$$

with $\mathcal{A}(I_- \cup J)$ acting blockwise only on $\mathcal{H}_{i_-^{\alpha_-}} \otimes \mathcal{H}_{j_L^{\alpha_-,\alpha_+}}$ and $\mathcal{A}(J \cup I_+)$ acting blockwise only on $\mathcal{H}_{j_R^{\alpha_-,\alpha_+}} \otimes \mathcal{H}_{i_+^{\alpha_+}}$, then

$$\mathcal{A}(J) \cong \bigoplus_{\alpha_-,\alpha_+} \mathcal{B}(\mathcal{H}_{j_L^{\alpha_-,\alpha_+}}) \otimes \mathcal{B}(\mathcal{H}_{j_R^{\alpha_-,\alpha_+}}),$$

and

$$\mathcal{H}_{I_- \cup J \cup I_+} \cong \bigoplus_{\alpha_-,\alpha_+} \mathcal{H}_{i_-^{\alpha_-}} \otimes \mathcal{H}_{j_L^{\alpha_-,\alpha_+}} \otimes \mathcal{H}_{j_R^{\alpha_-,\alpha_+}} \otimes \mathcal{H}_{i_+^{\alpha_+}}.$$

Proof. Complete reducibility gives the displayed decompositions. Commuting with the two transported cut actions forces strip operators to preserve the sector pair (α_-, α_+) and act only on the multiplicity space; Schur's lemma kills the off-diagonal intertwiners and leaves the direct-sum algebra above. Taking invariants across both cuts in the glued tripartition again uses Schur's lemma and leaves exactly the matching sector pairs. The left/right split of the multiplicity spaces is additional structure; it is not forced by the center transfer alone. QED.

Corollary 5.2b (Exact or controlled four-term null modular relation on an inherited strip model). On one fixed finite-dimensional strip model satisfying Proposition 5.2a, define

$$\mathfrak{M}_{I_-:J:I_+} := \{\sigma : \ I(I_- : I_+ \mid J)_\sigma = 0\},$$

and

$$\delta^{\mathrm{M}}_{I_-:J:I_+}(\varepsilon) := \sup\left\{ \inf_{\sigma \in \mathfrak{M}_{I_-:J:I_+}} \|\rho - \sigma\|_1 : I(I_- : I_+ \mid J)_\rho \le \varepsilon \right\}.$$

For any strip state η, let

$$\Delta K_J(\eta) := K_{I_- \cup J \cup I_+}(\eta) - K_{I_- \cup J}(\eta) - K_{J \cup I_+}(\eta) + K_J(\eta).$$

If the strip reference state ω is exact Markov, then $\Delta K_J(\omega)$ is central, and on the canonical inherited HJPW model one may take

$$\Delta K_J(\omega) = 0.$$

Equivalently, there exists a central operator $K_{\partial,J}(\omega) \in Z(\mathcal{A}(J))$ such that

$$K_{I_-\cup J\cup I_+}(\omega) = K_{I_-\cup J}(\omega) + K_{J\cup I_+}(\omega) - K_J(\omega) + K_{\partial,J}(\omega).$$

If instead $I(I_- : I_+ \mid J)_\omega \le \varepsilon$, choose $\widetilde{\omega}_J \in \mathfrak{M}_{I_-:J:I_+}$ with

$$\|\omega - \widetilde{\omega}_J\|_1 \le \delta^{\mathrm{M}}_{I_-:J:I_+}(\varepsilon).$$

Define

$$\mathfrak{D}_J(\omega, \widetilde{\omega}_J) := \Delta K_J(\omega) - \Delta K_J(\widetilde{\omega}_J).$$

Then

$$K_{I_-\cup J\cup I_+}(\omega) = K_{I_-\cup J}(\omega) + K_{J\cup I_+}(\omega) - K_J(\omega) + K_{\partial,J}(\widetilde{\omega}_J) + \mathfrak{D}_J(\omega, \widetilde{\omega}_J),$$

with $K_{\partial,J}(\widetilde{\omega}_J) = \Delta K_J(\widetilde{\omega}_J) \in Z(\mathcal{A}(J))$. Every bounded observable on $I_- \cup J$ or $J \cup I_+$ then differs from the exact-Markov reference by at most

$$\|O\|_\infty \, \delta^{\mathrm{M}}_{I_-:J:I_+}(\varepsilon).$$

If the relevant strip marginals are uniformly faithful with lower spectral bound $\lambda_* > 0$, then

$$\|\mathfrak{D}_J(\omega, \widetilde{\omega}_J)\|_\infty \le 4\lambda_*^{-1} \, \delta^{\mathrm{M}}_{I_-:J:I_+}(\varepsilon).$$

Independently, Fawzi–Renner gives a recovered comparison state with trace-norm error

$$r_{\mathrm{FR}}(\varepsilon) = 2\sqrt{1 - e^{-\varepsilon}} \le 2\sqrt{\varepsilon}.$$

Thus the exact four-term strip relation is available only at exact Markovity or in a controlled strip family on one fixed inherited strip model with $\delta^{\mathrm{M}}_{I_-:J:I_+}(\varepsilon_J) \to 0$, while $\mathfrak{D}_J$ is carried explicitly at finite stage.

Proof. On the inherited split of Proposition 5.2a, the exact-Markov case is the same HJPW block calculation as in the spatial collar theorem: the four logarithms cancel blockwise on the canonical model, and any residual bookkeeping term is central. The controlled replacement is the same fixed-model compactness argument used for ordinary collars, after relabeling $A, B, D \mapsto I_-, J, I_+$; the operator-norm bound on $\mathfrak{D}_J$ is the corresponding relabeling of the modular-transport estimate. QED.

Definition (Renormalized null modular functional). For a null interval I on generator Ω, let $K_\partial(I, \Omega)$ denote the central endpoint-label term singled out by Corollary 5.2b on the inherited strip model, or by its controlled exact-Markov replacement when that model is used as reference. Define

$$\widetilde{K}[I, \Omega] := K[I, \Omega] - K_\partial(I, \Omega), \qquad \widetilde{K}_a(\Omega) := \widetilde{K}[(a, \infty), \Omega].$$

Proposition 5.2c (Endpoint-Lipschitz null modular families and weak tail generator). For renormalized modular Hamiltonians on one null generator,

$$\widetilde{K}[I, \Omega] := K[I, \Omega] - K_\partial(I, \Omega), \qquad \widetilde{K}_a(\Omega) := \widetilde{K}[(a, \infty), \Omega],$$

the branch-internal endpoint-control coming from the local finite-constraint MaxEnt branch gives:

$$\big|\langle \psi, (\widetilde{K}[(a', b'), \Omega] - \widetilde{K}[(a, b), \Omega])\phi\rangle\big| \le C_{\psi,\phi,\Omega}\big(|a' - a| + |b' - b|\big)$$

for bounded intervals in a compact endpoint window, and

$$\big|\langle \psi, (\widetilde{K}_{a'}(\Omega) - \widetilde{K}_a(\Omega))\phi\rangle\big| \le C_{\psi,\phi,\Omega}|a' - a|$$

for half-lines. Hence

$$f_{\psi,\phi}(a) := \langle \psi, \widetilde{K}_a(\Omega)\phi \rangle$$

is locally Lipschitz and therefore absolutely continuous, and its distributional derivative defines the weak tail generator

$$\langle \psi, q(a,\Omega)\phi \rangle := -\frac{1}{2\pi}\,\partial_a f_{\psi,\phi}(a).$$

For $a < b$,

$$\langle \psi, (\widetilde{K}_b(\Omega) - \widetilde{K}_a(\Omega))\phi \rangle = -2\pi \int_a^b \langle \psi, q(v,\Omega)\phi \rangle\, dv.$$

At this stage $q(a,\Omega)$ is a weak tail generator, not yet a local operator-valued density; its identification with the positive self-adjoint Borchers generator occurs only after Corollary 5.2e and Lemma 5.2f.

Proof. The derived endpoint-control estimate applies to the renormalized family after removal of the central endpoint term. For bounded intervals, the symmetric-difference length is bounded by $|a' - a| + |b' - b|$; for half-lines it is exactly $|a' - a|$. Therefore the matrix-element functions are Lipschitz in the endpoints. A Lipschitz function on a compact interval is absolutely continuous, so $f_{\psi,\phi}(a)$ has an L^∞_{loc} derivative and obeys the fundamental theorem of calculus. Defining q as the rescaled negative derivative gives the displayed integral relation. QED.

Downstream c.12–c.13 boundary. The fixed-cutoff bridge established above is the end of this subsection's proved content.

Lemma 5.2d (Downstream density-upgrade template). If, in addition, the weak tail generator $q(a,\Omega)$ is weakly differentiable in a and both $f_{\psi,\phi}(a)$ and $q_{\psi,\phi}(a)$ vanish at $+\infty$, then one may define a density

$$\langle \psi, p(a,\Omega)\phi \rangle := -\partial_a \langle \psi, q(a,\Omega)\phi \rangle = \frac{1}{2\pi}\,\partial_a^2 \langle \psi, \widetilde{K}_a(\Omega)\phi \rangle$$

and obtain the half-line formula

$$\langle \psi, \widetilde{K}_a(\Omega)\phi \rangle = 2\pi \int_a^\infty (v-a)\,\langle \psi, p(v,\Omega)\phi \rangle\, dv.$$

This is a downstream density-upgrade step and is not proved from the fixed-cutoff strip arguments above.

Null half-line blow-up net. Fix a smooth cut point and choose affine coordinate v on generator Ω so that the cut sits at $v = 0$. For $a \geq 0$, write

$$H_a := (a, \infty), \qquad \mathcal{M}_a(\Omega) := \overline{\bigvee_{a<c<d<\infty} \mathcal{A}((c,d),\Omega)}.$$

By isotony of the null interval net,

$$a \leq b \quad \Longrightarrow \quad \mathcal{M}_b(\Omega) \subseteq \mathcal{M}_a(\Omega).$$

Corollary 5.2e (Derived half-sided modular inclusion on null half-lines). On the scaling-limit geometric-cap branch of Theorem 4.2, the blow-up modular action near a smooth entangling cut acts on the null coordinate by

$$v \mapsto e^{-2\pi t} v.$$

Therefore, for every $a \geq 0$,

$$\sigma_t^\omega(\mathcal{M}_a(\Omega)) = \mathcal{M}_{e^{-2\pi t} a}(\Omega).$$

Hence, for every $a > 0$,

$$\sigma_t^\omega(\mathcal{M}_a(\Omega)) \subseteq \mathcal{M}_a(\Omega) \qquad (t \le 0),$$

so the inclusion $\mathcal{M}_a(\Omega) \subset \mathcal{M}_0(\Omega)$ is half-sided modular. After the harmless convention change $t \mapsto -t$, this is the standard positive-time half-sided-inclusion form. The half-sided modular inclusion is therefore derived from the null-interval structure, isotony, and the scaling-limit geometric action rather than imported separately.

Proof. Blow up the cap modular flow of Theorem 4.2 near a smooth cut and restrict to the chosen null generator. In the tangent limit the cap-preserving flow becomes the null dilation $v \mapsto e^{-2\pi t}v$. For every bounded interval $(c, d) \subset H_a$, this sends $\mathcal{A}((c, d), \Omega)$ to $\mathcal{A}((e^{-2\pi t}c, e^{-2\pi t}d), \Omega)$, and taking the von Neumann closure of the interval net yields the displayed identity for $\mathcal{M}_a(\Omega)$. If $t \le 0$, then $e^{-2\pi t}a \ge a$, so $H_{e^{-2\pi t}a} \subseteq H_a$, and isotony gives $\mathcal{M}_{e^{-2\pi t}a}(\Omega) \subseteq \mathcal{M}_a(\Omega)$. QED.

Lemma 5.2f (Positive null-translation generator). For the derived half-sided modular inclusion

$$\mathcal{M}_a(\Omega) \subset \mathcal{M}_0(\Omega) \qquad (a > 0),$$

let $\Delta_0(\Omega)$ be the modular operator of the standard pair $(\mathcal{M}_0(\Omega), \omega)$ and define $K_0(\Omega) := -\log \Delta_0(\Omega)$. Borchers–Wiesbrock then yields a unique strongly continuous one-parameter unitary group

$$U_\Omega(a) = e^{iaP_\Omega}, \qquad a \in \mathbb{R},$$

such that

$$U_\Omega(a)\omega = \omega, \qquad U_\Omega(a)\mathcal{M}_b(\Omega)U_\Omega(a)^* = \mathcal{M}_{a+b}(\Omega) \quad (a, b \ge 0),$$

and whose generator P_Ω is positive and self-adjoint on the Stone domain

$$D(P_\Omega) := \left\{ \psi \in \mathcal{H} : \lim_{a \to 0} \frac{U_\Omega(a)\psi - \psi}{ia} \text{ exists} \right\}.$$

Moreover,

$$\Delta_0(\Omega)^{it} U_\Omega(a) \Delta_0(\Omega)^{-it} = U_\Omega(e^{-2\pi t}a), \qquad \Delta_0(\Omega)^{it} P_\Omega \Delta_0(\Omega)^{-it} = e^{-2\pi t} P_\Omega,$$

and on the common invariant analytic core of $K_0(\Omega)$ and P_Ω,

$$[K_0(\Omega), P_\Omega] = -\, i\, 2\pi P_\Omega.$$

If $\Delta_a(\Omega)$ denotes the modular operator of $(\mathcal{M}_a(\Omega), \omega)$ and $K_a(\Omega) := -\log \Delta_a(\Omega)$, then

$$\Delta_a(\Omega) = U_\Omega(a)\Delta_0(\Omega)U_\Omega(a)^*, \qquad K_a(\Omega) = U_\Omega(a)K_0(\Omega)U_\Omega(a)^*,$$

so

$$K_a(\Omega) = K_0(\Omega) - 2\pi a\, P_\Omega$$

as a quadratic-form identity on $D(K_0(\Omega)) \cap D(P_\Omega)$. Equivalently,

$$\langle \psi, (K_b(\Omega) - K_a(\Omega))\phi \rangle = -2\pi(b - a)\langle \psi, P_\Omega \phi \rangle$$

for all $\psi, \phi \in D(K_0(\Omega)) \cap D(P_\Omega)$. In the canonical normalization of Corollary 5.2b, the weak endpoint derivative from Proposition 5.2c is therefore exactly the Borchers generator. Bounded-interval modular-Hamiltonian formulas remain downstream and require the separate interval-preserving branch recorded in Theorem 5.2g.

Proof. Corollary 5.2e gives the required half-sided modular inclusion. Borchers–Wiesbrock then yields the unique strongly continuous unitary group $U_\Omega(a)$, its positive self-adjoint generator

P_Ω, and the affine covariance relation with the modular group; Stone's theorem gives the displayed domain formula. Differentiating

$$\Delta_0(\Omega)^{it} U_\Omega(a) \Delta_0(\Omega)^{-it} = U_\Omega(e^{-2\pi t} a)$$

first at $a = 0$ and then at $t = 0$ on the common analytic core gives

$$[K_0(\Omega), P_\Omega] = -\, i\, 2\pi P_\Omega.$$

Because $U_\Omega(a)\omega = \omega$ and $U_\Omega(a)\mathcal{M}_0(\Omega)U_\Omega(a)^* = \mathcal{M}_a(\Omega)$, uniqueness of modular data for the translated standard pair gives the displayed formulas for $\Delta_a(\Omega)$ and $K_a(\Omega)$. Differentiating $K_a(\Omega) = U_\Omega(a)K_0(\Omega)U_\Omega(a)^*$ in a yields $dK_a/da = -2\pi P_\Omega$ as a quadratic-form identity, and integrating from 0 to a gives the stated half-line modular-Hamiltonian relation. QED.

Theorem 5.2g (The half-line generator is the local null-stress charge). In the canonical normalization of the preceding half-line construction,

$$\langle \psi, P_\Omega \phi \rangle = -\frac{1}{2\pi} \frac{d}{da}\Big|_{a=0^+} \langle \psi, \widetilde{K}_a(\Omega)\phi \rangle$$

for all $\psi, \phi \in D(K_0(\Omega)) \cap D(P_\Omega)$. In the effective spacetime description of that same renormalized half-line family, the local null-stress charge is defined by the identical endpoint derivative,

$$Q_{kk}(\Omega) := -\frac{1}{2\pi} \frac{d}{da}\Big|_{a=0^+} \widetilde{K}_a^{\mathrm{eff}}(\Omega),$$

so

$$P_\Omega = Q_{kk}(\Omega)$$

as a quadratic-form identity on the common domain. The only imported continuum input is the standard local null modular-Hamiltonian formula for that same half-line family. Bounded-interval transport through the affine-covariant kernel $g_I(v)$ remains a separate projective-branch step, and reconstruction of a full tensor from the directional charges remains subject to the null-invisible metric ambiguity.

Boundary of the conditional null bridge. The null bridge is conditional rather than automatic. Its explicit burden is:

- transferred cut-center data alone give the central sector decomposition of the null strip, not yet the left/right HJPW split;
- the extra decomposition-inheritance condition of Proposition 5.2a is exactly what upgrades those sectors to the same collar-type block structure used in the spatial branch;
- the fixed-cutoff null-strip bridge is exact for exact Markov strip states on that inherited decomposition, and otherwise is replaced by a controlled limit with the carried defect operator $\mathfrak{D}_J$;
- the local finite-constraint MaxEnt branch gives endpoint-Lipschitz control strong enough to define a weak tail generator for renormalized half-lines, and the scaling-limit geometric-cap branch of Theorem 4.2 then derives half-sided modular inclusion on the half-line blow-up net;
- Borchers–Wiesbrock therefore supplies a genuine positive null-translation generator P_Ω inside the bridge itself, together with its Stone domain and the half-line modular-Hamiltonian relation $K_a = K_0 - 2\pi a P_\Omega$;
- the half-line generator/charge identification is fixed inside the bridge itself, while bounded-interval transport and the later tensor upgrade remain downstream.

This is the route from null-strip factorization to the later D5 Einstein branch, with the half-line generator/charge identification proved inside the null modular bridge itself.

3.6.3 Modular energy as stress-tensor charge (UV CFT)

The effective-theory input here is no longer a second operator added on top of the null bridge. In the canonical normalization of the preceding null-half-line construction, the positive null generator is the endpoint derivative of the renormalized half-line modular family:

$$\langle \psi, P_\Omega \phi \rangle = -\frac{1}{2\pi} \frac{d}{da}\Big|_{a=0^+} \langle \psi, \widetilde{K}_a(\Omega)\phi \rangle$$

for all $\psi, \phi \in D(K_0(\Omega)) \cap D(P_\Omega)$.

Now pass to the effective spacetime description of that same half-line family. There the local null-stress charge is defined by the identical endpoint derivative,

$$Q_{kk}(\Omega) := -\frac{1}{2\pi} \frac{d}{da}\Big|_{a=0^+} \widetilde{K}_a^{\mathrm{eff}}(\Omega),$$

so on the common quadratic-form domain

$$P_\Omega = Q_{kk}(\Omega).$$

Thus the null generator is exactly the local null-stress charge in the effective spacetime description; this step is not a separate scaling-limit premise. When the effective description admits a local stress tensor, for example in a UV CFT regime or any local Lorentzian regime with local modular Hamiltonians, the same operator is the standard null-stress charge associated with the chosen null generator.

For comparison, in a CFT vacuum on a ball one has the familiar local formula

$$H_\zeta = \int_\Sigma T_{ab} \zeta^b \, d\Sigma^a,$$

so the present identification is the null-half-line version of that same modular-energy locality rather than an additional EFT postulate.

What still lies downstream is narrower than before. The separate interval-preserving projective branch is still needed to transport this half-line statement to bounded null intervals with affine-covariant kernel $g_I(v)$, and the null-to-tensor upgrade still determines T_{ab} only up to the familiar ambiguity

$$T_{ab} \mapsto T_{ab} + \phi g_{ab}.$$

But the generator/charge identification itself is fixed inside the null bridge.

3.6.4 Localized generalized entropy from Markov + MaxEnt

Using the collar decomposition and the collar double-scaling hypothesis, supplied either as a stated collar-limit premise or derived via Theorem 2.5 from the local-Gibbs form together with exponential mixing, the state takes the Markov normal form. MaxEnt selection maximizes entropy within each edge sector, producing

$$\rho_C = \bigoplus_\alpha p_\alpha \left(\rho^\alpha_{\mathrm{bulk},C} \otimes \frac{\mathbf{1}^\alpha_{\mathrm{edge}}}{d_\alpha} \right).$$

The entropy splits as

$$S(\rho_C) = H(p_\alpha) + \sum_\alpha p_\alpha S(\rho^\alpha_{\mathrm{bulk},C}) + \sum_\alpha p_\alpha \log d_\alpha.$$

Convention: Throughout this paper, "log" denotes the natural logarithm (ln), so entropies are measured in **nats** (1 nat = 1/ln 2 ≈ 1.443 bits). This is standard in thermodynamics and QFT; the Bekenstein-Hawking formula S = A/4G uses nats. When clarity requires it, we write $\log_2$ explicitly for bits.

Define

$$S_{\text{bulk}}(C) := H(p_\alpha) + \sum_\alpha p_\alpha S(\rho^\alpha_{\text{bulk},C}),$$

and the central area operator

$$L_C := \sum_\alpha (\log d_\alpha) P_\alpha.$$

Then

$$S_{\text{gen}}(C) := \text{Tr}(\rho L_C) + S_{\text{bulk}}(C).$$

Deriving Newton's constant from edge entropy density.

Rather than normalize L_C by fiat, we *derive* the relation to G from the UV edge structure. In the collar double-scaling limit, the edge contribution becomes extensive along the entangling surface $\Sigma = \partial C$:

$$\text{Tr}(\rho L_C) \approx N_\Sigma \cdot \bar{\ell}(t), \qquad \bar{\ell}(t) := \sum_\alpha p_\alpha \log d_\alpha,$$

where N_Σ is the number of UV cut elements covering Σ and $\bar{\ell}(t)$ is the **single-cell edge entropy** from the heat-kernel distribution (Theorem 6.20). Similarly, the geometric area is extensive:

$$A(C) \approx N_\Sigma \cdot a_{\text{cell}},$$

where a_{cell} is the area per UV cut element in the emergent metric.

Matching these expressions gives the **derived formula for Newton's constant**:

$$G = \frac{a_{\text{cell}}}{4\,\bar{\ell}(t)}$$

where:

- a_{cell} is fixed operationally from the UV correlation/mixing length ξ via $a_{\text{cell}} \sim \xi^2$ (from the exponential mixing hypothesis),
- $\bar{\ell}(t) = \sum_R p_R(t) \log d_R$ is computed from the heat-kernel edge distribution with $p_R \propto d_R e^{-t\lambda_R}$.

Explicitly:

$$\bar{\ell}(t) = \frac{\sum_R d_R e^{-t\lambda_R} \log d_R}{\sum_R d_R e^{-t\lambda_R}}.$$

Thus G is the inverse edge-entropy density per geometric area, computable from the UV regulator and the reference-state Gibbs parameter t, rather than a normalization convention.

3.6.5 Fixed-cap generalized-entropy stationarity from MaxEnt

MaxEnt selection implies that for variations preserving cap labels (fixed size and charges),

$$\delta S_{\rm gen}(C) = 0.$$

Using the split above and the first law for the bulk term,

$$\delta S_{\rm gen}(C) = \delta\langle L_C\rangle + \delta\langle K_{\rm bulk}\rangle.$$

3.6.6 Einstein equation from fixed-cap generalized-entropy stationarity

In the d=4 scaling regime, the small-ball bridge is internal to the derived gravity chain once the separate bounded-interval projective branch is included. The only extra standard geometric input is the fixed-volume area-variation identity for a small geodesic ball. The modular kernel itself is not imported from a separate EFT law: on the extracted prime geometric subnet, a sufficiently small cap is the tangent causal diamond D_ℓ, whose preserving conformal Killing field is

$$\xi_{D_\ell} = \frac{1}{2\ell}\Big((\ell^2 - r^2 - t^2)\,\partial_t - 2t\,x^i\partial_i\Big).$$

On the t=0 slice its lapse is

$$\xi_{D_\ell}\cdot n = \frac{\ell^2 - r^2}{2\ell}.$$

The bounded-interval kernel comes from combining the D3 geometric cap generator with the separate interval-preserving projective branch used downstream of the D4 half-line stress bridge, so the D3 cap-modular theorem and the D4 stress bridge together yield

$$\delta S_{\rm bulk}(C) = 2\pi\int_{B_\ell}\frac{\ell^2 - r^2}{2\ell}\,\delta\langle T_{00}\rangle\,d^3x + \delta\langle E^{(\eta)}_{C,\ell}\rangle.$$

If $\delta\langle T_{00}\rangle$ is approximately constant across the ball and the carried remainder is $o(\ell^4)$, then

$$\delta S_{\rm bulk}(C) = \frac{8\pi^2\ell^4}{15}\,\delta\langle T_{00}\rangle + O(\ell^5\partial T) + o(\ell^4).$$

Stationarity of generalized entropy therefore gives the observer-covariant scalar relation

$$\delta\Big[(G_{ab} + \Lambda g_{ab})u^a u^b\Big] = 8\pi G\,u^a u^b\,\delta\langle T_{ab}\rangle$$

for the local diamond rest-frame four-velocity u^a. In the adapted rest frame $u^a = (1,0,0,0)$, this is

$$\delta(G_{00} + \Lambda g_{00}) = 8\pi G\,\delta\langle T_{00}\rangle.$$

3.6.7 Overlaps supply all timelike directions

This scalar-to-tensor upgrade is internal once the Lorentz branch is in place. Overlapping observers through the same bulk point realize every local timelike four-velocity u. Define

$$Y_{ab} := G_{ab} + \Lambda g_{ab} - 8\pi G\,\langle T_{ab}\rangle.$$

The rest-frame relation of the previous subsection says

$$u^a u^b\,\delta Y_{ab} = 0$$

for every such u. In local inertial coordinates,

$$u^a = \gamma(1, v^i), \qquad |\vec{v}| < 1,$$

so

$$0 = u^a u^b \, \delta Y_{ab} = \gamma^2 \Big(\delta Y_{00} + 2v^i \delta Y_{0i} + v^i v^j \delta Y_{ij} \Big)$$

for all $|\vec{v}| < 1$. The polynomial therefore vanishes coefficientwise, hence

$$\delta Y_{00} = 0, \qquad \delta Y_{0i} = 0, \qquad \delta Y_{ij} = 0.$$

So the full tensor variation vanishes:

$$\delta Y_{ab} = 0.$$

Along any connected scaling branch of reference states, Y_{ab} is therefore constant. The only purely local freedom left by the null reconstruction is the metric term, and fixing that term on one maximally symmetric reference state gives

$$G_{ab} + \Lambda g_{ab} = 8\pi G \, \langle T_{ab} \rangle.$$

Thus the tensor upgrade does not require a separate EFT handoff once the D3 geometric-cap theorem, the D4 null bridge, and the overlap-complete timelike family are in place.

3.6.8 Non-tunable numerical constants

The gravity chain yields specific numerical constants as rigid outputs of the axiom chain.

The 2π KMS normalization. From Theorem 4.2, once the scaling-limit geometric cap pair has been emitted and ordered cut-pair rigidity holds on it, cap modular flow is fixed by the KMS condition to the standard 2π normalization. This is the same rigidity that fixes Unruh/Hawking temperature normalization. The period is determined by the **T2** scaling-limit package together with those theorem-local hypotheses.

The geometric coefficient $\Omega_{d-2}/(d^2-1)$. This coefficient appears in both (a) the CFT-ball modular Hamiltonian weight integral and (b) the geometric area-variation identity. It is an exact integral identity:

$$\int_{B_\ell^{d-1}} \frac{\ell^2 - r^2}{2\ell} \, d^{d-1}x = \frac{\Omega_{d-2} \, \ell^d}{d^2 - 1}.$$

In $d = 4$:

$$\frac{\Omega_2}{4^2 - 1} = \frac{4\pi}{15} \approx 0.8377580409572781.$$

This is the reason prefactors cancel cleanly when going from $\delta S_{\text{gen}} = 0$ to the Einstein equation (leaving $8\pi G$ with the 2π fixed by Theorem 4.2).

What is predicted. The framework cleanly separates:

- **Non-tunable constants**: 2π (KMS period), $\Omega_{d-2}/(d^2-1)$ (geometric coefficient), the existence of the Einstein form.
- **Micro-dependent constants**: G (Newton's constant) is the conversion between edge entropy and geometric area (a density of edge degrees of freedom per geometric area), which is model-dependent. Λ is not fixed by the local reference-state variation data alone; it closes only after the global screen-capacity identification is supplied. These require specifying the microscopic model.

3.6.9 Quantitative Markov error and controlled corrections

The role of this subsection is to keep three distinct quantities separate:

1. the raw collar conditional mutual information

$$\varepsilon_\delta := I(A_\delta : D_\delta \mid B_\delta)_\omega;$$

2. the constructive recovery error

$$r_{\mathrm{FR}}(\varepsilon_\delta) := 2\sqrt{1 - e^{-\varepsilon_\delta}} \le 2\sqrt{\varepsilon_\delta};$$

3. the exact-Markov replacement error on one fixed faithful collar model,

$$\eta_\delta^{\mathrm{M}} := 4\lambda_*^{-1}\, \delta^{\mathrm{M}}_{A_\delta:B_\delta:D_\delta}(\varepsilon_\delta),$$

   where $\lambda_* > 0$ is the lower spectral bound needed to compare modular Hamiltonians.

The exact identity for the modular defect expectation is unchanged:

$$\langle \Delta K_\delta \rangle_\omega = -I(A_\delta : D_\delta \mid B_\delta)_\omega = -\varepsilon_\delta,$$

with

$$\Delta K_\delta := K_{A_\delta B_\delta D_\delta} - K_{A_\delta B_\delta} - K_{B_\delta D_\delta} + K_{B_\delta}.$$

What changes is the interpretation of later "exact" collar formulas. For each fixed finite-dimensional collar one may choose an exact Markov reference state σ_δ with

$$\|\omega_\delta - \sigma_\delta\|_1 \le \delta^{\mathrm{M}}_{A_\delta:B_\delta:D_\delta}(\varepsilon_\delta),$$

and a constructive recovered comparison state $\omega_\delta^{\mathrm{rec}}$ with

$$\|\omega_\delta - \omega_\delta^{\mathrm{rec}}\|_1 \le r_{\mathrm{FR}}(\varepsilon_\delta).$$

These two comparisons do different jobs:

Bounded-observable control. For every bounded observable X supported on $A_\delta \cup B_\delta$,

$$|\mathrm{Tr}[X(\omega_\delta - \omega_\delta^{\mathrm{rec}})]| \le \|X\|_\infty\, r_{\mathrm{FR}}(\varepsilon_\delta),$$

and

$$|\mathrm{Tr}[X(\omega_\delta - \sigma_\delta)]| \le \|X\|_\infty\, \delta^{\mathrm{M}}_{A_\delta:B_\delta:D_\delta}(\varepsilon_\delta).$$

The first bound is constructive and dimension-free; the second is the fixed-collar route to the exact Markov set.

Modular-additivity control. If the relevant marginals are uniformly faithful with lower spectral bound $\lambda_* > 0$, then

$$\|\Delta K_\delta(\omega) - \Delta K_\delta(\sigma_\delta)\|_\infty \le \eta_\delta^{\mathrm{M}}.$$

Since $\Delta K_\delta(\sigma_\delta)$ is central by exact Markovity, the finite-stage modular defect is within η_δ^{M} of the exact splice or additivity value.

Therefore the manuscript's exact collar identities arise as limits with a carried remainder

$$\eta_\delta := r_{\mathrm{FR}}(\varepsilon_\delta) + \eta_\delta^{\mathrm{M}},$$

together with the separate long-wavelength derivative remainders present in the small-ball expansion. Exact identities at finite collar width require exact Markovity; otherwise one works with η_δ-controlled approximations and then lets $\eta_\delta \to 0$ in the controlled collar limit.

Modified Einstein relation with an explicit carried remainder. In the small-ball rest-frame calculation, write

$$\delta S_C = \frac{8\pi^2\ell^4}{15}\,\delta\langle T_{00}\rangle + \delta\langle E_C^{(\delta)}\rangle + O(\ell^5\partial T),$$

where $E_C^{(\delta)}$ packages the carried collar remainder. Then the derived fixed-cap generalized-entropy stationarity theorem gives

$$\delta(G_{00} + \Lambda g_{00}) = 8\pi G\,\delta\langle T_{00}\rangle + \mathcal{E}_\delta,$$

with

$$\mathcal{E}_\delta := \frac{15G}{\pi\ell^4}\,\delta\langle E_C^{(\delta)}\rangle + O(\ell\,\partial T),$$

and, by the bounds above,

$$\left|\delta\langle E_C^{(\delta)}\rangle\right| \le C_C\,\eta_\delta$$

for some collar-dependent constant C_C on the fixed faithful model.

If one wishes to package this carried remainder as an effective anomalous energy density, the natural definition is

$$\delta\langle T_{00}^{\mathrm{anom}}\rangle := \frac{15}{8\pi^2\ell^4}\,\delta\langle E_C^{(\delta)}\rangle.$$

Then

$$\delta(G_{00} + \Lambda g_{00}) = 8\pi G\,\delta(\langle T_{00}\rangle + \langle T_{00}^{\mathrm{anom}}\rangle) + O(\ell\,\partial T).$$

This is a bookkeeping definition of the finite-stage remainder, not an upgrade of the recovered-core theorem. Its significance is exactly that it is controlled:

$$\left|\delta\langle T_{00}^{\mathrm{anom}}\rangle\right| \le \frac{15\,C_C}{8\pi^2\ell^4}\,\eta_\delta.$$

The framework carries the Markov error through three distinct controls:

1. $r_{\mathrm{FR}}(\varepsilon_\delta)$ controls bounded observables at one stage;
2. $\delta^{\mathrm{M}}_{A_\delta:B_\delta:D_\delta}(\varepsilon_\delta)$ controls convergence to the exact Markov normal form on a fixed collar;
3. η_δ^{M} controls modular-Hamiltonian replacements once faithfulness is assumed;
4. the Einstein-branch remainder is a carried $O(\eta_\delta)$ term, not a silent exact identity at finite cutoff.

This is the concrete bridge from "axioms about screens" to "precision GR predictions plus a disciplined carried remainder".

3.6.10 Focusing/QNEC internalization via relative entropy

Once the fixed-cutoff null modular structure of Section 5.2 is combined with the exact half-line generator/charge identification and the bounded-interval projective branch, focusing constraints follow from information-theoretic principles in the same scaling-limit branch rather than from a separate fixed-cutoff postulate.

Derivation chain. QNEC and focusing are supported conditionally within the same branch:

- Fixed-cutoff null modular bridge (§5.2) $\Longrightarrow$ exact-or-controlled strip additivity and a weak tail generator
- Derived half-sided modular pair + exact half-line generator/charge identification $\Longrightarrow$ local null-stress charge on half-lines and [K, P] = i 2π P
- Relative entropy monotonicity $\Longrightarrow$ QNEC
- Einstein (Thm 5.1) + Raychaudhuri $\Longrightarrow$ QFC for S_gen

Relative entropy monotonicity argument. The key input is the monotonicity of relative entropy under partial trace, which is pure information theory:

$$S(\rho_{AB}\|\sigma_{AB}) \geq S(\rho_A\|\sigma_A).$$

For null deformations parameterized by λ, consider nested null regions $R(\lambda) \subset R(\lambda')$ obtained by varying the entangling cut along v. The modular Hamiltonian K_λ generates the modular flow, and relative entropy satisfies convexity:

$$\frac{d^2}{d\lambda^2} S(\rho_\lambda\|\sigma_\lambda) \geq 0.$$

Proposition 5.10a (Conditional internal QNEC). Under the fixed-cutoff null bridge of §5.2 together with the downstream density-upgrade and bounded-interval projective premises, the second null variation of von Neumann entropy satisfies

$$\frac{d^2 S_{\text{bulk}}}{d\lambda^2} \leq 2\pi\langle T_{kk}(\lambda)\rangle,$$

with the 2π normalization fixed by Theorem 4.2.

Proof. Use the derived half-sided modular-inclusion statement of Corollary 5.2e and its Borchers-Wiesbrock consequence from Lemma 5.2f. This gives the translation structure with $[K, P] = -i2\pi P$.

Consider the relative entropy $S(\rho_\lambda\|\omega_\lambda)$ between the state ρ restricted to $R(\lambda)$ and the reference state ω. Monotonicity under restriction to smaller regions ($\lambda' > \lambda$) gives:

$$S(\rho_{R(\lambda)}\|\omega_{R(\lambda)}) \leq S(\rho_{R(\lambda')}\|\omega_{R(\lambda')}).$$

Using the first law $\delta S = \delta\langle K\rangle$ and the geometric half-line modular form supplied by Section 5.2, expand to second order in the deformation. The convexity of relative entropy yields the QNEC inequality. The bound saturates for coherent states in the standard way. QED.

Corollary 5.10b (QFC for generalized entropy). With the central area operator L_C from EC/MaxEnt (Section 5.4), define

$$S_{\text{gen}} = \text{Tr}(\rho L_C) + S_{\text{bulk}}.$$

Given the conditional Einstein branch (Theorem 5.1) and the classical Raychaudhuri identity for null congruences, the Quantum Focusing Conjecture (QFC) follows within the same scaling regime: the generalized expansion Θ_{gen} is non-increasing along null generators.

Proof sketch. The Raychaudhuri equation relates expansion evolution to R_{kk}. Einstein's equation gives $R_{kk} = 8\pi G(T_{kk} - \frac{1}{2}g_{kk}T)$. For null k, this simplifies to $R_{kk} = 8\pi G T_{kk}$. The QNEC (Prop 5.10a) then bounds the bulk entropy production, ensuring $d\Theta_{\text{gen}}/d\lambda \leq 0$. QED.

Significance. This section shows how the Recoverable Generalized Entropy focusing package is internally supported inside the same null-modular branch once the stress-tensor identification and Einstein relation are in place. That package remains part of the stated axiom set.

Theorem 5.1 (Observer-consistency implies a conditional scaling-limit Einstein branch). Under the five OPH axioms, the collar and null-strip hypotheses of Sections 2 and 5, the hypotheses of Theorem 4.2, the derived fixed-cap generalized-entropy stationarity theorem for admissible fixed-cap MaxEnt variations on the realized cap-label-preserving MaxEnt family, the half-line generator/charge identification of Section 5.2 together with the bounded-interval projective branch used there, and admissible fixed-cap MaxEnt variations about a maximally symmetric reference state, the fixed-cap generalized-entropy stationarity condition implies the rest-frame first-variation relation

$$\delta(G_{00} + \Lambda g_{00}) = 8\pi G\, \delta\langle T_{00}\rangle.$$

If this relation holds for all local directions and reference states in the locally Lorentzian scaling regime, overlap consistency upgrades it to the semiclassical Einstein equation modulo the expected Λg_{ab} ambiguity.

Proof sketch. The cap first law identifies δS_C with $\delta\langle K_C\rangle$, while the conditional null bridge together with the bounded-interval projective branch relates the bulk modular variation to the required null stress-tensor charge on the diamond. Since the reference state is maximally symmetric, the fixed-volume small-ball area identity enters at first order in the perturbation, and the derived fixed-cap generalized-entropy stationarity theorem for admissible fixed-cap MaxEnt variations on the realized cap-label-preserving MaxEnt family yields the displayed rest-frame relation. Overlap consistency then supplies all local directions, and the null-to-tensor lemma upgrades the relation to the tensor equation modulo Λg_{ab}. QED.

3.6.11 Black-hole structural statements and continuation-level spectroscopy

This subsection mixes two claim tiers and is organized accordingly. The retained structural black-hole package is a four-step chain: fixed-cutoff edge-center collar decomposition on the exact-Markov or idealized exact-recoverability horizon carrier, the small-CMI recoverability reading of interior encoding on that same carrier, Hawking/KMS normalization on the geometric modular branch, and the discrete area spectrum together with the Schwarzschild transition identity $\Delta E = k_B T_H \ln(d'/d)$ for an actual sector change $d \to d'$. Any clean comb structure, QNM selector, Page-type linewidth estimate, PBH burst template, Kerr/LIGO horizon spectroscopy template, or Page-curve/island closure remains continuation-level and uses extra premises stated below.

Derived structural statements.

Area eigenvalues from edge sectors. The central area operator (Section 5.4) is

$$L_C = \sum_\alpha (\log d_\alpha) P_\alpha,$$

where $d_\alpha \in \mathbb{N}$ is the dimension of the edge Hilbert space in sector α. With the normalization $\mathrm{Tr}(\rho L_C) = \langle A\rangle/4G$, the area eigenvalues are

$$A_\alpha = 4G \log d_\alpha = 4\ell_p^2 \ln d_\alpha,$$

where $\ell_p^2 = \hbar G/c^3$ is the Planck area. Since d_α is a positive integer, **areas are discretely spaced** with logarithmic gaps.

Hawking emission energy quantization. For a Schwarzschild black hole with $A(M) = 16\pi G^2 M^2/c^4$, a transition between sectors $d \to d'$ changes the area by

$$\Delta A = 4\ell_p^2 \ln(d'/d).$$

The corresponding energy at infinity is $\Delta E = c^2 \Delta M$, with $\Delta M = \Delta A/(dA/dM)$. This gives

$$\Delta E = \frac{\hbar c^3}{8\pi G M} \ln(d'/d).$$

Using the Hawking temperature $T_H = \hbar c^3/(8\pi G k_B M)$, whose 2π normalization is fixed by the BW_{S^2} tangent-limit normalization of Theorem 4.2:

$$\Delta E = k_B T_H \ln(d'/d).$$

Continuation-level spectroscopy templates. The remainder of this subsection is not part of the recovered core. It records what follows only if one adds extra discrete-horizon selection rules and, where stated, standard semiclassical inputs such as evaporation-power models, QNM/transition identifications, or greybody matching.

Integer transitions. If dominant transitions multiply the edge dimension by an integer k (i.e., $d'/d = k$), the spectrum becomes a discrete comb:

$$\Delta E_k = k_B T_H \ln k, \qquad \Delta f_k = \frac{c^3}{16\pi^2 G M} \ln k.$$

Structural condition: comb vs. generic discreteness. The log-integer *comb* structure requires the additional dynamical assumption that integer-multiplication transitions (d $\to$ kd) dominate. If generic transitions between arbitrary integers dominate instead, the set of ln(d'/d) values becomes a dense log-rational set that may appear quasi-continuous after folding in linewidths and astrophysical effects. What follows directly from the axioms is the *discrete area spectrum*; the clean comb pattern is conditional on the selection rule.

Continuation-level template (Discrete Hawking spectrum). If the additional integer-transition selection rule is realized, the Hawking emission spectrum consists of discrete lines with spacing $\Delta E_k = k_B T_H \ln k$, where k is an integer characterizing the dominant sector transitions, instead of a continuous thermal profile.

Mass-independent fractional linewidth (continuation-level estimate). Using Page's semiclassical calculation for emission power $P(M) = p_0 \hbar c^6/(G^2 M^2)$ with $p_0 \approx 2 \times 10^{-4}$, the emission rate is $\dot{N} \approx P/\langle E \rangle$ where $\langle E \rangle = a\, k_B T_H$ with $a \sim \mathcal{O}(1-10)$. The natural linewidth $\Gamma \sim \hbar \dot{N}$ divided by the level spacing gives:

$$\frac{\Gamma}{\Delta E_k} \approx \frac{64\pi^2 p_0}{a \ln k} \approx 3 - 5\%$$

Within this continuation branch, the emission lines are narrow (few-percent fractional width) and the estimated fraction is mass-independent.

Connection to quasinormal modes (interpretive continuation). The highly-damped Schwarzschild quasinormal modes have asymptotic real part (Motl, 2002):

$$\mathrm{Re}\,\omega \to \frac{c^3}{8\pi GM}\ln 3.$$

This matches exactly the $k = 3$ transition frequency $\Delta E_3/\hbar$. If one adopts a Bohr-type identification between quantum transition frequencies and asymptotic QNM frequencies, this selects

$$\Delta A = 4\ell_p^2 \ln 3 \approx 4.39\,\ell_p^2$$

as the fundamental area quantum.

Conditionality statement. The area quantization follows from the edge-sector structure (derived). The $k = 3$ selection requires the additional interpretive identification with QNM frequencies (not derived from axioms). The linewidth prediction uses standard semiclassical inputs.

Numerical examples. For Δf_k = ($c^3/16\pi^2$GM) ln k:

- **M = 30 M⊙**: k=2 at 29.7 Hz, k=3 at 47.1 Hz
- **M = 1 M⊙**: k=2 at 891 Hz, k=3 at 1412 Hz
- **M = 10^{12} kg (primordial)**: k=2 at 7.3 MeV, k=3 at 11.6 MeV

Within the integer-transition continuation, these frequencies track $k_B T_H \ln k$ exactly and are in principle distinguishable from a continuous thermal spectrum.

Continuation-level PBH burst search template.

Within the discrete-horizon continuation, the Hawking comb would provide a distinctive gamma-ray template. A template-level discriminant is **log-integer energy ratios**: if two emission lines are observed at energies E_2 and E_3, their ratio must satisfy

$$\frac{E_3}{E_2} = \frac{\ln 3}{\ln 2} \approx 1.585$$

exactly, independent of black hole mass. Within that continuation template, the ratio is fixed once the log-integer rule is assumed.

Available instruments and energy coverage. The $k = 2$ line energy $E_2 = k_B T_H \ln 2$ determines which instruments can see a given BH mass:

Instrument	Energy band	BH mass range (k=2 in band)
Fermi GBM (BGO)	0.15–40 MeV	2×10^{11}–5×10^{13} kg
Fermi LAT	0.1–300 GeV	2×10^{7}–7×10^{10} kg
H.E.S.S.	0.1–100 TeV	7×10^{4}–7×10^{7} kg
LHAASO-WCDA	1–15 TeV	5×10^{5}–7×10^{6} kg

Detector resolution vs. intrinsic linewidth. The continuation-level linewidth estimate is 3–5% (mass-independent). Current detector energy resolutions:

- Fermi GBM: $< 10\%$ (0.1–1 MeV), $\sim 4\%$ at 10 MeV (BGO)
- Fermi LAT: $< 10\%$ (1–100 GeV)
- H.E.S.S.: $\sim 15\%$ (TeV)
- LHAASO-WCDA: $\sim 33\%$ (TeV)

Within this template, the comb could be resolvable with GBM/LAT; at TeV energies it would appear as moderately broad bumps rather than sharp lines.

Search protocol. A dedicated OPH-comb search would:

1. Select burst-like candidates (10–120 s time windows, matching existing PBH burst search protocols).
2. Fit each candidate with null model (smooth continuum) vs. OPH comb model (peaks at $E_k = E_0 \ln k$ convolved with detector response).
3. Scan over the single scale parameter $E_0 = k_B T_H$ (equivalently, BH mass).
4. Require at least two lines satisfying log-integer ratio to claim detection.
5. Correct significance for trials (time windows $\times$ sky positions $\times$ E_0 scan).

Observational context. Dedicated PBH burst searches (H.E.S.S., LHAASO) report **no significant bursts**. An OPH-specific comb-template analysis of archival data would:

- Set upper limits on OPH-comb PBH burst rates
- Demonstrate direct testability of the discrete spectrum prediction
- Provide constraints comparable to or stronger than generic PBH burst limits

Data availability. Fermi GBM provides public Time-Tagged Event (TTE) burst data; Fermi LAT provides public photon event lists with documented analysis workflows. H.E.S.S. has a small public test data release.

Continuation-level GW horizon spectroscopy template for Kerr remnants.

The same continuation-level discrete-horizon branch extends to gravitational wave observables. For Kerr black holes, the thermodynamic first law is $\delta M = T_H \delta S + \Omega_H \delta J$, so the entropy change for absorbing a quantum with frequency ω and azimuthal number m is:

$$\delta S = \frac{\hbar(\omega - m\Omega_H)}{k_B T_H}.$$

In the edge-sector framework, $\delta S = \ln(d'/d)$, so the discreteness condition becomes:

$$\hbar(\omega - m\Omega_H) = k_B T_H \ln k, \qquad k \in \{2, 3, 4, \ldots\}$$

Under those additional discrete-horizon premises, this gives the **GW horizon spectroscopy comb**: a continuation-level set of discrete resonant frequencies where the horizon can efficiently absorb or emit energy.

Kerr line frequencies. For a remnant with mass M and dimensionless spin $\chi = a_*/M$, define the spin correction factor:

$$g(\chi) = \frac{2\sqrt{1-\chi^2}}{1+\sqrt{1-\chi^2}}, \qquad \Omega_H(M,\chi) = \frac{c^3}{2GM} \cdot \frac{\chi}{1+\sqrt{1-\chi^2}}.$$

The line frequencies are:

$$f_{k,m}(M,\chi) = \frac{m\,\Omega_H(M,\chi)}{2\pi} + \frac{c^3}{16\pi^2 GM}\, g(\chi)\, \ln k$$

Within this continuation template, once LIGO/Virgo infers (M, χ) for a remnant, the line pattern is fixed by the inferred remnant parameters and the assumed discrete-horizon rule. This is not a recovered-core theorem.

Line weights from GR envelope + discretization. In this continuation template, the line strengths are modeled by matching to the known GR greybody absorption spectrum in the semiclassical limit. The discretization rule gives bin width $\Delta\omega_k \approx \omega_T \ln(1 + 1/k)$ where $\omega_T = k_B T_H/\hbar$. The net line weight (absorption minus stimulated emission) is:

$$W^{\text{net}}_{k,\ell m} = \Gamma^{\text{GR}}_{\ell m}(\omega_{k,m}) \cdot \Delta\omega_k \cdot \frac{k-1}{k}$$

where $\Gamma^{\text{GR}}_{\ell m}$ is the standard GR greybody factor and the $(k-1)/k$ factor arises from KMS detailed balance with $e^{(\omega - m\Omega_H)/T_H} = k$.

Universal stacking coordinate. Define the dimensionless rescaled frequency:

$$x := \frac{GM}{c^3 g(\chi)}(\omega - m\Omega_H).$$

Then the predicted line locations collapse to universal constants:

$$x_k = \frac{\ln k}{8\pi} \qquad (k = 2, 3, 4, \ldots)$$

Numerically: $x_2 = 0.02758$, $x_3 = 0.04371$, $x_4 = 0.05516$, $x_5 = 0.06404$.

Stacking test. Multiple BBH events can be mapped to this universal x coordinate and stacked. If the comb is real, peaks align across events with different (M, χ); detector noise does not stack coherently.

Comparison to existing work. Prior area-quantization searches [36] used parameterized models with one free spacing constant. The OPH prediction is more constrained: multiple lines with exact $\ln k$ ratios, plus the $(k-1)/k$ weight hierarchy from detailed balance.

Numerical example (GW170608). Remnant parameters: $M_f \approx 18.0 M_\odot$, $\chi_f \approx 0.69$. For $m = 2$, the horizon rotation frequency is $m\Omega_H/(2\pi) \approx 719$ Hz. The **thermal comb spacing** (the part that encodes the area quantization) is:

k	$\Delta f_k := \frac{c^3 g(\chi)}{16\pi^2 GM} \ln k$ (Hz)	Relative weight $(k-1)/k$
2	41.6	0.500
3	65.9	0.667
4	83.2	0.750
5	96.5	0.800
6	107.5	0.833

The full physical frequencies are $f_{k,2} = 719 + \Delta f_k$ Hz (i.e., 760–827 Hz), outside LIGO's most sensitive band for this remnant. However, the **stacking analysis** uses the rescaled coordinate $x = GM(\omega - m\Omega_H)/(c^3 g(\chi))$, which maps the thermal spacing to universal constants $x_k = \ln k/8\pi$ regardless of the rotation offset.

Template-matching criterion. After rescaling by (M, χ), spectral features in this continuation template must satisfy $f_k/f_2 = \ln k/\ln 2$ exactly, independent of remnant parameters. Absence of coherent stacking at the predicted x_k values would challenge the discrete-horizon continuation template, not by itself the derived area-spectrum statement.

3.6.12 Classical mechanics from emergent GR

Once the Einstein equation is established, the framework inherits standard GR consequences. This section makes explicit how classical mechanics emerges.

Stress-energy conservation is automatic. The contracted Bianchi identity is geometric:

$$\nabla^a G_{ab} = 0.$$

Combined with the Einstein equation, this implies:

$$\nabla^a \langle T_{ab} \rangle = 0.$$

Geodesic motion from dust limit. For pressureless classical matter ("dust"), $T^{ab} = \rho\, u^a u^b$. Conservation yields:

$$\nabla_a(\rho u^a u^b) = 0 \quad \Rightarrow \quad u^b \nabla_a(\rho u^a) + \rho\, u^a \nabla_a u^b = 0.$$

Projecting orthogonally to u^b using $h^b{}_c = \delta^b{}_c + u^b u_c$ kills the first term, giving:

$$\rho\, u^a \nabla_a u^b = 0 \quad \Rightarrow \quad u^a \nabla_a u^b = 0.$$

This is the geodesic equation: free classical bodies follow spacetime geodesics.

Newtonian limit from weak-field GR. Take the weak-field, slow-motion limit with metric:

$$g_{00} \approx -(1 + 2\Phi/c^2), \qquad g_{0i} \approx 0, \qquad g_{ij} \approx \delta_{ij}(1 - 2\Phi/c^2),$$

and velocities $|\mathbf{v}| \ll c$. Then $G_{00} \approx 2\nabla^2\Phi/c^2$ (leading order), and $T_{00} \approx \rho c^2$. The Einstein equation reduces to:

$$\nabla^2 \Phi = 4\pi G \rho.$$

Geodesic motion reduces to:

$$\ddot{\mathbf{x}} = -\nabla \Phi.$$

These are Newton's gravitational law and Newton's second law. Classical mechanics is recovered as a controlled limit of the emergent GR dynamics.

Precision classical predictions. Once the field equation is fixed to Einstein form, the framework inherits the standard GR precision toolbox (post-Newtonian expansion, lensing, time delay, etc.), with no free "shape" parameters beyond G and Λ.

Selected precision predictions (in the regime where the GR derivation applies):

Light bending by mass M: For impact parameter b,

$$\Delta\theta = \frac{4GM}{c^2 b}.$$

For the Sun with $b \approx R_\odot$: $\Delta\theta \approx 1.751$ arcsec.

Mercury perihelion advance: Per orbit,

$$\Delta\varpi = \frac{6\pi GM}{a(1-e^2)c^2}.$$

Using Mercury's orbital parameters: $\Delta\varpi \approx 42.98$ arcsec/century.

Gravitational redshift: Between two radii in a static potential,

$$\frac{\Delta\nu}{\nu} \approx \frac{\Delta\Phi}{c^2}.$$

For the Sun (surface to infinity): $z \approx 2.12 \times 10^{-6}$.

These predictions are fixed functions of G and known source parameters, and are confirmed observationally to high precision. The framework inherits them automatically once the Einstein equation is derived.

3.6.13 Precision gravity predictions and experimental bounds

The gravity sector makes symmetry-protected exact-zero predictions that can be confronted with the tightest available experimental bounds. This section translates the theoretical predictions into the specific observables that experiments actually constrain.

Speed of gravitational waves. The derived GR regime implies massless gravitons propagating on the same null cones as photons:

$$\frac{c_{\mathrm{GW}} - c}{c} = 0 \text{ exactly.}$$

Current bound (GW170817 + GRB 170817A multi-messenger):

$$-3 \times 10^{-15} < \frac{c_{\mathrm{GW}} - c}{c} < +7 \times 10^{-16} \quad (90\% \text{ credibility}).$$

For a source at $\sim$ 40 Mpc, this fractional difference corresponds to only a few seconds of propagation-time mismatch across $\sim 10^8$ years of travel.

Graviton mass. The gauge redundancy (diffeomorphism invariance) forbids a hard mass term:

$$m_g = 0 \text{ exactly.}$$

Current bound (GW dispersion analysis, PDG 2025):

$$m_g \leq 1.76 \times 10^{-23} \text{ eV}/c^2 \quad (90\% \text{ credibility}).$$

This corresponds to a reduced Compton wavelength $\bar{\lambda}_C \gtrsim 1.6 \times 10^{16}$ m, i.e., order $\sim$ 1.6 light-years.

No dipole radiation. Many modified gravity theories predict extra channels (scalar/vector) producing dipolar radiation at (-1)PN order. The derived GR limit predicts no such channel.

Current bound (GW170817 inspiral phasing, PDG 2025):

$$-4 \times 10^{-6} < \delta\hat{p}_{-2} < 2 \times 10^{-5} \quad (90\% \text{ credibility}).$$

Only tensor polarizations. The GR outcome means only the two tensor (helicity-2) modes propagate. Pure non-tensor hypotheses are disfavored by observational constraints, and mixed tensor-scalar/vector models are tightly constrained.

Equivalence principle tests. Additional null checks from the derived GR structure:

- Universality of free fall (space tests): precision ~10^{-15}
- Nordtvedt parameter ($\eta = 4\beta$ - γ): (0.47 $\pm$ 0.55) $\times$ 10^{-4}
- Binary pulsar radiative damping (PSR J0737-3039): 0.999963 $\pm$ 0.000063

3.6.14 Theory-side error propagation from Markov bounds

The framework provides exact-zero predictions and quantitative control over how well those predictions hold. The Markov/recovery machinery can be propagated through the entire GR emergence chain.

The key quantitative hook. From Theorem 3.1, if the target state satisfies

$$I(A_k : C_k \mid B_k) \leq \varepsilon_k,$$

then recovery maps exist with trace-distance error

$$\delta_k = 2\sqrt{\ln 2 \cdot \varepsilon_k}.$$

Trace distance gives immediate bounds on observable errors. Using the standard dual norm inequality:

$$|\langle O\rangle_\rho - \langle O\rangle_\sigma| \le \|O\|_\infty \|\rho - \sigma\|_1 = 2\|O\|_\infty D(\rho,\sigma),$$

where $D(\rho,\sigma) = \frac{1}{2}\|\rho - \sigma\|_1$ is the trace distance.

Exponential decay from the mixing hypothesis. The mixing assumption (Section 2.3) provides:

$$I_\omega(A_\delta : D_\delta \mid B_\delta) \le c \cdot |\partial C|_{\mathrm{UV}} \cdot e^{-\delta/\xi}.$$

Combining these gives an explicit precision dial:

$$\delta_{\mathrm{step}} \lesssim 2\sqrt{\ln 2 \cdot c \cdot |\partial C|_{\mathrm{UV}}} \cdot e^{-\delta/(2\xi)}.$$

What precision requires. To match the GW speed bound ($\sim 10^{-15}$ fractional accuracy), the recovery-map error must satisfy:

$$\delta \lesssim 10^{-15} \quad \Rightarrow \quad \varepsilon \lesssim \frac{(\delta/2)^2}{\ln 2} \approx 3.6 \times 10^{-31}.$$

This is extremely small, but achievable: with a macroscopic boundary ($|\partial C|_{\mathrm{UV}} \sim 10^{35}$ for a meter-scale boundary at Planck UV scale), the exponential decay $e^{-\delta/\xi}$ with $\delta/\xi \sim$ a few hundred easily pushes below 10^{-31} once the prefactor is included.

Precision summary. The framework provides:

1. Exact-zero predictions ($m_g = 0$, $c_{\mathrm{GW}} = c$) from symmetry protection.
2. Translation of those zeros into the specific observables experiments constrain.
3. Explicit bounds on how far derived geometric statements can drift, using the conditional mutual information $\to$ trace distance $\to$ observable error chain.

This is the concrete path from "axioms about screens" to "precision GR predictions with quantitative error control."

3.6.15 Dark-sector response from the modular anomaly

This subsection studies the benchmark scale obtained by adding a specific deep-IR response ansatz on top of the modular-anomaly term. Neither the observational dark-matter identification nor MOND/RAR-like dynamics are derived here.

The modular anomaly term T_{ab}^{anom} derived in Section 5.9 supplies one structural ingredient for a possible dark-sector continuation, without introducing new particle species.

The identification. The anomalous stress-energy contribution

$$\langle T_{00}^{\mathrm{anom}}\rangle = \frac{15}{8\pi^2} \cdot \frac{\delta\langle K_C^{(\mathrm{anom})}\rangle}{\ell^4}$$

is "dark" by construction: it arises from information-theoretic/gravitational structure (modular Markov imperfections), not from Standard Model fields. It gravitates but does not couple electromagnetically. These properties explain why the term is explored as a dark-sector ingredient, but they do not by themselves close the observational identification.

Connection to the cosmological parameter package. The framework makes Λ a global capacity parameter and fixes the same D6 static-patch radius:

$$\Lambda = \frac{3\pi}{GN_{\rm scr}}, \quad r_{\rm dS} = \sqrt{\frac{3}{\Lambda}}.$$

This imports the D6 static-patch scale into the continuation surface. Any galaxy-scale continuation built from the anomaly term would therefore be an IR phenomenon, appearing only when accelerations are small and distances are large.

Acceleration benchmark. If one assumes the relevant deep-IR response is controlled only by $r_{\rm dS}$, c, and the fixed anomaly prefactor, then the benchmark scale must:

1. Vanish if $r_{\rm dS} \to \infty$ (infinite capacity, no de Sitter static-patch scale)
2. Be controlled by $r_{\rm dS}$ as the only new IR scale
3. Carry non-tunable coefficients from the derivation

The anomaly enters with prefactor $\frac{15}{8\pi^2}$. The corresponding benchmark acceleration scale constructible from (Λ, c) is:

$$\boxed{a_0^{(\rm OPH)} := \frac{15}{8\pi^2} \cdot c^2 \sqrt{\frac{\Lambda}{3}} = \frac{15}{8\pi^2} \cdot \frac{c^2}{r_{\rm dS}}}$$

Normalization estimate. Using Planck 2018 ΛCDM parameters ($H_0 \approx 67.4$ km/s/Mpc, $\Omega_\Lambda \approx 0.685$):

- $\Lambda \approx 1.09 \times 10^{-52}$ m^{-2}
- $r_{\rm dS} \approx 1.66 \times 10^{26}$ m
- Therefore:

$$\boxed{a_0^{(\rm OPH)} \approx 1.03 \times 10^{-10} \text{ m/s}^2}$$

For comparison, observational fits to galaxy regularities (RAR/MDAR/MOND phenomenology) quote $a_0 \sim 1.2 \times 10^{-10}$ m/s^2. This numerical proximity does not by itself derive galaxy dynamics.

One illustrative response ansatz. If one further assumes that $T_{00}^{\rm anom}$ is the dominant deep-IR source and that the response organizes into a MOND/RAR-like law, the Newtonian limit could be written as:

$$\nabla^2 \Phi = 4\pi G(\rho_b + \rho_{\rm anom}),$$

i.e., baryons plus an effective extra density. Under the same extra ansatz the radial acceleration relation (RAR) could be written as:

$$g_{\rm obs} \approx g_b + \sqrt{a_0 \cdot g_b}, \quad g_{\rm DM} := g_{\rm obs} - g_b \approx \sqrt{a_0 \cdot g_b}.$$

With $a_0 = a_0^{(\rm OPH)}$ fixed, that same ansatz would imply:

(i) Baryonic Tully-Fisher relation.

$$V^4 \approx G \cdot M_b \cdot a_0^{(\rm OPH)}$$

where V is the asymptotic rotation velocity and M_b is baryonic mass.

(ii) Flat rotation curves. For a point mass M_b:

$$g_{\mathrm{DM}}(r) = \frac{\sqrt{GM_b\, a_0^{(\mathrm{OPH})}}}{r} \quad \Rightarrow \quad M_{\mathrm{DM}}(r) \propto r$$

i.e., inferred dark mass grows linearly with radius, producing flat rotation curves.

(iii) Characteristic surface density.

$$\Sigma_0^{(\mathrm{OPH})} = \frac{a_0^{(\mathrm{OPH})}}{2\pi G} \approx 0.25\ \mathrm{kg/m^2} \approx 120\, M_\odot/\mathrm{pc}^2.$$

This lies in the range of observed central halo surface densities, but it is not a derived halo theorem.

Status. What is grounded in the framework developed here:

- The modular anomaly term exists with fixed coefficient $\frac{15}{8\pi^2}$
- Λ and r_{dS} are determined by screen capacity

What remains missing for theorem-level closure:

- A controlled nonrelativistic limit from the anomaly term to galaxy observables
- A derived response law selecting the MOND/RAR functional rather than alternative IR behavior
- Lensing, cluster, and Bullet-Cluster phenomenology
- Cosmological abundance and structure-formation checks
- Environment-dependence and stability control

This branch contains structural ingredients and an IR benchmark, but it does not yet derive the precise galaxy-scale response from the recovered core.

Falsifiability. If later work cannot supply the missing derivation steps, or if any resulting continuation is incompatible with galaxy, lensing, cluster, Bullet-Cluster, or cosmological data, then the modular-anomaly dark-sector continuation retracts. The recovered core theorem package does not.

3.6.16 De Sitter holography: static patch vs boundary-at-infinity

A natural question arises: how does this framework relate to the "unsolved problem" of de Sitter holography?

What the usual dS holography problem is. When people say "dS holography is unsolved," they typically mean that we do not have anything as sharp as AdS/CFT, where the bulk has a timelike asymptotic boundary supporting a well-defined dual CFT with a precise dictionary. For de Sitter, there is no asymptotic timelike boundary in the static patch where one can simply place the dual theory. The classic dS/CFT proposal at future infinity has familiar difficulties, including non-unitarity worries and complex conformal weights.

Static-patch/horizon-screen setup. The framework begins with an observer's static patch and its horizon screen S^2, building a net of subregion algebras on that screen. At finite cutoff those algebras are type-I regulators. The Lorentz statement, when invoked, is a scaling-limit statement about the refinement-limit observer net, and that limit may leave the regulator class. On the theorem-local branch of Theorem 3.5.2 where the extracted prime geometric cap pair has been emitted, the realized scaling-limit cap modular action is geometric and, in the non-type-I case of

interest, generally outer. That identification is *not* derived from MaxEnt alone; it is exactly the explicit **T2** plus extraction-and-rigidity boundary.

This is therefore a fundamental fork away from AdS/CFT-style holography:

AdS/CFT	This framework
Codimension-1 boundary at infinity	Codimension-2 horizon screen (S^2)
Single global boundary theory	Observer-dependent patches that overlap
Dual CFT required	Only algebras + consistency conditions
Negative Λ	Positive Λ natural

This aligns with the static-patch/complementarity intuition in the dS literature, where the fundamental description is patch-based and different static patches are related by consistency rules, not by a single global boundary theory.

The mechanism: Λ as global capacity, not local physics. A key structural result is that null modular data reconstruct the stress tensor only up to an additive metric term. This is the statement that vacuum-energy or cosmological-constant shifts are invisible to the local null-data route. The Einstein equation derived from the fixed-cap generalized-entropy stationarity theorem is therefore fixed only up to Λg_{ab}.

Theorem boundary: cosmological-constant / screen-capacity closure stack. On the D5→D6 branch, assume the local Einstein equation is fixed only modulo Λg_{ab}, the external input

$$N_{\rm scr} = \log \dim \mathcal{H}_{\rm tot},$$

the D6 screen-capacity identification

$$N_{\rm scr} = S_{\rm dS},$$

the standard de Sitter entropy relation

$$S_{\rm dS} = \frac{A_{\rm dS}}{4G} = \frac{3\pi}{G\Lambda},$$

and the standard de Sitter static-patch formulas

$$r_{\rm dS} = \sqrt{\frac{3}{\Lambda}}, \qquad t_\Lambda = \frac{r_{\rm dS}}{c}.$$

Then:

1. local null data determine the Einstein branch only modulo Λg_{ab};
2. the same branch closes globally as
$$G_{ab} + \frac{3\pi}{G N_{\rm scr}}\, g_{ab} = 8\pi G \,\langle T_{ab}\rangle;$$
3. the same D6 closure fixes
$$S_{\rm dS} = N_{\rm scr}, \qquad A_{\rm dS} = 4G N_{\rm scr}, \qquad r_{\rm dS} = \sqrt{\frac{3}{\Lambda}}, \qquad t_\Lambda = \frac{r_{\rm dS}}{c};$$
4. the observed cosmic age remains a downstream FLRW benchmark rather than an additional theorem output.

Thus the cosmological-constant package is one local/global theorem stack rather than a split local-plus-global story.

Reason. The local null-data route is blind to metric-term shifts, so the Einstein branch is fixed locally only modulo Λg_{ab}. The external input N_{scr}, together with the screen-capacity identification and the imported de Sitter entropy relation, then yields

$$\Lambda = \frac{3\pi}{GN_{\mathrm{scr}}},$$

which closes the same Einstein branch globally and fixes the displayed static-patch package. The observed cosmic age is a later FLRW comparison quantity rather than part of this theorem stack.

The imported premises are exactly the external input N_{scr}, the D6 identification $N_{\mathrm{scr}} = S_{\mathrm{dS}}$, the de Sitter entropy relation, and the standard static-patch formulas. What remains external is the screen-capacity identification itself: the local null-data route does not derive $N_{\mathrm{scr}} = S_{\mathrm{dS}}$ from the bare local axioms. This logic is compatible with the BW scaling branch but does not depend on the scaling-limit algebra remaining type I.

What this solves vs. what it assumes. The model does **not** solve the classic "give me a unitary CFT at future infinity" problem. It does not aim there. It also does **not** prove that every refinement-stable MaxEnt branch lands in the static-patch geometric modular branch with emitted cap pair and standard modular action. What it does provide is a coherent route to **patch holography** in which de Sitter static patches are natural:

1. the fundamental object is a horizon screen in a static-patch description;
2. Λ is a capacity parameter tied to finite Hilbert-space dimension, not a locally reconstructible vacuum-energy term;
3. Einstein-like dynamics emerge up to Λg_{ab};
4. on the explicit **T2** plus extraction-and-rigidity branch, the realized scaling-limit cap modular action is geometric and may be outer on a non-type-I observer algebra.

BW-side status. The Lorentz side stays at the explicit conditionality boundary recorded elsewhere in the manuscript. The internal extension route is a two-object scaffold on the geometric subnet: first a realized scaling-limit cap-pair extraction from transported marginals, then ordered cut-pair rigidity on that realized limit. The pressure point is the first object. Until that realized scaling-limit geometric cap pair exists, any stronger vacuum or canonical-cap phrasing remains theorem-external shorthand for the **T2** plus extraction-and-rigidity branch and does not become a fully internalized static-patch theorem.

Many observers, one Λ. In this framework, each timelike observer is associated with a horizon patch rather than a single global description. The de Sitter parameter Λ is the shared global capacity constraint across overlap-consistent descriptions.

Summary. The model gets de Sitter by moving the holographic screen from "infinity" to an observer's horizon and by elevating de Sitter entropy, i.e. finite screen capacity, to a fundamental input. The usual dS-holography obstacles are precisely the ones avoided by refusing the boundary-at-infinity viewpoint. This is not a claim of a solved dS/CFT dual. It is a static-patch holography program with an explicit local/global split and an explicit extracted-geometric-subnet conditionality boundary.

3.7 Gauge Reconstruction and Standard Model Structure

3.7.1 Edge sector category and gauge group reconstruction

At any fixed UV cutoff, edge-center completion provides finite tensor categories of transportable edge charges. The local MaxEnt / collar-mixing package established earlier controls only fixed-cutoff recoverability, modular-support localization, and carried error terms on the realized branch. It is logically separate from the question whether transportable edge sectors survive refinement, and it supplies no theorem that the refinement-limit sector category is trivial or nontrivial. A later nontrivial transportable colimit is therefore treated here as additional gauge data rather than as something decided by the mixing estimate itself. Whenever compact gauge reconstruction is invoked, that transportable refinement-directed sector system is exactly the content of **T6**.

Assume therefore that in the EFT regime there is a directed system $(\mathsf{Sect}_r)_r$ of transportable edge-sector categories with refinement functors, and write

$$\mathsf{Sect}_\infty := \varinjlim_r \mathsf{Sect}_r,$$

for the directed colimit retaining the sectors and intertwiners that persist in that transportable system, with objectwise finite-dimensional fibers. The theorem below is neutral about whether Sect_∞ is trivial or nontrivial: if **T6** supplied only the tensor unit, the reconstructed compact group would be the trivial group.

This gives a tensor category Sect_∞ of edge charges:

- objects: transportable sector labels that persist in the assumed directed system,
- morphisms: intertwiners between sectors,
- tensor product: fusion by collar concatenation, $\alpha \otimes \beta = \bigoplus_\gamma N_{\alpha\beta}{}^{\gamma}\, \gamma$,
- duals: orientation reversal $\alpha \leftrightarrow \bar{\alpha}$,
- symmetric braiding in the EFT regime (no anyonic statistics in 3+1D).

Let $\mathcal{F} : \mathsf{Sect}_\infty \to \mathsf{Hilb}_{\mathrm{fd}}$ be the bosonic fiber functor that sends each sector to its edge multiplicity space. The fibers are finite-dimensional objectwise even though Sect_∞ can have infinitely many simple objects. The theorem begins only once that transportable colimit and its fiber functor are given; those data are not consequences of Axiom 3 alone.

Theorem 6.1 (Tannaka/DR reconstruction). If Sect_∞ is a rigid symmetric C^* tensor category with a faithful bosonic fiber functor $\mathcal{F}$, then

$$G := \mathrm{Aut}_\otimes(\mathcal{F})$$

is a compact group and $\mathsf{Sect}_\infty \simeq \mathrm{Rep}(G)$. In particular G is unique up to isomorphism, and this group is a compact subgroup of a product of unitary groups.

Proof. The MaxEnt/local-Gibbs/collar-mixing package is not used here to prove existence of the transportable colimit or to decide whether it is trivial or nontrivial; those data are supplied by the directed-colimit premise **T6**. Fix a small skeleton of Sect_∞. For every object X, a monoidal natural automorphism $\eta \in \mathrm{Aut}_\otimes(\mathcal{F})$ has a unitary component $\eta_X \in U(\mathcal{F}(X))$, so

$$\mathrm{Aut}_\otimes(\mathcal{F}) \hookrightarrow \prod_X U(\mathcal{F}(X)), \qquad \eta \mapsto (\eta_X)_X.$$

For each intertwiner $f : X \to Y$, naturality gives the closed relation $\mathcal{F}(f)\eta_X = \eta_Y \mathcal{F}(f)$. For each pair X, Y, the monoidal structure isomorphism $J_{X,Y}$ of $\mathcal{F}$ gives the closed relation

$$\eta_{X\otimes Y} = J_{X,Y}\,(\eta_X \otimes \eta_Y)\,J_{X,Y}^{-1}, \qquad \eta_{\mathbf{1}} = \mathrm{id}_{\mathbb{C}}.$$

Hence $G = \mathrm{Aut}_{\otimes}(\mathcal{F})$ is a closed subgroup of the compact product $\prod_X U(\mathcal{F}(X))$, so G is compact. The remaining DR/Tannaka hypotheses are exactly the stated ones: Sect_∞ is rigid, symmetric, and C^*, and $\mathcal{F}$ is a faithful bosonic fiber functor with finite-dimensional values. Therefore the Doplicher–Roberts/Tannaka reconstruction theorem applies and gives a symmetric C^*-tensor equivalence $\mathsf{Sect}_\infty \simeq \mathrm{Rep}(G)$ [20, 21]. If another compact group G' gave the same category through a faithful bosonic fiber functor, the induced symmetric tensor equivalence $\mathrm{Rep}(G) \simeq \mathrm{Rep}(G')$ compatible with the forgetful functors would identify both groups as the automorphism group of the same fiber functor. Thus $G \cong G'$, so the reconstruction is unique up to isomorphism. QED.

Corollary 6.1 (field algebra reconstruction, conditional). If in the small-region limit the edge sectors are localized and transportable in the DHR sense (i.e., charges can be moved between patches without changing their fusion), then there exists a field algebra $\mathcal{F}$ and a compact group G such that $\mathcal{A} = \mathcal{F}^G$. This is the Doplicher–Roberts reconstruction of local gauge symmetry from sectors. QED.

Proposition 6.1a (Transportability from gluing obstruction on the central branch). On the central-defect branch, the gluing framework identifies the explicit loop-coherence criterion that accompanies DHR transportability in the later ordinary compact-gauge route. In the gluing framework (Section 3), transportability is the statement that charges can be moved between patches without changing fusion rules. The obstruction to path-independent transport is the Čech-type central cocycle z_{ijk} from the central-defect subbranch.

Explicitly: the gluing framework gives a gauge-invariant Čech 2-cocycle class $[z]$ built from triple overlaps (Section 6.6). Transportability holds iff this class vanishes:

$$\text{DHR transportable} \iff [z] = 0 \iff \text{loop-coherent gluing.}$$

Proof. Transportability means charges can be moved along any path without affecting the result. In gluing language, this is path-independent parallel transport of edge labels. Lemma 6.12 shows that loop-coherent global gluing exists iff $[z] = 0$. But loop-coherent gluing is exactly path-independent transport, so the equivalence holds. QED.

Corollary. On the central branch, the DHR-transportability condition in Corollary 6.1 is tracked explicitly by the equivalent loop-coherence condition $[z] = 0$. This does not make transportability theorem-internal to OPH; it makes the central gluing-side criterion explicit. On the genuinely noncentral branch, Corollary 3.4a gives the parallel fixed-cutoff criterion in terms of the higher-gauge class q_Σ, but that branch is handled by crossed-module data rather than by the ordinary DR field-algebra corollary.

3.7.2 Selecting the SM factors (derived from MAR)

Theorem 6.1 yields *some* compact G. Axiom 5 (MAR, Minimal Admissible Realization) uniquely determines which G is realized.

Axiom 5 (MAR): Minimal Admissible Realization. Among all OPH-realizable sector packages $\mathfrak{S}$ consisting of the connected Lie gauge-sector image relevant in the low-energy EFT, its admissible light chiral matter content, and one Higgs doublet, and which are (i) loop-coherent / transportable (vanishing relevant obstruction: $[z] = 0$ on the central branch or $q_\Sigma = 0$ on

the higher-gauge branch), (ii) anomaly-free, (iii) refinement-stable with light chiral matter, (iv) single-Higgs Yukawa-completable with one connected abelian charge factor acting nontrivially on the coupled carrier, (v) intrinsically CP-capable, (vi) weak-sector UV-completable, the realized low-energy package is the lexicographically minimal one under

$$C(\mathfrak{S}) = (\chi_{\mathrm{cpl}},\ N_{\mathrm{nonab}},\ N_c,\ N_g).$$

The complete formal statement, admissibility definitions, and proof route are given in the rest of Section 6.2.

Here χ_{cpl} is the **coupled edge capacity**: the dimension of the minimal unitary carrier containing a common irreducible block on which the admissible pseudoreal and complex nonabelian charge types both act nontrivially. This is intentionally stronger than the abstract minimal faithful representation dimension. For $S(U(3) \times U(2))$, the block-diagonal action on $\mathbb{C}^3 \oplus \mathbb{C}^2$ is faithful of dimension 5, but it is not coupled and therefore does not enter MAR. The object minimized by MAR is the sector package $\mathfrak{S}$, not the bare tensor category by itself. MAR is thus an explicit structural-economy selector on the admissible class, not a theorem derived from the earlier axioms.

Note on transport obstruction. The loop-coherent gluing condition is an explicit premise of the gauge derivation. On the central branch this is $[z] = 0$ (Proposition 6.1a); on the genuinely noncentral branch it is $q_\Sigma = 0$ (Corollary 3.4a). In either form it ensures that the reconstructed compact group acts as a genuine global gauge symmetry.

What MAR derives. Product structure, minimal sector content, and coupled edge-capacity minimality are consequences of MAR applied to the admissible class:

- **Product structure**: follows from the minimal coupled carrier $\mathbb{C}^3 \otimes \mathbb{C}^2$, which enforces commuting color and weak actions.
- **Minimal sector content**: the pseudoreal doublet and complex triplet are the minimal nonabelian representations satisfying the admissibility conditions, while the connected abelian charge factor is an explicit admissibility input.
- **Coupled edge-capacity minimality** (formerly S): is the first component of MAR's complexity vector.

With MAR stated, the SM derivation proceeds via standard lemmas:

Lemma 6.2 (Product factorization implies product group). If $\mathsf{Sect} \simeq \mathrm{Rep}(G)$ and $\mathsf{Sect} \simeq \mathsf{Sect}_1 \boxtimes \mathsf{Sect}_2$, then

$$G \cong G_1 \times G_2, \qquad \mathsf{Sect}_i \simeq \mathrm{Rep}(G_i).$$

QED.

Lemma 6.3 (SU(2) from a pseudoreal doublet). Let H be a positive-dimensional compact connected Lie group with a faithful 2D pseudoreal unitary representation V. Then the semisimple part of H contains an $SU(2)$ factor acting as the fundamental doublet. Finite or disconnected counterexamples are not part of the theorem package, which only applies the lemma to the identity component on the relevant nonabelian image. QED.

Lemma 6.4 (SU(3) from an irreducible triplet). Let H be a positive-dimensional compact connected Lie group with a faithful irreducible complex 3D unitary representation W. Then the semisimple image contains an $SU(3)$ factor acting as the fundamental triplet. Finite or disconnected counterexamples are not part of the theorem package, which only applies the lemma to the identity component on the relevant nonabelian image. QED.

Lemma 6.5 (Connected abelian factor criterion). If the admissible sector package contains a connected abelian charge factor acting nontrivially on the coupled carrier, then the identity component of the abelianized reconstructed group contains a one-torus. Under the single connected abelian-factor admissibility premise, this factor is $U(1)$. QED.

Proposition 6.6 (physical group quotient). If the realized matter spectrum has hypercharges quantized in sixths, then the kernel acting trivially on all realized sectors is Z_6, so

$$G_{\text{phys}} = \frac{\mathrm{SU}(3) \times \mathrm{SU}(2) \times \mathrm{U}(1)}{\mathbb{Z}_6}.$$

QED.

Proposition 6.6a (SM from MAR). Under Axiom 5 (MAR):

- Admissibility conditions (iii)–(iv) require both a pseudoreal nonabelian charge type and a complex nonabelian charge type, and include one connected abelian charge factor acting nontrivially on the coupled carrier (Lemma 6.7, Corollary 6.8).
- The minimal faithful pseudoreal representation is the doublet ($\chi = 2$), giving $SU(2)$. No intrinsically complex 2D irrep qualifies in the connected compact Lie class, because the connected derived image of any irreducible 2D unitary representation lies in $SU(2)$, so the nonabelian action is pseudoreal up to abelian twist. The first intrinsically complex case is therefore the triplet ($\chi = 3$), giving $SU(3)$.
- The minimal coupled carrier for both is $\mathbb{C}^3 \otimes \mathbb{C}^2$, giving coupled edge capacity $\chi_{\text{cpl}} = 6$.
- The block-diagonal faithful representation $\mathbb{C}^3 \oplus \mathbb{C}^2$ of $S(U(3) \times U(2))$ has dimension 5, but it is not coupled and therefore is not the MAR minimizer.
- The maximal compact subgroup of $U(6)$ acting on $\mathbb{C}^3 \otimes \mathbb{C}^2$ with commuting actions is $(SU(3) \times SU(2) \times U(1))/(\text{finite center})$.
- The commutant of $SU(3) \times SU(2)$ inside $U(6)$ is exactly $U(1)$, so any connected abelian factor acting nontrivially on the coupled carrier is necessarily a single $U(1)$, and no additional continuous factors appear without increasing χ_{cpl}.
- Product structure is not separately assumed: it follows from the tensor product structure of the minimal coupled carrier.

Combined with Proposition 6.6 (hypercharges quantized in sixths from the realized spectrum), this yields:

$$G_{\text{phys}} = \frac{SU(3) \times SU(2) \times U(1)}{Z_6}.$$

The full proof route is given in the remainder of Section 6.2.

3.7.3 Refinement stability and unprotected relevant operators

The refinement-stability used later in the gauge branch is contained in the third OPH axiom. Because the same finite local constraint family is preserved across cutoffs, the realized UV states lie in one common finite-dimensional MaxEnt family rather than in a new coupling space at every refinement step. The selected stable branch of this family is therefore not an extra axiom beyond the MaxEnt branch itself.

Concretely, if

$$\omega_\ell(\lambda) = Z_\ell(\lambda)^{-1} \exp\left(-\sum_x \sum_a \lambda_a O_a(x) - \sum_b \mu_b Q_b \right),$$

then any refinement channel $\Phi_{\ell\to L}$ compatible with Axiom 3 acts on the realized branch by an induced finite-dimensional map

$$\Phi_{\ell\to L}\big(\omega_\ell(\lambda)\big) = \omega_L\big(R_{\ell\to L}(\lambda)\big).$$

The resulting refinement-stable branch is simply a trajectory or invariant subset of this finite-dimensional multiplier map. Accordingly, when later sections speak of a refinement-stable directed colimit of sectors, the refinement-stable qualifier means persistence along this MaxEnt branch itself; the remaining extra ingredients are transportability, symmetry, and the bosonic fiber-functor clauses, and the MaxEnt/mixing package does not decide whether the resulting colimit is trivial or nontrivial.

Relevant operators that are neither symmetry-forbidden nor retained in the selected constraint family are precisely the directions that try to push the flow off that stable branch.

Lemma 6.7 (refinement-stable MaxEnt branch forbids unprotected relevant operators). Assume the local finite-constraint MaxEnt/refinement branch of the third OPH axiom. Let $\mathcal{O}$ be a gauge-invariant Lorentz-scalar relevant deformation in the emergent EFT sense ($\Delta < 4$ in $3+1$D), allowed by symmetry and absent from the retained constraint family. Then the selected branch can keep the coupling of $\mathcal{O}$ at zero only if symmetry forbids that direction or the constraint family explicitly retains it. Otherwise generic refinement induces a nonzero component along that direction and the flow leaves the selected branch. This is a branch-persistence statement, not a universal entropy-ordering theorem for arbitrary off-branch phases.

Proof sketch. Linearize the induced refinement map $R_{\ell\to L}$ on the finite-dimensional multiplier space around the selected branch. A relevant operator gives an unstable direction with scaling exponent $y > 0$. If $\mathcal{O}$ is not fixed by symmetry and is not part of the retained constraint family, generic UV mismatch produces a nonzero component along that direction. Because $y > 0$, repeated coarse-graining amplifies the component and pushes the flow off the selected branch unless one fine-tunes it away at every scale or protects it by symmetry or by the declared constraint family. QED.

Corollary 6.8 (chirality selector). A gauge-invariant Dirac mass term is a relevant scalar. If both chiralities exist in conjugate representations, the mass term is allowed and will be generated under refinement unless symmetry-forbidden or explicitly retained as a protected constraint. Therefore keeping light fermions on the selected refinement-stable branch requires chiral matter content or an explicit protecting mechanism. QED.

3.7.4 Generation number from CP violation and refinement stability (derived from MAR)

Anomaly cancellation is generation-by-generation, so it does not fix the number of generations. The admissibility conditions constrain the window; MAR selects the minimum.

Proposition 6.9 (The number of generations is N_g = 3). Under (i) intrinsic CP violation in the quark sector, (ii) UV-completability of SU(2)_L (asymptotic freedom at one loop), (iii) MaxEnt/refinement stability selecting minimal viable spectrum, and (iv) the derived N_c = 3 from Theorem 6.14, the generation number is

$$N_g = 3.$$

Inputs.

1. **Intrinsic CP violation exists** in the quark sector (empirical fact; also, the framework treats "intrinsic CP violation" as a selector input).

2. **UV-completability proxy**: SU(2)_L is asymptotically free at one loop in the emergent EFT.
3. **MaxEnt + refinement stability** penalizes unnecessary unfixed flavor structure, selecting the minimal viable spectrum.
4. Use the derived N_c = 3 from Theorem 6.14.

Step 1: CP violation lower bound. The number of physical CP-violating phases in an N_g × N_g CKM matrix is:

$$\#(\text{CP phases}) = \frac{(N_g - 1)(N_g - 2)}{2}.$$

- For N_g = 1, 2: this is 0 → **no intrinsic CP violation possible**.
- For N_g = 3: this is 1 → **intrinsic CP violation possible**.

So intrinsic CP violation requires:

$$N_g \geq 3.$$

Step 2: SU(2) asymptotic freedom upper bound. The one-loop coefficient is:

$$b_{\mathrm{SU}(2)} = \frac{22}{3} - \frac{1}{3} N_g (N_c + 1) - \frac{1}{6},$$

where the final $-1/6$ is the contribution of one complex Higgs doublet. Asymptotic freedom means $b_{\mathrm{SU}(2)} > 0$, i.e.,

$$N_g (N_c + 1) < \frac{43}{2}.$$

With N_c = 3, we have N_c + 1 = 4, so:

$$4N_g < \frac{43}{2} \quad \Rightarrow \quad N_g \leq 5.$$

Combining: 3 ≤ N_g ≤ 5.

Step 3: MAR selection. Given the allowed window {3, 4, 5}, MAR (fourth component of the complexity vector $C(\mathfrak{S})$) selects the smallest viable choice:

$$N_g = 3.$$

QED.

Why this is convincing.

- It predicts a **single integer**.
- It uses **two admissibility conditions** (CP violation exists; weak sector is UV-completable) plus MAR's lexicographic minimality.
- It is not a fit to a continuous number.
- Under the stated gauge-selection hypotheses, this is a derived result, not conditional.

3.7.5 Hilbert-space formulation of gluing data

Let {P_i} be a good cover of the screen. For each patch, fix a representation

$$\pi_i : \mathcal{A}_i \to \mathcal{B}(\mathcal{H}_i).$$

For each overlap, choose a unitary intertwiner

$$U_{ij} : \mathcal{H}_j \to \mathcal{H}_i$$

such that for all $O \in \mathcal{A}_{ij}$,

$$\pi_i(O) = U_{ij}\pi_j(O)U_{ij}^\dagger.$$

Normalize U_ii = 1 and U_ji = U_ij†.

Lemma 6.10 (centrality on triple overlaps). On a triple overlap define

$$\Omega_{ijk} := U_{ij}U_{jk}U_{ki}.$$

For all $O \in \mathcal{A}_{ijk}$,

$$\Omega_{ijk}\pi_i(O) = \pi_i(O)\Omega_{ijk}.$$

Proof. Conjugation by U_ki sends π_i(O) to π_k(O), by U_jk to π_j(O), by U_ij back to π_i(O). Thus conjugation by Ω_ijk fixes π_i(O), so Ω_ijk commutes with π_i(O). QED.

Lemma 6.11 (gauge behavior). If Ũ_ij = V_i U_ij V_j† with V_i acting trivially on overlap observables, then

$$\tilde{\Omega}_{ijk} = V_i\Omega_{ijk}V_i^\dagger.$$

In particular, if Ω_ijk is central, its class is gauge invariant. QED.

3.7.6 Loop obstruction class (central defect)

Assume the defect is central and write

$$\varphi_{ij} := \mathrm{Ad}(U_{ij}) \quad \text{on } \mathcal{A}_{ij}.$$

This is the abelian truncation of the full 2-group obstruction in Section 3.4.

Then there exist central unitaries z_{ijk} such that

$$\varphi_{ij}\varphi_{jk}\varphi_{ki} = \mathrm{Ad}(z_{ijk}).$$

Theorem 6.12 (loop-coherent gluing iff vanishing obstruction). The family $\{z_{ijk}\}$ is a Čech 2-cocycle, and its class $[z]$ is gauge invariant. On any quadruple overlap P_{ijkl},

$$z_{jkl}z_{ikl}^{-1}z_{ijl}z_{ijk}^{-1} = 1.$$

A loop-coherent global gluing exists iff [z] = 0.

Proof. Compare two parenthesizations of ϕ_ij ϕ_jk ϕ_kl ϕ_li on a quadruple overlap to obtain the cocycle condition above. Gauge changes shift z by a coboundary. If [z]=0, rephase by a 1-cochain to eliminate defects and obtain path-independent transport. Conversely, loop-coherent gluing implies z_ijk = 1. QED.

3.7.7 EFT reduction to anomaly cancellation

Assume ExtEFT: a low-energy 3+1D chiral gauge theory exists with group G. Then the obstruction class $[z]$ is structurally analogous to the EFT 't Hooft anomaly class and is expected to map to it after a separate anomaly-descent construction. That map is not constructed in this manuscript, so $[z] = 0$ is used here as an internal gluing/transportability condition rather than a proved equivalence to EFT anomaly cancellation.

3.7.8 Hypercharge from anomaly freedom and Yukawas

Theorem 6.13 (Hypercharge from anomaly freedom and Yukawas). Assume gauge group SU(N_c) × SU(2) × U(1)_Y and one generation of left-handed Weyl fermions (Q, u^c, d^c, L, e^c), with a Higgs doublet H and Yukawa terms

$$QHu^c, \qquad QH^\dagger d^c, \qquad LH^\dagger e^c.$$

Then anomaly freedom and Yukawa invariance fix the hypercharges up to an overall normalization, yielding the Standard Model pattern for N_c = 3.

Proof. Yukawa invariance gives

$$Y_u = -(Y_Q + Y_H), \quad Y_d = -Y_Q + Y_H, \quad Y_e = -Y_L + Y_H.$$

Anomaly cancellation yields

$$\begin{aligned} SU(2)^2U(1): &\quad N_cY_Q + Y_L = 0, \\ \text{grav}^2U(1): &\quad 2N_cY_Q + N_cY_u + N_cY_d + 2Y_L + Y_e = 0. \end{aligned}$$

Solving gives

$$Y_L = -N_cY_Q, \quad Y_H = N_cY_Q, \quad Y_u = -(N_c+1)Y_Q, \quad Y_d = (N_c-1)Y_Q, \quad Y_e = 2N_cY_Q.$$

With these relations, SU(N_c)^{2}U(1) and U(1)^3 anomalies vanish automatically. Fixing the normalization by Q = T_3 + Y and Q(ν_L)=0 gives

$$Y_Q = \frac{1}{2N_c}.$$

For N_c = 3,

$$Y_Q = \frac{1}{6}, \quad Y_L = -\frac{1}{2}, \quad Y_e = 1, \quad Y_u = -\frac{2}{3}, \quad Y_d = \frac{1}{3}, \quad Y_H = \frac{1}{2}.$$

Without Yukawas, the cubic anomaly leaves two discrete branches (Y_u, Y_d exchange). Yukawa invariance selects the branch with a single Higgs doublet. QED.

Corollary 6.13a (Exact rational hypercharges). With the derived N_c = 3, the hypercharge assignments are uniquely fixed to exact rational values:

$$Y_Q = \tfrac{1}{6}, \quad Y_L = -\tfrac{1}{2}, \quad Y_u = -\tfrac{2}{3}, \quad Y_d = \tfrac{1}{3}, \quad Y_e = 1, \quad Y_H = \tfrac{1}{2}.$$

Why this is convincing.

- These are **exact rationals**, not approximate numbers.

- Their ratios are fixed by anomaly freedom + Yukawa invariance, and the absolute lattice is fixed by the standard normalization, with no continuous parameters to adjust.
- This high-precision set of numbers strongly constrains the realized matter package and matches observation exactly.

3.7.9 Witten anomaly and the number of colors

Theorem 6.14 (The number of colors is N_c = 3). Under the gauge structure SU(N_c) × SU(2)_L × U(1)_Y with one left-handed quark doublet Q per color and one left-handed lepton doublet L per generation, the global SU(2) anomaly (Witten, 1982) requires N_c to be odd. Under MAR (lexicographic minimization of $C(\mathfrak{S})$, where N_c is the third component), this yields:

$$N_c = 3.$$

Inputs.

1. Low-energy gauge group contains an SU(2)_L factor and an SU(N_c) color factor.
2. The matter content per generation includes:
   - one left-handed quark doublet Q which is an SU(2) doublet and carries color,
   - one left-handed lepton doublet L which is an SU(2) doublet and color singlet.
3. **Witten's global SU(2) anomaly constraint** (Witten, 1982): the number of left-handed SU(2) doublets must be even.
4. **MAR** (Selection Axiom): among allowed values, the third component of the complexity vector $C(\mathfrak{S})$ is minimized.

Proof. Count SU(2) doublets per generation:

- Quark doublets: N_c copies (one per color),
- Lepton doublets: 1 copy.

Total doublets per generation:

$$N_c + 1.$$

Witten anomaly cancellation requires this to be even:

$$N_c + 1 \equiv 0 \pmod 2 \quad \Rightarrow \quad N_c \text{ is odd.}$$

The Witten constraint alone allows N_c ∈ {1, 3, 5, 7, ...}. N_c = 1 fails admissibility (SU(1) is trivial, no complex nonabelian charge type). MAR (input 4) then selects N_c = 3 as the smallest nontrivial value. QED.

Status. The Witten anomaly derives N_c odd. The specific value N_c = 3 follows from MAR applied to the admissible class. Under the stated gauge-selection hypotheses, this is derived, not assumed.

Why this is convincing.

- It predicts a **single integer** given the minimality selector.
- The odd constraint is independent of continuous parameters, RG running, masses, or Yukawa values.
- It cannot be adjusted without changing the basic notion of electroweak doublets and color replication.

3.7.10 Bond-dimension gatekeeping

In tensor-network or code realizations, gauge actions act on edge factors of size χ, so emergent compact gauge groups embed in $U(\chi)$. This shows a capacity constraint: accommodating SU(3) color and SU(2) weak factors shows $\chi \geq 6$ in the minimal case, consistent with the MAR-derived gauge group.

3.7.11 Inevitability of photon and graviton

The model requires photons and gravitons.

Photon inevitability chain:

1. Gauge-as-gluing states that overlap identifications have redundancy forming a local groupoid.
2. Theorem 2.3 (edge-center completion) decomposes collar Hilbert spaces into sectors labeled by boundary gauge representations.
3. Theorem 6.1 (Tannaka/DR reconstruction) recovers a compact gauge group G from the transportable bosonic edge-sector category defined by these fusion rules.
4. Corollary 6.1 (conditional on DHR transportability) reconstructs a field algebra with G as a local gauge symmetry.
5. For the Standard Model, G includes U(1)_em after electroweak symmetry breaking.
6. A gauge boson is the quantum of the gauge field. Once U(1)_em emerges from overlap redundancy, its gauge field exists, and its quantum (the photon) must exist.

The photon is not postulated. It is forced by the axioms through the chain above. The photon mediates the correlations between charged excitations in different patches; it is how the U(1) redundancy structure propagates through the algebra net.

Graviton inevitability chain:

1. Theorem 4.2 (BW_{S^2}) shows that on the stated OPH Bisognano–Wichmann refinement branch, and under collar Markov locality plus the controlled refinement-limit premises, modular flow on caps becomes geometric conformal dilation.
2. Theorem 4.3 identifies the induced kinematic group as $\mathrm{Conf}^+(S^2) \cong \mathrm{PSL}(2,\mathbb{C}) \cong \mathrm{SO}^+(3,1)$, the Lorentz group.
3. Theorem 5.1 shows that, under the stated scaling-limit null-stress and reference-state premises, the condition $\delta S_gen = 0$ implies the rest-frame first-variation Einstein relation, with overlap consistency upgrading it to the semiclassical Einstein equations in the EFT regime.
4. The metric tensor emerges as the compression of modular flow data, and its dynamics are fixed by the derived fixed-cap generalized-entropy stationarity theorem.
5. A dynamical metric in a quantum theory requires a spin-2 quantum field. Its quantum (the graviton) must exist.

The graviton is not postulated. It is forced by the axioms through the chain above. Diffeomorphism invariance emerges because the bulk spacetime description is a compression of screen data; different coordinate descriptions are redundancies in how that compression is presented.

3.7.12 Quotient-protected charge quantization

Theorem 6.19 (Charge quantization and no fractional color singlets). If the global gauge group is

$$G_{\text{phys}} = \frac{\text{SU}(3) \times \text{SU}(2) \times \text{U}(1)}{Z_6},$$

as derived in Proposition 6.6, then

All color-singlet states have integer electric charge.

Equivalently: no stable isolated particles with charges like $\pm 1/3$ can exist as color singlets.

Proof. The Z_6 quotient identifies the center elements (e^{2πi/3}, -1, e^{iπ/3}) ∈ SU(3) × SU(2) × U(1) with the identity. For a color-singlet state ($\tau = 0$), the SU(3) factor acts trivially. The remaining identification requires the SU(2) × U(1) quantum numbers to satisfy

$$(-1)^{2j} \cdot e^{i\pi n/3} = 1,$$

where j is the SU(2) spin and n = 6Y is the integer hypercharge label. This gives n ≡ -6j (mod 6), i.e., n ≡ 0 (mod 6) for integer j and n ≡ 3 (mod 6) for half-integer j. Equivalently: Y is integer when j is integer, and Y is half-integer when j is half-integer.

After electroweak breaking, Q = T_3 + Y. For integer j, $T_3 \in \mathbb{Z}$ and Y ∈ ℤ, so Q ∈ ℤ. For half-integer j, $T_3 \in \mathbb{Z} + 1/2$ and Y ∈ ℤ + 1/2, so Q = (half-integer) + (half-integer) ∈ ℤ. In both cases, Q ∈ ℤ. QED.

Experimental status: No fractionally charged color-singlet particles have been observed. Three independent high-precision bounds confirm this:

1. **Neutrality of matter** (PDG 2024): The proton-electron charge sum satisfies

   $$|q_p + q_e|/e < 1 \times 10^{-21},$$

   confirming charge quantization to 21 decimal places.

2. **Fractional charge searches in bulk matter**: Silicone oil drop experiments limit fractionally charged particle abundance to

   $$\text{(fractionally charged particles)/nucleon} \lesssim 10^{-22}.$$

3. **Collider searches** (CMS, PRL 134, 2025): Exclusions for stable particles with $q \in [e/3, 0.9e]$ up to masses ∼ 640 GeV (95% CL).

Existing searches are fully consistent with these structural consequences.

These are exact symmetry- or quotient-protected outputs: the empirical content is that violations remain excluded up to available experimental sensitivity.

3.7.13 Coupling extraction from edge-sector probabilities

The edge-center completion (Theorem 2.3) yields sector probabilities p_α on collar boundaries. These probabilities encode the renormalized gauge coupling through a heat-kernel/Laplacian weighting law.

Abelian case (Z_n). For a Z_n gauge theory, the edge sectors are labeled by charge q ∈ {0, 1, ..., n-1}. The correct "Casimir" eigenvalue is the Laplacian eigenvalue of the boundary random walk:

$$\lambda_q = 4 \sin^2 \left(\frac{\pi q}{n} \right).$$

Note: only in the limit n → ∞ and q ≪ n does λ_q ≈ $(2\pi q/n)^2 \propto q^2$. For finite n, the exact form is essential.

The sector probabilities follow a heat-kernel law:

$$p_q \propto e^{-t(\mu)\lambda_q},$$

where t(μ) is the "modular time" parameter encoding the scale. The extraction formula is:

$$t(\mu) = -\frac{\log(p_q/p_0)}{\lambda_q}, \qquad g_{\text{ent}}^2(\mu) = \frac{t(\mu)}{2\pi}.$$

Consistency requires that t extracted from different charges q agrees; this has been verified numerically (see Section 6.14).

Electric-center measurement. The edge sectors are measured using the *electric-center* prescription. For a region A and boundary vertex v ∈ ∂A, define the restricted star operator:

$$Q_v^{(A)} = \prod_{\ell \in \text{star}(v) \cap A} X_\ell^{\pm 1},$$

where X_ℓ is the shift operator on link ℓ. The sector projectors are:

$$P_{v,q} = \frac{1}{n} \sum_{m=0}^{n-1} \omega^{-mq} \left(Q_v^{(A)}\right)^m, \qquad \omega = e^{2\pi i/n},$$

and the probabilities are p_{v,q} = ⟨P_{v,q}⟩. This electric-center operator, built from X's rather than Z's, correctly captures the boundary gauge charge/flux that labels entanglement edge sectors.

Non-abelian generalization. For SU(N) gauge theories, the edge sectors are labeled by irreducible representations with probabilities:

$$p_j \propto d_j \, e^{-t(\mu)C_2(j)},$$

where d_j is the dimension and $C_2(j)$ the quadratic Casimir. Extraction:

$$t(\mu) = -\frac{\log(p_j/p_0)}{C_2(j)}, \qquad g_{\text{ent}}^2(\mu) = \frac{t(\mu)}{2\pi}.$$

Theoretical derivation. The fixed-cutoff same-overlap edge law is carried by the regulated microphysics package once the declared overlap-sector projectors, reversible thermal branch, and quasi-local local-Gibbs generator of Theorem 2.6 are in hand. On this synthesis surface, the point is to summarize that fixed-cutoff source package and to state the refinement/Peter–Weyl lift boundary explicitly rather than to relocate the leaf-level source surface.

Theorem 6.20 (Fixed-cutoff edge-sector law and refinement lift boundary). Under the OPH axioms, the fixed-cutoff overlap-gauge realization of Sections 2.3 and 3.2, and the quasi-local local-Gibbs form supplied by Theorem 2.6, the regulated same-overlap edge-sector probability distribution satisfies:

$$p_R = \frac{d_R \, e^{-t\lambda_R}}{\sum_{R'} d_{R'} \, e^{-t\lambda_{R'}}}$$

where λ_R is the Laplacian eigenvalue on the R-isotypic component and t is determined by the collar Gibbs parameter.

Proof.

Step 1 (Edge Hilbert space). From the fixed-cutoff overlap-gauge realization, the edge degrees of freedom at a boundary circle Σ = ∂C live in a Hilbert space transforming under the gauge group G. Microscopically this is a finite-dimensional quantum-link edge space on the declared overlap interface; the effective representation-theoretic description used only for the refinement lift is modeled by $L^2(G)$. By the Peter–Weyl theorem [30]:

$$L^2(G) \cong \bigoplus_R V_R \otimes V_R^*$$

where V_R is the carrier space of irrep R.

Step 2 (Gauge invariance). The Gauss law constrains physical states. For an entanglement cut at Σ, the physical edge Hilbert space decomposes as $\mathcal{H}$_{edge}^{phys} = ⊕_R W_R where W_R contains states with flux in representation R.

Step 3 (Natural Hamiltonian). From the local-Gibbs form, the MaxEnt generator restricted to edge modes takes the form H_{edge} = Σ_R h_R P_R where P_R is the projector onto the R-sector. The key claim is that h_R = λ_R.

Justification: For a compact simple factor, the group Laplacian Δ_G = -Σ_a (T^a)^2 is the unique bi-invariant second-order differential operator up to scale. For a product group $G = \prod_i G_i$, the most general bi-invariant second-order operator is a positive linear combination $\sum_i c_i \Delta_{G_i}$, with one coefficient per factor. Any other gauge-invariant local choice would require higher derivatives, violating locality. For finite groups, the Cayley graph Laplacian plays the same role: λ_R = |S| - (1/d_R) Σ_{s∈S} χ_R(s).

Step 4 (MaxEnt selection). MaxEnt selects the Gibbs state:

$$\rho_{\text{edge}} = \frac{1}{Z} e^{-tH_{\text{edge}}} = \frac{1}{Z} \sum_R e^{-t\lambda_R} P_R.$$

Step 5 (Sector probabilities). The probability of sector R is p_R = Tr(ρ_{edge} P_R). The effective dimension for entanglement is d_R (not d_R^2) because we trace over one side of the cut. This gives:

$$p_R = \frac{d_R\, e^{-t\lambda_R}}{Z}.$$

QED.

Why the entropy rank is d_R (instead of d_R^2). The full edge space has dimension d_R^2 in sector R (from $V_R \otimes V_R^*$). Entanglement entropy, however, measures correlations *across* the cut. After tracing over one side, the reduced density matrix has effective rank d_R. Mathematically: in the Markov normal form, the edge factor on one side contributes $\log d_R$ to the entropy.

Status. The fixed-cutoff same-overlap derivation is the theorem-bearing part of the edge-law package. The quasi-local local-Gibbs generator used here is derived from Theorem 2.6: if MaxEnt constraints are expectations of finitely many quasi-local operators, the entropy maximizer is automatically a Gibbs state with a quasi-local generator. The additional representation-theoretic step summarized here is the compact-group / Peter–Weyl refinement lift, together with its factorwise Laplacian identification. What remains open above the fixed-cutoff source package is refinement-stable universality/uniqueness of that lift, not the existence of the fixed-cutoff same-overlap Casimir law itself.

Normalization anchor: 2D Yang-Mills. The parameter t can be exactly matched to a conventional coupling in 2D Yang-Mills, where the physical Hamiltonian is literally the group Laplacian:

$$H = \frac{g^2}{2}\Delta_G, \qquad \Delta_G \chi_R = -C_2(R)\chi_R \quad \Rightarrow \quad E_R = \frac{g^2}{2}C_2(R).$$

Euclidean evolution for "time" A (the area of a cylinder in 2D YM) gives weight(R) $\propto$ exp(-A E_R) = exp(-g^2 A C_2(R)/2). Comparing with the heat-kernel expansion K_t(U) = Σ_R d_R χ_R(U) e^{-t C_2(R)} yields the exact identification:

$$t_{\text{phys}} = \frac{g^2 A}{2} \quad \text{(in 2D YM, no ambiguity).}$$

This shows that the Laplacian + MaxEnt → heat-kernel structure is exact in a solvable continuum Yang-Mills case. The coefficient in front of C_2 is fixed. In any regime where the edge theory reduces to an effective 2D YM with known "Euclidean thickness" A_eff:

$$g^2(\mu) = \frac{2}{A_{\text{eff}}(\mu)} \cdot \frac{\Delta_R(\mu)}{C_2(R)},$$

and the RHS must be R-independent. This R-independence is an internal precision consistency test; the formula itself is the normalization map that connects t to the conventional gauge coupling.

3.7.14 Numerical validation of the heat-kernel law

The heat-kernel/Laplacian weighting of edge sectors has been validated in explicit 2D Z_n gauge models on closed geometries.

Model. A 2×2 periodic lattice gauge theory (8 links) with Z_n link Hilbert spaces and Hamiltonian:

$$H = -K\sum_p \mathrm{Re}(B_p) - h\sum_\ell \mathrm{Re}(X_\ell) - \Gamma\sum_v \mathrm{Re}(A_v),$$

where X_ℓ is the Z_n shift on link ℓ, B_p is the oriented plaquette operator (product of Z's around plaquette p), and A_v is the oriented star/Gauss operator (outgoing X, incoming X†). With K = 1 and Γ = 5, the ground state satisfies ⟨A_v⟩ = 1 at all vertices to numerical precision.

Region and edge operator. Region A consists of links whose tail has x = 0 ("half-lattice" cut). At each boundary vertex v, the electric-center edge charge is the restricted star Q_v^{(A)} = ∏_{ℓ ∈ star(v) ∩ A} X_ℓ^{±1}.

Results for Z_2. With $\lambda_1 = 4\sin^2(\pi/2) = 4$:

h	p_0	p_1	t	g_ent
0.5	0.8266	0.1734	0.391	0.249
1.0	0.9612	0.0388	0.803	0.357
2.0	0.9917	0.0083	1.194	0.436

Results for Z_3 (overconstrained test). With $\lambda_1 = \lambda_2 = 4\sin^2(\pi/3) = 3$:

h	p_0	p_1	p_2	t(q=1)	t(q=2)	g_ent	m_plaq
0.2	0.4395	0.2803	0.2803	0.1500	0.1500	0.154	2.22
0.5	0.7509	0.1245	0.1245	0.5989	0.5989	0.309	1.75
1.0	0.9606	0.0197	0.0197	1.2956	1.2956	0.454	4.07

h	p_0	p_1	p_2	t(q=1)	t(q=2)	g_ent	m_plaq
1.5	0.9851	0.0074	0.0074	1.6288	1.6288	0.509	7.06
2.0	0.9921	0.0039	0.0039	1.8440	1.8440	0.542	10.10

The equality $p_1 = p_2$ is exact (charge conjugation symmetry in Z_3). The equality t_{q=1} = t_{q=2} is the crucial **overconstrained** check: at h = 1.0, extracting t from q = 1 and q = 2 independently gives t_{q=1} ≈ 1.2956389318579 and t_{q=2} ≈ 1.2956389318521. The agreement to ~10^{-14} (machine precision) confirms that the edge distribution genuinely follows the heat-kernel/Laplacian form.

Region-choice robustness. At h = 1, the extracted g_ent is nearly independent of region size:

- 2 links (one vertex's outgoing links): g_ent ≈ 0.453
- 4 links (half-lattice): g_ent ≈ 0.454
- 6 links (three vertices): g_ent ≈ 0.453

This locality confirms that the coupling is dominated by physics near the cut, not global bookkeeping, exactly what is expected if this behaves like a local QFT observable.

Results for Z_5 (golden ratio test). The Z_5 case provides a stringent test because the Laplacian eigenvalues have a distinctive ratio involving the golden ratio $\phi = (1+\sqrt{5})/2$:

$$\lambda_q = 4\sin^2\left(\frac{\pi q}{5}\right), \qquad \frac{\lambda_2}{\lambda_1} = \frac{\sin^2(72^\circ)}{\sin^2(36^\circ)} = \phi^2 \approx 2.618.$$

This ratio distinguishes the Laplacian law from naive alternatives: a linear model (λ_q ∝ q) would predict ratio 2, while a quadratic model (λ_q ∝ q^2) would predict ratio 4.

Simulations on a 2×2 torus in the dual/flux basis (125 states in the zero-winding sector) give:

h	Measured ratio $\ln(p_2/p_0)/\ln(p_1/p_0)$	Deviation from ϕ^2
0.5	2.25	14%
1.0	2.51	4%
2.0	2.619	< 0.1%

In the weak-field limit (h → 0, strong magnetic coupling), the simulation converges to the golden ratio squared. This confirms that the vacuum entanglement spectrum encodes the precise geometric structure of the gauge group Laplacian.

Significance. This validates the mathematical law (sector probabilities weighted by Laplacian eigenvalues) in explicit 2D gauge-invariant models with non-flat sector distributions. The Z_3 and Z_5 tests are structurally identical to SU(2)/SU(3): multiple irreps overconstrain the slope, and agreement confirms the mechanism works before jumping to nonabelian groups.

Results for S_3 (first nonabelian test). The abelian tests above use charge-sector projectors that reduce to Fourier modes. For nonabelian groups, the edge-sector projector must be generalized to character projectors:

$$P_{v,R} = \frac{d_R}{|G|}\sum_{h\in G}\chi_R(h^{-1})Q_v^{(A)}(h),$$

where d_R is the dimension of irrep R, χ_R is its character, and Q_v^{(A)}(h) is the restricted gauge action at boundary vertex v acting only on links in region A.

For S_3 (the smallest nonabelian group, order 6), there are three irreps: trivial (d=1), sign (d=1), and standard (d=2). The Cayley-graph Laplacian eigenvalues for the transposition generating set are:

$$\lambda_{\text{triv}} = 0, \qquad \lambda_{\text{sign}} = 6, \qquad \lambda_{\text{std}} = 3.$$

Exact reduction on one plaquette. For the single-plaquette model (4 links), imposing Gauss's law at all vertices means the physical wavefunction depends only on the plaquette holonomy's conjugacy class. Since S_3 has exactly 3 conjugacy classes, the gauge-invariant Hilbert space is 3-dimensional, spanned by the character states $\{|\chi_R\rangle\}$. In this basis, the edge-sector probabilities are exactly p_R = $|$c_R$|^2$ where $|\psi_0\rangle = \Sigma$_R c_R $|\chi$_R$\rangle$. This is not an approximation; it is an exact identity for the one-plaquette gauge-invariant sector.

The heat-kernel ansatz predicts p_R $\propto$ d_R exp(-t λ_R). Extracting t independently from the sign and standard irreps provides an overconstrained test: the ratio λ_sign/λ_std = 6/3 = 2 is a parameter-free prediction. Results from a single-plaquette S_3 lattice gauge model (K=1, Γ=5):

h	p_triv	p_sign	p_std	t (sign)	t (std)	Δt/t	log-ratio
0.5	0.909	0.0013	0.089	1.09	1.01	8.4%	2.17
1.0	0.980	7.5×10^{-5}	0.020	1.58	1.54	2.8%	2.06
2.0	0.996	4.3×10^{-6}	0.004	2.06	2.04	1.0%	2.02
5.0	0.9993	1.0×10^{-7}	0.00066	2.68	2.67	0.3%	2.006
12	0.9999	3.0×10^{-9}	0.00011	3.27	3.27	0.1%	2.002
100	1.0000	6.1×10^{-13}	2.0×10^{-6}	4.69	4.69	0.009%	2.0002

The "Δt/t" column shows the fractional difference (t_sign - t_std) / $\bar{t}$. The "log-ratio" column shows log(p_sign/p_0) / log(p_std/(2 p_0)), which should equal λ_sign/λ_std = 2 if the heat-kernel form holds exactly.

As h increases, both diagnostics converge: Δt/t drops below 10^{-4} and the log-ratio approaches 2.000. This is exactly the expected behavior: finite-size corrections are largest at strong coupling; the heat-kernel form becomes exact as the perturbative regime is approached.

This is the first nonabelian validation of the edge-sector extraction mechanism. The structure (character projectors, Laplacian eigenvalues from the group's Cayley graph, overconstrained t extraction) is identical to what will be used for SU(2) and SU(3).

Parameter-free predictions for SU(2) and SU(3). The heat-kernel law yields exact, parameter-free ratio predictions that require no scheme matching. Define the "Casimir log-gap":

$$\Delta_R \equiv \ln\left(\frac{p_0}{d_0}\right) - \ln\left(\frac{p_R}{d_R}\right) = t\, C_2(R).$$

Ratios of Δ_R cancel all unknowns (t, partition function):

$$\frac{\Delta_{R_1}}{\Delta_{R_2}} = \frac{C_2(R_1)}{C_2(R_2)} \quad \text{(exact, parameter-free).}$$

SU(2) predictions. Irreps labeled by spin j have d_j = 2j+1 and C_2(j) = j(j+1). The framework predicts:

- $\Delta_1/\Delta_1/_2 = 2/(3/4) = \mathbf{8/3} \approx \mathbf{2.667}$

- $\Delta_{3/2}/\Delta_{1/2} = (15/4)/(3/4) = \mathbf{5}$
- $\Delta_{3/2}/\Delta_1 = (15/4)/2 = \mathbf{15/8 = 1.875}$

SU(3) predictions. Irreps labeled by Dynkin indices (p,q) have $C_2(p,q) = (p^2 + q^2 + pq + 3p + 3q)/3$. Using the fundamental **3** = (1,0) with $C_2 = 4/3$ as the reference:

- $\Delta_8/\Delta_3 = 3/(4/3) = \mathbf{9/4 = 2.25}$
- $\Delta_6/\Delta_3 = (10/3)/(4/3) = \mathbf{5/2 = 2.5}$
- $\Delta_{10}/\Delta_3 = 6/(4/3) = \mathbf{9/2 = 4.5}$
- $\Delta_{15}/\Delta_3 = (16/3)/(4/3) = \mathbf{4}$
- $\Delta_{27}/\Delta_3 = 8/(4/3) = \mathbf{6}$

These are the SU(2)/SU(3) analogs of the Z_5 golden-ratio test: exact rational numbers fixed entirely by group theory, with no adjustable parameters.

Preliminary SU(3) results. A one-plaquette SU(3) "quantum link" model (finite truncated irrep basis, n_max = 12, $\kappa = 2$) has been used to extract t from 14 different irreps simultaneously. The results show internal consistency at the 1-3% level:

bare g^2	extracted t (mean±std)	g_ent	gap
0.3	0.314±0.0005	0.224	1.92
0.5	0.539±0.0025	0.293	1.83
0.8	0.896±0.012	0.378	1.72
1.0	1.144±0.025	0.427	1.64

The standard deviation across irreps provides a built-in error estimate. This is a QCD proton-physics surrogate rather than a full proton calculation; it lacks dynamical quarks and operates on a single plaquette. It nevertheless demonstrates that the nonabelian extraction machinery produces self-consistent outputs without tuning.

Extracting the normalization factor A_eff. The 2D YM anchor (Section 6.13) gives $t = g^2 A / 2$, so the "effective Euclidean thickness" is

$$A_{\text{eff}} = \frac{2t}{g^2}.$$

Computing this from the SU(3) table:

bare g^2	extracted t	A_eff
0.3	0.314	2.093
0.5	0.539	2.156
0.8	0.896	2.240
1.0	1.144	2.288

Mean: A_eff ≈ 2.19 with point-to-point scatter ~4%.

Extrapolation to weak coupling. The systematic drift in A_eff shows fitting $A_eff(g^2) = A_0 + a \cdot g^2$. A weighted linear fit gives:

$$A_0 = 2.004 \pm 0.012$$

with $\chi^2/\text{dof} \approx 0.09$, indicating excellent consistency. This strongly shows that, in this toy UV completion, the "missing normalization" converges to A_eff $\to 2$ as $g^2 \to 0$.

The normalization factor behaves like a quasi-constant and extrapolates to a simple value (≈ 2) in the weak-coupling limit. This provides a concrete path to absolute coupling predictions: once A_eff is determined from microphysics, the conversion $g^2 = 2t/\text{A_eff}$ fixes the gauge coupling without additional free parameters.

Internal validation summary. The heat-kernel law has been validated with increasing precision across multiple gauge groups:

- $\mathbf{Z_3}$: Overconstrained t extraction (q=1 vs q=2), precision ~10^{-14}
- $\mathbf{Z_5}$: Golden ratio squared ($\lambda_2/\lambda_1 = \phi^2$), precision 0.04%
- $\mathbf{S_3}$: Casimir log-ratio (λ_sign/λ_std = 2), precision 0.01%
- **SU(3)**: 14-irrep simultaneous extraction, precision 1-3%

The Z_3 test achieves machine precision because it is exactly overconstrained. The Z_5 and S_3 tests converge to their predicted ratios as coupling decreases. This provides strong internal validation of the mechanism "MaxEnt + Laplacian => heat-kernel sector weights" before applying it to physical gauge groups.

3.7.15 IBM Quantum Cloud hardware benchmarks

The lattice calculations above are internal numerical validations. The IBM Quantum Cloud benchmarks add a hardware check of whether OPH-motivated reduced-sector structures survive preparation and readout on real superconducting-qubit devices. These runs are small-sector consistency benchmarks rather than a standalone confirmation of the full framework. Their value is that they probe the same recoverability and heat-kernel observables on physical hardware and in simulation.

Detailed circuits, representative raw outputs, and rerun instructions are public in the project write-up `extra/IBM_QUANTUM_CLOUD.md` and the associated code/data bundle `code/ibm_quantum_cloud/`.

- **Stage 1 (Markov/recoverability benchmark).** On `ibm_marrakesh` and `ibm_fez`, the structured state reconstructs below both controls in conditional mutual information and above both controls in Petz fidelity. On `ibm_marrakesh`, the observed ordering is $0.2309 < 0.3890 < 0.9474$ for CMI and $0.9297 > 0.8649 > 0.6066$ for Petz fidelity; on `ibm_fez`, it is $0.1498 < 0.4992 < 0.9166$ and $0.9479 > 0.8117 > 0.6449$. This matches the predicted ordering on two independent real backends.
- **$\mathbf{Z_3}$ hardware sanity check.** The overconstrained Z_3 extraction passes cleanly on hardware: across prepared $t = 0.30, 0.60, 0.90$, the mean extracted t remains within about 0.02 of the target, the two independent extractions agree to within 0.0142, 0.0034, and 0.0043, and leakage stays below 0.1%. This shows that the reduced-sector preparation and readout path is internally coherent on-device before sharper ratio claims are interpreted.
- **$\mathbf{Z_5}$ exact-ratio test.** The parameter-free OPH target is $\Delta_2/\Delta_1 = \varphi^2 \approx 2.618$. Across repeated sweeps on `ibm_marrakesh` and one replication on `ibm_fez`, the best `ibm_marrakesh` points land within 0.8%, 1.2%, and 1.5% of the target, while a focused high-shot rerun at $t = 0.90$ lands within 2.6%. The `ibm_fez` replication is noisier but remains in the same neighborhood rather than producing a clean contradictory ratio.
- **$\mathbf{S_3}$ nonabelian ratio test.** The first nonabelian hardware run revealed a real layout-dependent bias. After reversing the qubit layout, the mitigated ratio moves from 1.8724 to 2.0299 against the OPH target 2.0000, and a direct confirmation run returns 2.0657. The

nonabelian target reappears once the hardware mapping is corrected, indicating an identifiable device/layout effect instead of a structural failure of the OPH prediction.

These IBM runs are best read as hardware benchmarks for the local edge-sector picture. They show that the predicted recoverability ordering survives on two real backends, the abelian exact-ratio tests land near the parameter-free targets, and the first nonabelian test returns to the expected ratio once a diagnosed layout bias is corrected. They add real-device evidence to the simulation-based checks, but they do not validate the full theorem stack by themselves.

3.8 Status, Tests, and Open Problems

3.8.1 Classification of results and remaining dependence

This section states which node of the program has been proved, which standard mathematics is imported, and which extra physical premises or external inputs remain load-bearing. The documentary node labels D1–D12 match the ledger in Section 3.2.6. The three-phase boundary is

$$\text{Phase I} = (D1\text{–}D5) \cup (D7\text{–}D9), \qquad \text{Phase II} = D6 \cup D10, \qquad \text{Phase III} = D12.$$

Node	Representative outputs in this manuscript	Standard mathematics used	Extra physical premises / external inputs required	Claim tier
D1	Theorem 3.1 schedule-independent normal form on the physical quotient together with observable-level confluence on the declared fixed-cutoff physical algebras for the recovery-derived repair relation	local-to-global Lyapunov descent on the inconsistency potential; local-diamond completion from overlap-associative center-sector / higher-gauge union-collar gluing; Newman's lemma	repair completeness; stated Petz support/CPTP control where that branch is used	Phase I structural theorem
D2	EC decomposition, exact/approximate Markov collar, and generalized-entropy split	HJPW; Petz; Fawzi–Renner	explicit recoverability control when exact identities are used; coarse-grained $A/(4G)$ dictionary	Phase I structural lemma/proposition layer

Node	Representative outputs in this manuscript	Standard mathematics used	Extra physical premises / external inputs required	Claim tier
D3	Theorems 4.2–4.3 Lorentz branch	Bisognano–Wichmann modular geometry; conformal-group classification	rotationally invariant constraints together with T2 and the theorem-local realized transported geometric cap-local system whose carried-collar schedule is derived from the transported fixed-local-collar Markov/faithfulness datum, and ordered cut-pair rigidity on the emitted scaling-limit geometric cap pair; the scaling-limit observer algebra may leave the finite type-I regulator class, so the theorem is automorphism-level there and $K_C = 2\pi B_C$ is only the special type-I limit form	Phase I conditional scaling-limit theorem
D4	Section 5.2 null modular bridge with exact-or-controlled strip additivity, endpoint-Lipschitz renormalized half-line families, the weak tail generator, the derived half-sided modular inclusion, and the explicit positive half-line null-translation generator on its Stone domain with affine half-line modular relations	Borchers–Wiesbrock positive-generator theorem for standard half-sided inclusions; Stone's theorem; distribution theory for half-line differentiation	the theorem-local null-cut center transfer, the extra inherited left/right strip-split condition needed for the spatial-collar-type tensor decomposition, and the exact-or-controlled Markov hypotheses of Section 5.2, plus T2; the same-half-line local modular-Hamiltonian input is imported on the continuum side only for the later generator/charge identification, the endpoint/propagation controls are internal to the local finite-constraint MaxEnt branch, the geometric scaling action on the null half-line blow-up net derives the half-sided modular pair, and bounded-interval formulas still require the separate interval-preserving projective branch	Phase I conditional bridge theorem

Node	Representative outputs in this manuscript	Standard mathematics used	Extra physical premises / external inputs required	Claim tier
D5	Theorem 5.1 rest-frame Einstein relation plus the conditional tensor upgrade of Corollary 5.1	fixed-volume area-variation identity for small geodesic balls; local quadratic-polarization argument for the tensor upgrade	the derived fixed-cap generalized-entropy stationarity theorem for admissible fixed-cap MaxEnt variations on the realized cap-label-preserving MaxEnt family, the internal small-ball bridge from the geometric cap generator, the locally Lorentzian $d = 4$ scaling regime, the separate interval-preserving projective branch used in the small-ball step, small-ball constancy assumptions, D4 carried remainder negligible at order ℓ^4, and the all-directions/all-reference-states premise used in the tensor upgrade	Phase I conditional scaling-limit theorem/corollary
D6	local/global theorem stack for Λ: null-invisible metric ambiguity, screen-capacity closure of the same Einstein branch, and the derived de Sitter static-patch parameter package, plus a legacy capacity-level neutrino side estimate	de Sitter entropy relation, static-patch radius/time formulas, and dimensional analysis	external input N_{scr} and the screen-capacity identification	Phase II input-dependent global closure / estimate
D7	Theorem 6.1 compact gauge reconstruction	Doplicher–Roberts / Tannaka reconstruction	bosonic symmetric transportable colimit and fiber-functor premises on the ordinary or central-defect branch; the genuinely noncentral fixed-cutoff branch is handled separately by crossed-module data, and the refinement-stable qualifier is internal to the local finite-constraint MaxEnt/refinement branch	Phase I conditional structural theorem
D8	product gauge structure up to finite quotient under MAR	compact Lie representation classification; Schur's lemma	the same ordinary or central-defect bosonic branch as D7, together with MAR, a connected admissible class, one connected abelian factor, and faithful action on the minimal coupled carrier	Phase I realized-branch theorem
D9	realized $\mathrm{SU}(3) \times \mathrm{SU}(2) \times \mathrm{U}(1)/\mathbb{Z}_6$, exact hypercharges, the counting chain $N_g = 3$ then $N_c = 3$, and no gauge-mediated proton decay as a product-group corollary	anomaly algebra; Witten anomaly; CKM CP counting	realized one-generation/one-Higgs package and MAR admissibility premises	Phase I realized-branch theorem/corollary chain

Node	Representative outputs in this manuscript	Standard mathematics used	Extra physical premises / external inputs required	Claim tier
D10	forward gauge-coupling closure and target-free source-only electroweak repair theorem of the D10 quantitative-closure branch	RG running and matching conventions	pixel ratio P, pixel closure, the source-only forward transmutation solve $\mathcal{F}(\alpha_U; P) = 0$, the fixed-cutoff edge heat-kernel / Casimir theorem on the microphysics surface together with the compact-group / Peter–Weyl lift, and the printed threshold conventions	Phase II quantitative-closure sector
D12	charged-lepton centered-readback / centered-to-affine frontier, textures, dark-sector, baryogenesis, spectroscopy, controlled large-N_{edge} string/worldsheet effective descriptions, and other downstream continuations	branch-specific EFT and phenomenological manipulations	explicit additional ansätze beyond D6, D9, or D10, including a distinct large-N_{edge} regime with fixed $\tau = tN_{\text{edge}}$ window and uniform genus-remainder control on the worldsheet lane	Phase III phenomenological continuation / program branch

The ledger implies the following reading rules.

1. Phase I, the recovered core carried by this manuscript, is D1–D5 together with D7–D9. These are the structural or realized-branch claims defended at theorem/corollary level once all stated premises are included.

2. Phase II contains the input-dependent global closure D6 of the D5 Einstein branch and the D10 quantitative-closure branch. Failure there does not by itself erase Phase I.

3. Phase III contains D12 continuations only. Nothing in D12 is theorem-level unless its extra ansätze are replaced by proved premises.

4. **Existence / strange-loop continuations**: self-simulation or strange-loop closure stories are continuation-level epilogues only. The internally available ingredient is a theorem-usable OPH state-and-law habitat built from overlap-consistent patch states together with finite Axiom-3 law coordinates. The recovered core does *not* derive an existence theorem, an internal closure map on an invariant admissible observer-supporting sector, or any uniqueness/stability result for such a map.

The corresponding low-energy effective-action form is

$$\mathcal{L}_{\text{eff}}^{\text{OPH}} \approx \sqrt{-g}\left[\frac{1}{16\pi G}(R - 2\Lambda) + \mathcal{L}_{\text{SM}}^{\text{realized branch}}\right] + \sum_i \frac{c_i}{M_*^{\Delta_i - 4}}\mathcal{O}_i.$$

Here $\mathcal{L}_{\text{SM}}^{\text{realized branch}}$ denotes the realized $\mathrm{SU}(3)\times\mathrm{SU}(2)\times\mathrm{U}(1)/\mathbb{Z}_6$ sector with the exact hypercharge lattice and the realized counting chain $N_g = 3$ then $N_c = 3$. Phase I secures the conditional Einstein branch and the realized Standard Model structural chain, D6 supplies the same global screen-capacity closure together with its derived de Sitter static-patch parameter package once the screen-capacity identification is adopted, and the higher-dimension operators absorb UV-sensitive or downstream corrections not fixed by the recovered core itself.

Symmetry-protected exact zeros such as the masslessness of the photon, gluons, and graviton, together with the absence of gauge-mediated proton decay, are corollaries inside Phase I. They do not define separate axiom-only nodes.

3.8.2 Structural assessment

- **Dynamics**: The GR chain requires modular covariance plus the fixed-cutoff null-surface modular bridge. Section 5.2 carries that bridge through the derived positive null-translation stage and the exact half-line generator/charge identification on the declared geometric scaling branch. The remaining inputs are modular covariance, the bounded-interval projective transport branch, and the tensor upgrade modulo the null-invisible metric term.
- **Gauge structure**: The gauge group is reconstructed from the transportable bosonic sector package, and the anomaly/gluing link is precise. Axiom 5 (MAR) fixes the SM factors; the CP/UV admissibility window and MAR give $N_g = 3$, and Witten's parity constraint with that realized generation count gives $N_c = 3$. On the central branch, Proposition 6.1a identifies the explicit loop-coherence criterion $[z] = 0$ that accompanies DHR transportability in the ordinary compact-group route. On the genuinely noncentral branch, Corollary 3.4a gives the parallel fixed-cutoff criterion $q_\Sigma = 0$, but that branch is handled separately by crossed-module data rather than by the ordinary DR field-algebra route. The chirality selector still has to be justified in explicit models.
- **Microscopic theory**: Quantum link models (Section 2.6) provide explicit fixed-cutoff realized presentations and give EC + Markov collars automatically; they are not proofs of a unique microscopic completion. Open items are (i) a microscopic derivation of Recoverable Generalized Entropy, and (ii) the continuum lift on the extracted geometric subnet, namely **T2** together with the realized transported geometric cap-local system, its derived carried-collar schedule above the transported fixed-local-collar Markov/faithfulness datum, and the later ordered cut-pair rigidity step.
- **Loop gluing beyond central defect**: At fixed cutoff the genuinely noncentral branch is closed by the crossed-module higher-gauge EC and transport package; what remains open is the refinement-limit transportable-sector lift together with quantitative matching to EFT anomalies.

3.8.3 Testable Items and Conditional Phenomenological Branches

The framework makes several directly testable statements, together with weaker continuation-level templates that can also be confronted with data:

GW horizon spectroscopy template (Section 5.11; continuation-level). If the extra discrete-horizon premises of Section 5.11 hold, the log-integer area spectrum gives discrete resonant frequencies for Kerr black hole horizons:

f_{k,m}(M,χ) = (m Ω_H)/(2π) + (c^3 g(χ))/(16π^2 GM) $\cdot$ ln(k) for k = 2, 3, 4, ...

After rescaling by remnant parameters, all events should stack at universal coordinates x_k = ln(k)/(8π). This is checkable with public LIGO/Virgo data as a continuation-level test of that extra branch. Absence of coherent stacking would challenge the discrete-horizon continuation, not the recovered-core package by itself.

Discrete Hawking template (Section 5.11; continuation-level). If the same discrete-horizon branch is realized, primordial black holes in the final evaporation stage should show comb structure at E_k/E$_2$ = ln(k)/ln(2). Current PBH burst searches (Fermi, H.E.S.S.) can constrain this with dedicated template analysis.

Casimir ratio precision (Section 8.1). Future lattice measurements of SU(3) edge-sector probabilities should confirm $\Delta_8/\Delta_3 = 9/4$ exactly, not 2.67 (dimension-only) or 5.06 (Casimir-squared). The full set of parameter-free SU(3) ratio predictions is:

- $\Delta_8/\Delta_3 = 9/4 = 2.25$

- $\Delta_6/\Delta_3 = 5/2 = 2.5$
- $\Delta_{10}/\Delta_3 = 9/2 = 4.5$
- $\Delta_{15}/\Delta_3 = 4$
- $\Delta_{27}/\Delta_3 = 6$

These exact rationals are fixed entirely by group theory (Casimir eigenvalue ratios), with no adjustable parameters. Any deviation identifies a measurement contradiction with the heat-kernel edge-sector mechanism.

Conditional $\mathbf{Z}_6$ center-label entropy scale (Section 6.18). Under the uniform sixfold center-label ensemble used in the flavor continuation, the associated entropy scale is $\log_2 6 \approx 2.585$ bits. This is a phenomenological ansatz tied to the quotient label set, not a theorem-level direct observable of the quotient alone.

Black-hole spectroscopy bookkeeping within the continuation template (Section 5.11). If one adopts the extra discrete-horizon selection rule, the semiclassical linewidth estimate, and the greybody matching used in Section 5.11, then the template carries secondary structure:

1. *Universal energy ratios*: E_k/E_2 = ln(k)/ln(2) exactly within the assumed log-integer rule. For example, E_3/E_2 = ln(3)/ln(2) ≈ 1.585. This arithmetic pattern of ratios distinguishes the template from other "quantized area" proposals that have different functional forms or free spacing parameters.

2. *Mass-independent fractional linewidth estimate*: The continuation-level linewidth estimate $\Gamma/\Delta E_k \approx 3$–$5\%$ is approximately independent of black hole mass.

3. *Fixed weight hierarchy*: Within the same template, line weights follow a (k-1)/k pattern from detailed balance in the log-integer transition rule, on top of the GR greybody envelope.

Inequality bounds on GR deviations (Section 5.8). The modular additivity defect satisfies the exact identity ⟨ΔK⟩ = -I(A:D|B), where I(A:D|B) is the conditional mutual information. Under the Markov/mixing assumptions, this defect is exponentially small in collar thickness:

|⟨ΔK⟩| ≤ 2|A| · η^{w/ξ}

This propagates into an explicit upper bound on how far the Einstein equation can deviate from GR in regimes where the emergence proof applies. Unlike typical beyond-GR frameworks that postulate corrections, OPH provides a quantitative ceiling: given the information-theoretic primitives, corrections decay exponentially with collar width. This "UV ignorance → rigorous inequality" structure is distinctive.

Shared excitation dictionary benchmark (Section 6.21). The same-family continuations are organized by one common excitation dictionary instead of a primitive one-number suppression law. The proof-facing data are the same-label gap/overlap scalars

$$(g_e, \omega_e), \qquad d_e = 1 - \omega_e, \qquad q_e = \sqrt{g_e d_e}, \qquad \eta_e = \log q_e - \frac{1}{3}\sum_f \log q_f,$$

equivalently the centered eta-class $[\eta_e]$ or the normalized weights

$$\mu_e := \frac{e^{\eta_e}}{\frac{1}{3}\sum_f e^{\eta_f}}.$$

Let s_e denote the realized sector-charge step carried by the arrow e. The internal integer data are the excitation multiplicities

$$m_{\gamma,e} \in \mathbb{Z}_{\geq 0}$$

with which a given readout path γ traverses the realized same-label arrows, equivalently the integers satisfying

$$s_\gamma = \sum_e m_{\gamma,e} s_e,$$

and the corresponding excitation action

$$A_\gamma = -\sum_e m_{\gamma,e} \log q_e.$$

Old "base-6 exponents" are only the compression

$$n_\gamma^{(6)} := A_\gamma / \log 6,$$

which becomes an actual defect count only after the extra sixfold-uniform collapse compatible with the realized $\mathbb{Z}_6$ quotient. The intrinsic neutrino lane factors directly through the same-label scalar certificate, the quark mass-side continuation compresses to $(\Delta_{ud}^{\text{overlap}}, \eta_Q^{\text{centered}})$ and the emitted ray $\mathcal{R}_{D12}^{ud}$, and the charged lane fixes centered logs once a charged source pair exists while the common-shift quotient is closed negatively. The exact lane-dependent theorem objects therefore sit above the dictionary: $\widehat{C}_e^{\text{cand}} \to \widehat{C}_e$ and then the post-promotion lift whose descended scalar is $\mu_{\text{phys}}(Y_e)$, with the physical identity-mode equalizer beneath that scalar, for charged leptons; the light-quark overlap-defect scalar $\Delta_{ud}^{\text{overlap}} = \frac{1}{6}\log(c_d/c_u)$ closes target-free, and on the emitted D12 mass ray this is equivalently the downstream one-scalar law `quark_d12_t1_value_law`, with `intrinsic_scale_law_D12` as the derived wrapper; a separate direct public descent theorem closes on the public physical quark frame class selected by P, where the exact physical sigma datum descends uniquely to target-free public data on that selected class, the affine mean law emits (g_u, g_d) algebraically, and the exact forward construction emits the exact sextet together with explicit exact forward Yukawas Y_u and Y_d; this is selected-class closure only and does not claim a global classification of all quark frame classes; a stronger class-uniform public descent theorem is separate and is not claimed here; while the weighted-cycle neutrino lane fixes the bridge-rigidity invariant $C_\nu = G^2 Q S^{-1/2} = 0.9994295999075177$ above the emitted proxy $P_\nu = 6.699825740519345$, with $B_\nu = P_\nu C_\nu = 6.696004159297337$ retained only as the paper-facing amplitude parameterization. The scalar benchmark $\varepsilon = 1/6$ survives only as the extra uniform sixfold collapse of this richer dictionary, so exponents $-\log(y_f)/\log 6$ are compare-only summaries of excitation action, not recovered-core outputs or independent flavor inputs.

No gauge-mediated proton decay (Section 6.11). If the gauge group is genuinely a product (from sector factorization), coupling unification is geometric (shared edge diffusion parameter) rather than simple-group embedding. This predicts unification-like coupling relations *without* GUT leptoquark bosons, hence no gauge-mediated proton decay. The combination "coupling unification + no proton decay" is a crisp discriminator against classic GUT predictions.

3.8.4 What is not predicted (gaps)

Section 3.8.1 fixes the three-phase boundary. The list below collects the items that remain outside Phase I even when they are discussed elsewhere in the manuscript.

Not predicted by the recovered core.

- **Flavor and Yukawa data**: Yukawa matrices, flavor hierarchies, charged-lepton continuation ansätze beyond exact centered readback, quark-texture exponents, CKM/PMNS closure, and related downstream flavor fits are not Phase-I outputs. The common proof-facing family base is the same-label excitation dictionary

$$(g_e, \omega_e) \longmapsto d_e = 1 - \omega_e, \qquad q_e = \sqrt{g_e d_e}, \qquad [\eta_e].$$

Its internal integer data are the realized traversal multiplicities

$$m_{\gamma,e} \in \mathbb{Z}_{\geq 0}$$

with

$$s_\gamma = \sum_e m_{\gamma,e} s_e.$$

The associated excitation action is

$$A_\gamma = -\sum_e m_{\gamma,e} \log q_e,$$

from which any compare-only base-6 compression is formed. The lane-specific readout boundary above that dictionary is explicit. On the lower D12 selector surface the same-label selector is fixed to σ_{ref}, and the mass side emits the one-parameter ray $D12^{\mathrm{mass}}_{ud}$. The target-free mass bridge closes as $\Delta^{\mathrm{overlap}}_{ud} = \frac{1}{6}\log(c_d/c_u)$, equivalently $\Theta^{\mathrm{mass}}_{ud} =$ `quark_d12_t1_value_law`, with $t_1 = \frac{5}{6}\ell_{ud}$, $\eta^{\mathrm{centered}}_Q = -\frac{1-x_2^2}{27}t_1$, $\kappa_Q = -t_1/54$, and the odd source package functorially, with `intrinsic_scale_law_D12` as the derived wrapper. A separate direct public descent theorem closes on the public physical quark frame class selected by P: the exact physical sigma datum descends uniquely to target-free public data on that selected class, the affine mean law emits (g_u, g_d) algebraically, and the exact forward construction emits the exact sextet together with explicit exact forward Yukawas Y_u and Y_d. That selected-class theorem hits the official PDG 2025 API running-quark surface exactly, with the top coordinate taken from the PDG cross-section entry. The bridge to the auxiliary direct-top PDG row remains open and is tracked in issue #207. Supporting exact surfaces are the same-family witness on `current_family_only` and the restricted common-refinement transport-frame exact chain. The theorem does not claim a global classification of all quark frame classes. A stronger class-uniform public descent theorem is separate and is not claimed here. The charged-lepton lane carries exact centered readback, a closed common-shift no-go, the live same-label q_e readback, a source-side determinant character for any fixed formal exponent vector $S_M = \sum_e M^{\mathrm{ch}}_e \log q_e$, a conditional determinant-line lift on theorem-grade physical charged data, and an algebraic mass readout from theorem-grade $A_{\mathrm{ch}}(P)$. The theorem lane does not emit a theorem-grade sector-isolated charged determinant exponent vector, and it does not identify a source-side determinant character with the physical charged determinant line. The charged source theorem is the determinant trace-lift attachment $3\mu(r) = S_M(r)$ on the charged determinant channel. The weighted-cycle neutrino theorem boundary is carried by the bridge-rigidity invariant $C_\nu = G^2 Q S^{-1/2} = 0.9994295999075177$ with $G =$ sum_gap, $Q =$ prod_qbar, and $S =$ solar_response_over_mstar, above the emitted proxy $P_\nu = 6.699825740519345$; $B_\nu = P_\nu C_\nu = 6.696004159297337$ is retained only as the paper-facing amplitude parameterization, and absolute attachment fixes $\lambda_\nu = (m_{\star,\mathrm{eV}}/q^{p_\nu}_{\mathrm{mean}})P_\nu C_\nu = 1.7237014208357415$, which emits the absolute family and splittings on that branch. The exact positive-segment adapter, the bridge corridor, and the reduced-correction audit remain diagnostic-only beneath this bridge-rigidity surface. The $\varepsilon = 1/6$ texture language is the extra uniform-center-label collapse of this richer dictionary and is not a recovered-core theorem.

- **Hadron masses and resonances**: the particle branch supplies the seeded-family, correlator, and stable-channel readout bridge, but promotable hadron rows remain execution-contract-frozen. The declared particle roadmap does not include hadron-mass closure, because the remaining step is one production backend export bundle followed by real nonperturbative production computation and production systematics rather than an additional symbolic theorem alone, and that execution burden is not a realistic target within the current project compute scope.

- **Neutrino data**: beyond any legacy, input-dependent capacity-level side estimate tied to N_{scr}, the recovered core does not derive neutrino masses or mixings. A continuation branch emits intrinsic masses, splittings, and PMNS data; its weighted-cycle branch reaches the physical PMNS/hierarchy regime, the normalized same-label overlap-defect weight section closes beneath that branch, and the weighted-cycle bridge-rigidity theorem emits $C_\nu = G^2 Q S^{-1/2} = 0.9994295999075177$ above the emitted proxy $P_\nu = 6.699825740519345$. Absolute attachment then fixes the paper-facing amplitude parameterization $B_\nu = P_\nu C_\nu = 6.696004159297337$, $\lambda_\nu = (m_{\star,\mathrm{eV}}/q_{\mathrm{mean}}^{p_\nu}) P_\nu C_\nu = 1.7237014208357415$, and therefore the absolute family $m_i = \lambda_\nu \hat{m}_i$, $\Delta m_{ij}^2 = \lambda_\nu^2 \widehat{\Delta m_{ij}^2}$. Because the positive selector segment is explicit, a compare-only two-parameter continuation adapter still hits the representative central splittings exactly on the same family, but it is strictly diagnostic-only beneath the bridge-rigidity invariant C_ν and its induced paper-facing amplitude B_ν. Together with the exact W/Z sidecar pair, the exact Higgs/top sidecar, the exact charged same-family witness, and the selected-class quark theorem with its exact supporting surfaces, the exact non-hadron output lane is complete even though the theorem-grade charged lane remains open and the quark theorem is stated only on the selected public frame class f_P. The attached weighted-cycle stack factors through $q_e = q_{\mathrm{mean}} qbar_e$, and the best closed constructive local object beneath that bridge-rigidity surface is the defect-weighted same-label edge family $q_e = \sqrt{g_e d_e}$ together with its induced μ_e family; that family is constructive beneath the bridge-rigid theorem surface and does not replace it.

- **Dark-sector phenomenology**: the sign, closure, and galaxy-scale response law of any modular-anomaly stress tensor are not derived by the recovered gravity chain.

- **Baryogenesis**: suppression-counting estimates are not a substitute for a derived out-of-equilibrium mechanism.

- **Black-hole spectroscopy and evaporation closure**: beyond the structural statements retained in Section 5.11 (edge-center obstruction to naive factorization, recoverability-style interior encoding at small CMI, Hawking/KMS normalization, discrete area spectrum, and the Schwarzschild transition identity $\Delta E = k_B T_H \ln(d'/d)$ for actual sector changes), the manuscript does not derive integer-transition combs, $k = 3$/QNM selection, Page-type linewidth estimates, PBH burst templates, Kerr/LIGO horizon spectroscopy combs, or any Page-curve/island closure.

- **String/worldsheet reorganization**: the exact heat-kernel bridge fixes the edge partition function on the two-dimensional Yang–Mills surface, and the compact paper gives a theorem-level criterion for when the standard Gross–Taylor large-N rewriting upgrades the large-N_{edge} branch to a controlled effective description once one fixes a distinct large-N_{edge} sequence, a compact fixed-$\tau = tN_{\mathrm{edge}}$ window, and uniform genus-remainder control. Any further lift to critical superstring structure remains a conjectural extension requiring extra worldsheet ingredients.

- **Other downstream continuations**: strong-CP proposals, proton-spin fractions, and similar late-stage continuations appearing on other OPH surfaces are not derived by the recovered core.

Core/Implementation Boundary.

- **Scaling-limit control**: the Lorentz/null-modular/Einstein chain depends on the explicit scaling-limit premises declared earlier in the manuscript; proving or constructing that regime is an open task.

- **Fermionic gauge branch**: the manuscript proves compact gauge reconstruction only in the bosonic internal-gauge branch; a full fermionic/super-Tannakian extension remains open.

- **External inputs**: N_{scr} remains external, and the numerical value of Λ is not a local recovered prediction because local null data determine the Einstein equation only up to a metric term ϕg_{ab}; fixing Λ requires the global capacity input N_{scr}. The local pixel ratio P is fixed only on the Phase-II outer/inner closure program summarized in Section 5; it is not part of Phase I.

- **Quantitative-closure branch**: the D10 numerical implementation depends on printed running, matching, and threshold conventions together with the integrated heat-kernel quantitative appendices. Its internal transmutation data are fixed forward from the OPH pixel ratio P through the source-only pixel-closure solve, not read back from measured low-energy couplings. The displayed carrier closes its own exact W/Z chart, and the promoted target-free source-only repair theorem emits the public electroweak quintet on the Phase II quantitative-closure branch. The freeze-once coherent repair surface is retained only as compare-only validation; on that frozen authoritative surface the canonical W/Z pair is hit exactly, but it remains below the target-free theorem.

- **Hadron execution boundary**: the hadron lane is not blocked by one small symbolic gap. The next object is one production backend export bundle with publication-complete manifest provenance and real stable-channel correlator arrays; the normalized production dump and then production statistical/systematic control only follow after that bundle lands. Because that remaining nonperturbative execution burden is not a realistic target within the current project compute scope, hadron rows remain outside the declared particle roadmap and outside the derived output surface.

The compact Higgs/top forward seed downstream of the D10 gauge core is a declared theorem on the same D10/D11 quantitative surface used by the emitted electroweak carrier. Its one-scalar seed law is $\sigma_{D11,\mathrm{HT}} = \alpha_U \cos(2\theta_{W0})/\sqrt{\pi}$, and the live forward path satisfies the exact fixed-ray factorization $\delta y_t/y_t^{\mathrm{core}} = -(9/16)\,\delta\lambda/\lambda^{\mathrm{core}} = \sigma_{D11,\mathrm{HT}}$ on that declared surface. The D11 Jacobian then emits the public forward Higgs/top pair. A separate compare-only inverse slice on the same D11 Jacobian can be solved to hit the canonical Higgs/top reference pair exactly, but that sidecar is not the live forward branch and does not change the theorem status of the public D11 rows.

Failures localize by tier. A failure in a Phase-III flavor, dark-sector, baryogenesis, spectroscopy, string, strong-CP, proton-spin, or similar continuation retracts that continuation only. A failure in D10 challenges the corresponding quantitative-closure branch. Only a failure in Phase I would overturn the recovered-core claim set.

3.8.5 Comparison with other unification approaches

Unified models attempting to tie together QFT, gravity, and SM structure tend to encounter a repeatable set of conceptual difficulties. This subsection examines how the observer-patch holography framework addresses these common pitfalls.

1. Subsystem factorization in gauge theory and gravity.

In gauge theories and gravity, the Hilbert space does not cleanly split as "inside $\otimes$ outside" across a cut. This infects entanglement entropy definitions, area terms, edge modes, and observable identification. Many unification attempts handwave this or patch it with conventions.

Resolution: The framework builds from a net of von Neumann algebras on patches plus overlap consistency. It does not start from naïve tensor factorization. The gauge-as-gluing + regulator package yields edge-center completion: a canonical block decomposition on collars where the center captures superselection data at the cut, and the state becomes (exactly or approximately) Markov across the collar. The entropy split $S(\rho_C) = S_bulk + \langle L_C \rangle$ follows from having a center with sector labels. This replaces the ad hoc "add an area term" move.

2. Modular Hamiltonian nonlocality.

Many entanglement-based gravity derivations depend on modular Hamiltonians that look like local stress-tensor charges (true only in special states/regions). In generic QFT states, modular Hamiltonians are nonlocal, making "first law of entanglement $\Rightarrow$ Einstein equation" arguments fragile.

Resolution: The Markov collar condition does heavy lifting: approximate Markov implies approximate modular additivity, with the defect controlled by conditional mutual information. This makes "modular locality" a controlled approximation. It is not treated as an assumption. On the extracted prime geometric subnet of Theorem 4.2, the controlled tangent-half-space comparison then locks modular flow to geometric dilations with rigid 2π normalization.

3. Lorentz invariance as a derived output.

Discrete microscopic models generally break Lorentz symmetry, and many unified proposals simply postulate Lorentz invariance in the IR.

Resolution: Lorentz kinematics are tied to geometric modular flow on caps. On the extracted prime geometric subnet, once modular flow acts as conformal transformations on S^2, we get $\mathrm{Conf}^+(S^2) \cong \mathrm{PSL}(2,\mathbb{C}) \cong \mathrm{SO}^+(3,1)$, so the Lorentz group appears as a theorem-level output of modular structure on that branch. No external spacetime symmetry axiom is added.

4. Dynamics beyond kinematics.

Many approaches produce emergent geometry/kinematics but stall at dynamics: why Einstein's equations (with the right coefficient) rather than some other geometric PDE?

Resolution: The framework combines the MaxEnt-selected fixed-cap generalized-entropy stationarity theorem, the derived $K_C = 2\pi B_C$ structure with rigid normalization, the internal null modular bridge identifying the half-line generator with the local null-stress charge, and the internal small-ball bridge from the geometric cap generator. The remaining D5 hypotheses are the explicit scaling-limit regularity, bounded-interval transport, and tensor-upgrade premises. It does not rely only on "assume a UV CFT."

5. Gauge symmetry origin and compactness.

Most unification stories pick a gauge group and work out consequences. Emergent-gauge approaches sometimes produce noncompact groups or uncontrolled redundancies.

Resolution: Gauge symmetry is recast as redundancy in overlap identifications (gauge-as-gluing). From edge sectors and fusion, a tensor category is reconstructed; Tannaka-Krein / Doplicher-Roberts reconstruction then yields a compact group G given the categorical hypotheses. "Gauge symmetry" names the gluing redundancy at the conceptual level. "Compact group" is

the mathematical form compatible with finite-dimensional sector/fiber-functor structure.

6. Massless photon and graviton usually hand-imposed.

Getting massless gauge bosons is easy if exact gauge invariance is assumed, but that restates the problem. Massless graviton is more delicate (mass terms, vDVZ discontinuity, strong coupling scales).

Resolution: Once gauge and diffeomorphism invariance are emergent redundancies of description (from gluing consistency / emergent geometry), hard mass terms are forbidden: "a coordinate system's Jacobian can't show up as a physical mass." These symmetry-protected zeros emerge from the same consistency machinery that gives the symmetries.

7. Global consistency, anomalies, and loop patching.

Building physics from local patches hits loop/holonomy problems: consistent gluing on a tree but obstructions around loops. These obstructions are often anomalies or global topological constraints.

Resolution: This is elevated to a first-class organizing principle: gluing data on overlaps define cocycles; central defects define a Čech obstruction class [z] (and more generally a 2-group/crossed-module cocycle for noncentral defects). "Global consistency exists iff the obstruction class vanishes" becomes the universal statement. Anomalies become "failure to glue" rather than a mysterious quantum pathology.

8. Charge quantization without a GUT.

Without embedding into a simple GUT group, explaining charge quantization (why all isolated color singlets are integer charged) is awkward. Standard lore requires grand unification or monopoles.

Resolution: The framework leans on global group structure (the Z_6 quotient) and derives congruence/selection rules for allowed representations/hypercharges. This gives a structural explanation for integer-charged color singlets without introducing the proton-decay channel of simple-GUT models.

9. Coupling unification usually forces proton decay.

Traditional simple-group unification introduces leptoquark gauge bosons (X, Y) mediating proton decay. Experiment keeps pushing limits up, pressuring minimal GUTs.

Resolution: "Unification" here is geometric/entropic (shared edge diffusion parameter, heat-kernel weights). It is not an embedding into a simple Lie group. If the reconstructed gauge group genuinely factorizes as a product (sector factorization selector), there are no mixed generators playing the X/Y role. "Unify couplings" and "unify groups" therefore come apart.

10. Cosmological constant locality.

The cosmological constant problem is a graveyard of unified theories: local QFT estimates are enormous, and tiny observed Λ seems to demand absurd fine tuning.

Resolution: From null modular data, T_ab is reconstructed only up to ϕg_ab. Local consistency conditions and null focusing are blind to vacuum-energy shifts, so the Einstein equation is fixed only up to Λg_ab. Λ becomes a global "capacity" parameter of the static patch (tied to log dim H_tot). It is not a locally computable quantity. This resolves the conceptual tension: local microphysics *cannot* fix Λ by structural information-theoretic reasons.

11. UV infinities and nonrenormalizability.

Unified programs struggle to give sharp, finite microscopic definitions. Formal continuum structures, infinite entropies, and regularization dependence abound.

Resolution: The regulator premises explicitly require local patch algebras to be type-I and finite-dimensional, with a MaxEnt branch whose generator is quasi-local and obeys a Lieb-Robinson bound. So the fixed-cutoff UV branch is interacting in the ordinary finite-range sense, and the fundamental degrees of freedom are finite and live on the screen. The framework does *not* claim a unique microscopic UV completion: physical uniqueness is only modulo gauge or implementation

hiding together with inert ancillary stabilization. The genuinely noncentral topological branch is also closed at fixed cutoff by the higher-gauge crossed-module collar theorem; the remaining burdens are the continuum BW lift and the refinement-limit transportable gauge branch.

12. Predictivity vs. parameter explosion.

Unified models often explode in parameters, sectors, or vacua, becoming hard to test directly because everything depends on choices.

Resolution: The framework compresses freedom into a "pixel area" (resolution) parameter and a total Hilbert space capacity (size) parameter, then derives structure from consistency (Lorentz, Einstein form, compact gauge group reconstruction, exact zeros, quantization patterns). The explicit MAR selector then picks the SM factors and sector factorization on the admissible class discussed in Section 6.2.

Structural pattern. The framework treats locality, Lorentz invariance, gauge symmetry, and gravity as consequences of consistency conditions among overlapping descriptions together with information-theoretic properties of states. Modular rigidity then supplies the familiar symmetry and dynamical structures.

Engineering deliverables. Certain problems can be stated as explicit closure tasks:

- Λ is structurally explained as a global capacity parameter and is not numerically predicted
- A full microphysical derivation of geometric modular action is still required

These are shared challenges across unification approaches. The framework provides an explicit map of where they live and what would resolve them.

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4 Consensus, Defects, and Implementation Hiding

The fixed-cutoff consensus package admits an equivalent finite patch-net formulation. Local patch descriptions agree on overlaps, local recovery-derived repair rules remove mismatches, and the physical output is the schedule-independent quotient normal form together with the induced terminal expectation functionals on the declared physical observable algebras. This section records the fixed-point, defect, and gauge-quotient statements in that language.

The consensus lane also carries a simple finite-candidate law-selection model: once one equips candidate repair rules with a fitness functional, replicator dynamics gives a clean toy picture in which more successful reconciliation laws dominate a competing pool. That selection picture is useful for the overall synthesis picture, but in the consensus paper it remains a finite-candidate monotonicity result, not part of the recovered-core gravity/gauge theorem surface, not a universality claim, and not a literal cosmological dynamics claim.

On its declared fixed-cutoff branch, *Reality as a Consensus Protocol: The Fixed-Point Computation That Implements Physics* [2] imports four core D1 surfaces into this synthesis surface:

1. **Asynchronous confluence.** For the declared accepted repair law, the local-fit contract makes Φ a Lyapunov functional and, together with the fixed-cutoff union-collar gluing package, yields a unique schedule-independent normal form under repair completeness.

2. **Cycle obstruction and higher-gauge defects.** On the abelian branch, global consistency holds exactly when cycle holonomy vanishes; on the genuinely noncentral branch, the crossed-module defect class
$$q_\Sigma \in \check{H}^2(N_\Sigma, H_\Sigma \to G_\Sigma)$$

labels the fixed-cutoff higher-gauge obstruction.

3. **Gauge quotient and observable-level confluence.** The repair law descends to the overlap-invariant quotient, and the induced terminal state on the declared physical observable algebras is unique there even when microscopic representatives differ by gauge or sector relabelings inside one quotient-local glued state.

4. **Record algebra and stability.** On the fixed-cutoff observer-accessible surface, central record projectors carry the Born/Lüders package directly, while approximate record projectors inherit explicit $(\varepsilon, \delta_{\mathrm{rec}})$ stability bounds on that same event surface.

The consensus paper is equally explicit about the imported repair-law data and what remains outside the fixed-cutoff theorem stack. The declared repair step already includes the touched-overlap local-fit contract and the union-collar gluing package on the physical quotient; the remaining burdens above that step are repair completeness and, on the Petz branch, the stated support/CPTP clause. On that same fixed-cutoff surface, exact normal-form computation is finite-state and decidable, approximate stability is only collar-local through the splice and record estimates, no global approximate-consensus theorem for long asynchronous noisy repair is claimed, and no universality theorem for a uniform family of growing patch nets and repair laws is claimed.

5 Particle-Spectrum Branch

The particle branch follows the same logical order as *Deriving the Particle Zoo from Observer Consistency* [4]. One first fixes the compact-gauge basis and realized branch, then identifies the realized Standard Model quotient, and only then asks how transport-stable overlap data organize the observed particle families.

Structural particle theorem package. On the realized compact-gauge branch, the structural carrier skeleton is fixed by the compact gauge and MAR-admissibility chain:

$$\frac{\mathrm{SU}(3) \times \mathrm{SU}(2) \times \mathrm{U}(1)}{\mathbb{Z}_6}, \qquad N_g = 3, \qquad N_c = 3.$$

This determines the exact structural zero rows carried by the gauge and gravity sectors: the photon, gluons, and graviton are massless structural carriers on the realized branch.

Proof route. The proof route is the one developed in *Recovering Relativity and the Standard Model from the OPH Package Rooted in Observer Consistency* [1] and *Deriving the Particle Zoo from Observer Consistency* [4]. Under the explicit transportable-colimit, symmetry, and bosonic fiber-functor premises of the compact paper, the surviving transportable edge-sector category reconstructs a compact internal symmetry group. MAR then selects the realized admissible branch, anomaly freedom fixes the hypercharge lattice, and the realized one-generation/one-Higgs package forces $N_g = 3$ and then $N_c = 3$. The particle branch builds on that structural skeleton and does not bypass it.

Sector	Claim tier	What is fixed here	Boundary to stronger claim
Structural carriers	realized-branch structural theorem	exact $m_\gamma = m_g = m_{\mathrm{grav}} = 0$; realized quotient; $N_g = 3$; $N_c = 3$	theorem-grade structural exactness on the particle side
Electroweak bosons	quantitative-closure branch + exact sidecar	target-free public W/Z rows together with the exact frozen-repair pair $M_W = 80.377$ GeV, $M_Z = 91.18797809193725$ GeV	exact only on the frozen authoritative repair surface; the public theorem rows remain the target-free electroweak outputs
Higgs/top stage	exact source-only D11 split theorem on the declared D10/D11 running, matching, and threshold surface + compare-only exact sidecar	shared scalar ρ_{HT}, source-only split selectors, and the declared D11 Jacobian together with an exact Higgs/top inverse pair	the split theorem is closed on the declared quantitative surface; the inverse pair is compare-only and does not replace the theorem rows
Quark family	selected-class theorem + exact supporting surfaces	exact running quark sextet together with explicit exact forward Yukawas on the public physical quark frame class f_P	selected-class closure only; the sextet matches the official PDG 2025 API running-quark surface exactly, with the top coordinate taken from the PDG cross-section entry; supporting exact surfaces are the same-family witness on `current_family_only` and the restricted common-refinement transport-frame exact chain, which also emits explicit exact forward Yukawas Y_u and Y_d; the target-free mass bridge $\Theta_{ud}^{\mathrm{mass}} =$ `quark_d12_t1_value_law` closes separately on the emitted D12 mass ray; σ_{ref} is a lower negative sheet statement beneath the selected-class theorem; no global classification of all quark frame classes is claimed; a stronger class-uniform public descent theorem is separate and is not claimed
Charged leptons	continuation branch + exact witness	exact same-family charged triple plus exact same-carrier centered readback, with the ordered three-point quadratic readout closed on that scope	same-family-only on the exact witness; theorem lane carries the closed common-shift no-go, then waits on $\widehat{C}_e^{\mathrm{cand}}$ and the post-promotion lift whose descended scalar is $\mu_{\mathrm{phys}}(Y_e)$, with the physical identity-mode equalizer beneath that scalar
Neutrinos	weighted-cycle continuation + emitted bridge invariant + diagnostic sidecars	emitted absolute masses/splittings on the weighted-cycle branch through the bridge-rigidity invariant $C_\nu = G^2 Q S^{-1/2}$ above the emitted proxy P_ν, with induced paper-facing amplitude B_ν	theorem-grade on the declared weighted-cycle branch; the exact positive-segment adapter, bridge corridor, and correction audit remain diagnostic-only and do not feed back into theorem state
Hadrons	nonperturbative continuation	stable-channel and readout architecture are defined	one production backend export bundle, then executed unquenching, runtime receipt, and production systematics

Claim-tier boundary. Structural carrier content and the realized Standard Model branch are theorem-bearing outputs. The electroweak D10 branch, the Higgs/top stage, quark rows, charged-lepton rows, neutrino rows, and hadron rows sit on their stated quantitative-closure, continuation, or nonperturbative claim tiers. On the target-free electroweak D10 lane, the logical order is forward: the outer/inner closure relation fixes P, and then P fixes $\alpha_U(P)$, equivalently $t_U(P)$ and $t_{\mathrm{tr}}(P)$, then the downstream scale data and public electroweak rows. Measured electroweak quantities

remain compare-only validation targets and do not feed back into the transmutation solve.

Outer/inner self-consistency for the pixel ratio. This synthesis paper is the canonical place where the local pixel ratio P is defined, so the compact and particle papers can cite one derivation rather than duplicate it. The construction has six steps.

1. **Outer pixel ratio.** The externally fixed local screen datum is the dimensionless area ratio
$$P := \frac{a_{\text{cell}}}{\ell_P^2}.$$

2. **Exact equilibrium benchmark.** The total/bulk/edge hierarchy has one exact self-similar balance point. Writing
$$x(C) := \frac{S_{\text{gen}}(C)}{S_{\text{bulk}}(C)} = 1 + \frac{\langle L_C \rangle}{S_{\text{bulk}}(C)},$$
the self-similar balance condition
$$\frac{S_{\text{gen}}(C)}{S_{\text{bulk}}(C)} = \frac{S_{\text{bulk}}(C)}{\langle L_C \rangle}$$
gives $x^2 - x - 1 = 0$ and hence $x = \varphi = (1+\sqrt{5})/2$. The realized branch must sit near, but not exactly at, this point because exact equilibrium is too symmetric to support durable records, structure, and dynamics.

3. **Outer detuning variable.** On the declared Phase-II closure surface we parametrize the externally fixed detuning by
$$\alpha_{\text{ext}}(P) := \frac{P - \varphi}{\sqrt{\pi}}, \qquad \text{equivalently} \qquad P = \varphi + \alpha_{\text{ext}}(P)\sqrt{\pi}.$$

4. **Source-only D10 anchor from the same pixel.** Let $I \subset \mathbb{R}_+$ be the physical pixel interval and let
$$F_{D10} : I \longrightarrow \mathcal{D}_{D10}$$
denote the declared D10 source map. For any trial $P \in I$, the forward D10 lane emits
$$P \longmapsto \alpha_U(P) \longmapsto (t_U(P), t_{\text{tr}}(P)) \longmapsto (t_2(P), t_3(P), v(P)) \longmapsto A_Z(P) = \alpha_{\text{em}}^{-1}(m_Z^2; P).$$

5. **Ward-projected Thomson transport.** Let T_Q be the declared Ward-projected $\mathrm{U}(1)_Q$ transport operator from the m_Z-anchor to the Thomson limit, using only emitted D10 data on that same branch. Define
$$A_T(P) := T_Q\big(A_Z(P), F_{D10}(P)\big) = \alpha_{\text{Th}}^{-1}(P).$$
The inner coupling used in the outer/inner closure is the coupling, not the inverse coupling:
$$\alpha_{\text{in}}(P) := \frac{1}{A_T(P)}.$$

6. **Closure.** The realized pixel ratio is the fixed point for which the outer detuning equals the inner observation coupling:
$$\alpha_{\text{ext}}(P) = \alpha_{\text{in}}(P).$$
Equivalently,
$$H(P) := P - \varphi - \frac{\sqrt{\pi}}{A_T(P)} = 0, \qquad \text{or} \qquad P = \varphi + \frac{\sqrt{\pi}}{A_T(P)} = \varphi + \alpha_{\text{in}}(P)\sqrt{\pi}.$$

Writing

$$G(P) := \varphi + \frac{\sqrt{\pi}}{A_T(P)},$$

the closure problem is the fixed-point condition $G(P) = P$. On any physical interval where this induced map is a self-map and a contraction, the closure root is locally unique.

The closure interpretation assigns one screen cell two coupled roles. On the outer side it is one cell of the screen whose displacement above the exact self-similar equilibrium φ sets the geometric detuning. On the inner side the same cell emits the smallest electromagnetic observation step available on the particle branch. The pixel ratio P is the size of that cell in Planck areas, and the realized P is the value at which those outer and inner readings agree. The fixed-point equation is the mathematical form of that consistency condition. The public code records a numerical closure witness; the interval-wide uniqueness certificate remains a separate mathematical burden, with a local numerical contraction certificate recorded as the machine-checkable hardening step. Once the fixed point $P_\star$ is selected, every downstream quantity is propagated forward from that same $P_\star$. A separate pending hardware note reports an optical-cavity check of the same fixed-point geometry; this is treated as corroborating engineering evidence. This is a consistency certificate on the declared closure surface. It is not a theorem that earlier local physics must uniquely generate the outer screen settings. The same closure surface also admits an inverse observational use case in which observers infer P and total screen capacity from inside the world even if those values are fixed externally at a later stage.

One clean refinement studies the same closure directly at zero momentum. That route would close the electromagnetic row without transporting from the m_Z anchor. The present paper keeps the fixed-point idea itself simple and treats that zero-momentum promotion as a separate quantitative task.

Exact non-hadron hit surface. For quick orientation, the exact non-hadron lanes are:
The explicit values behind these exact-hit lanes are the exact frozen W/Z pair, the exact Higgs/top inverse pair, the exact same-family charged triple, the exact same-family quark sextet, and the emitted weighted-cycle neutrino mass triple with emitted weighted-cycle absolute splittings recorded in Ref. [4]'s exact-output surface. The public Higgs/top pair comes from the source-only split theorem on the declared D10/D11 surface rather than from the compare-only inverse slice, while the repo-wide exact top row is also carried by the selected-class quark theorem.

Extended derivation. The structural-to-family route and the full per-sector claim-tier audit are developed in *Deriving the Particle Zoo from Observer Consistency* [4].

6 Screen Microphysics, Records, and Observer Continuation

The screen-microphysics lane is the engineering side of the paper corpus. It asks whether some microscopic model might exist in principle and gives one explicit regulated reference architecture: a finite gauge-register or quantum-link style screen model in which local registers, patches, overlaps, edge observables, records, repair instruments, and observer-facing interfaces are all made concrete at fixed cutoff.

This lane has five theorem-bearing components. First, the reference architecture itself is explicit. Second, the regulated patch-net embedding theorem is closed on its fixed-cutoff object/local-interface surface, and the edge-sector thermal/Casimir branch carries an explicit fixed-cutoff law with one remaining compact lift above that theorem surface. Third, the fixed-cutoff

measurement/Born-rule package is closed. Fourth, the fixed-cutoff Bell/CHSH package is closed on the declared two-wing surface. Fifth, the checkpoint/restoration observer package is closed. The external burdens on this lane are the final packet-level quotient closure where one wants an autonomous exported dynamics, the compact heat-kernel lift, global consensus analysis, refinement-stable universality, and the continuum/gravity lift.

Reference-architecture takeaway. The synthesis-level point is that the regulated screen-side architecture supplies a concrete habitat in which local observables, overlap observables, record registers, and synchronization maps can all be written without pretending to identify a unique final UV completion.

Imported theorem from Ref. [3] (Fixed-cutoff record algebra and Born-Lüders package). For each completed compare/write/verify slice of the regulated microphysics, the declared pointer and overlap-sector projectors generate a finite commutative central record algebra

$$\mathcal{Z}_{\mathrm{rec}}.$$

Every observer-accessible event E in that algebra has probability

$$\mathbb{P}(E) = \mathrm{Tr}(\rho P_E),$$

and conditioning on E gives the operational post-measurement state

$$\rho|_E = \frac{P_E \rho P_E}{\mathrm{Tr}(\rho P_E)}.$$

If a practical readout instead uses projectors Q_a with commuting central reference projectors $\widehat{Q}_a$ on the same declared slots and

$$\delta_{\mathrm{rec}} = \max_a \|Q_a - \widehat{Q}_a\|,$$

then

$$\|[Q_a, Q_b]\| \le 4\,\delta_{\mathrm{rec}},$$

and any accessible-state perturbation $\|\widetilde{\rho} - \rho\|_1 \le \varepsilon$ changes each declared elementary record-event probability by at most $\varepsilon + \delta_{\mathrm{rec}}$.

Imported theorem from Ref. [3] (Fixed-cutoff Bell / CHSH package). On the declared fixed-cutoff physical observable algebra, fix commuting left/right wing subalgebras, central binary setting registers, binary projective readouts on each wing, and a physical source state on the joint wing algebra. Then the compare slice carries the joint law

$$p(a, b \mid x, y) = \mathrm{Tr}\big(\rho_{LR}\, P^{(L)}_{a|x} P^{(R)}_{b|y}\big),$$

the local marginals are independent of the remote setting, the Bell correlators satisfy

$$E(x, y) = \mathrm{Tr}\big(\rho_{LR}\, A_x B_y\big),$$

and the CHSH combination obeys

$$\big|E(0,0) + E(0,1) + E(1,0) - E(1,1)\big| \le 2\sqrt{2}.$$

If the source family contains an explicit two-qubit branch with the stated Pauli readouts, the same fixed-cutoff surface saturates $2\sqrt{2}$ exactly on that branch. The source-state premise is explicit: the theorem derives the Bell law and the Tsirelson bound from the stated fixed-cutoff operator surface, while Bell-pair preparation and two-qubit factor structure remain stated branch premises.

Imported theorem from Ref. [3] (Checkpoint/restoration and observer backup). For an observer patch P_O, let the checkpoint data consist of the observer-facing record algebra, the accessible local state, the future update schedule, and the externally visible overlap interface data. Exact restoration reproduces the full future law of observer-accessible events, while an ε-accurate restoration changes that future law by at most ε in total variation.

Observer-facing conclusion. The theorem-level content is modest but sharp: the paper corpus supplies a fixed-cutoff measurement interface, a fixed-cutoff checkpoint/restoration package, and an operational observer-identity criterion on observer-accessible event algebras. Stronger substrate-selection or strange-loop closure claims remain outside the recovered theorem package.

Extended derivation. The fixed-cutoff edge-law, measurement, Born-rule, Bell/CHSH, and checkpoint/restoration packages are developed in *Screen Microphysics, Patches, and Observer Synchronization in OPH* [3].

7 Conditional Worldsheet/String Branch

The synthesis paper carries the string/worldsheet branch on the same theorem boundary used across the current paper corpus: a genuine theorem-level bridge from the edge-sector theorem surface to the two-dimensional Yang–Mills partition function, followed by a controlled large-N_{edge} effective-description theorem on the stated branch.

Imported theorem (Heat-kernel edge-sector bridge). Import the fixed-cutoff same-overlap edge-law package from *Screen Microphysics, Patches, and Observer Synchronization in OPH* [3]. Under the same overlap-gauge realization and local-Gibbs/MaxEnt edge branch, the edge-sector weights satisfy

$$p_R(t) \propto d_R e^{-tC_2(R)}.$$

Consequently the edge partition function matches the standard two-dimensional Yang-Mills heat-kernel form.

Imported proof route. The microphysics source surface carries the fixed-cutoff thermal/Casimir law on the declared overlap interface. Peter–Weyl decomposition [30] then identifies the resulting sum with the standard heat-kernel expression for two-dimensional Yang-Mills on the compact-group refinement lift.

Exact bridge identity. Writing the closed edge partition function as

$$Z_{\text{edge}}(t) = \sum_R d_R^2 e^{-tC_2(R)},$$

Peter–Weyl identifies it with the compact-group heat kernel at the identity,

$$Z_{\text{edge}}(t) = K_t(1).$$

The Chapman–Kolmogorov gluing law for K_t is the precise sense in which collar sewing matches the two-dimensional Yang–Mills heat-kernel surface before any large-N_{edge} continuation is invoked.

Controlled large-N_{edge} branch. Fix a distinct large-N_{edge} realization, with $N_{\text{edge}} \neq N_c = 3$, and the fixed-$\tau$ variable

$$\tau = tN_{\text{edge}}$$

on a compact window I. If the resulting edge free energy satisfies the compact paper's large-N_{edge} criterion, with remainder control on I, then the Gross-Taylor rewriting [29] is a controlled theorem-level worldsheet effective description of the edge dynamics on that branch. Critical-superstring claims, worldsheet CFT closure, anomaly cancellation, and full massless-spectrum matching remain outside the recovered core theorem package.

Higher-gauge continuation slot. The genuinely noncentral crossed-module gluing data on cuts provide the natural continuation-level slot for B-field / gerbe data in any string-style reorganization of the OPH edge sector. This is a structural analogy only; it is not yet an open/closed-string boundary formalism or a critical-string theorem.

8 Cross-Lane Theorem Boundary and Open Directions

The state of the current paper corpus is:

1. The fixed-cutoff collar, higher-gauge, and consensus theorem packages are part of the closed finite-regulator surface. The genuinely noncentral topological branch is not fixed-cutoff gap, and the quotient-level repair normal form is explicit on the finite patch-net surface. The consensus lane is classified there as a finite-state exact theorem package rather than as a universality claim.

2. The screen-microphysics lane has an explicit regulated reference architecture together with closed measurement/Born-rule, Bell/CHSH, and checkpoint/restoration packages. Its open fronts are packet-level quotient closure where a declared observable family is meant to carry an autonomous exported dynamics, the compact heat-kernel lift, refinement-stable universality, global consensus closure, and the continuum/gravity lift.

3. Physical UV uniqueness is quotient-level and ancilla-stable: the theorem-grade object is the family of terminal expectation functionals on the declared physical observable algebras, fixed modulo implementation hiding and inert ancillary refinement, not a unique microscopic representative.

4. The Lorentz/null-modular/Einstein chain remains a controlled refinement/scaling-limit branch. In the canonical premise ledger, the D3 side carries **T2** together with the theorem-local realized transported geometric cap-local system whose carried-collar schedule sits above the transported fixed-local-collar Markov/faithfulness datum, and then ordered cut-pair rigidity on the extracted prime geometric subnet; on that branch the finite cap regulators are type-I, but the scaling-limit observer algebra may be non-type-I, so the theorem is automorphism-level there and $K_C = 2\pi B_C$ is only the special type-I limit form; stronger BW / canonical-cap phrasing remains theorem-external unless the extraction-and-rigidity package is supplied. The D4 side still uses theorem-local null-cut center transfer, the extra inherited left/right strip-split condition needed for the spatial-collar-type tensor decomposition, the exact-or-controlled Markov hypotheses on one fixed inherited strip model, and the geometric scaling action on the null half-line blow-up net that derives the half-sided modular pair, while the same-half-line local modular-Hamiltonian input and the separate bounded-interval

projective branch enter only later; on that D4 surface the exact-or-controlled strip additivity, the endpoint-Lipschitz renormalized half-line family, the weak tail generator, the derived half-sided modular inclusion, and the positive null-translation generator on its Stone domain with the affine half-line modular relation are internal to the bridge itself; and the D5 side adds the derived fixed-cap generalized-entropy stationarity theorem for admissible fixed-cap MaxEnt variations on the realized cap-label-preserving MaxEnt family, the internal small-ball bridge from the geometric cap generator together with the separate bounded-interval projective branch, the locally Lorentzian $d = 4$ scaling regime, the small-ball constancy and D4 carried-remainder hypotheses, and the all-directions/all-reference-states tensor-upgrade premise; on that same gravity lane the D5→D6 cosmological-capacity theorem stack is explicit: the local Einstein equation is fixed only modulo Λg_{ab}, the external screen-capacity input N_{scr} together with the screen-capacity identification $N_{\mathrm{scr}} = S_{\mathrm{dS}}$ and the standard de Sitter entropy relation

$$S_{\mathrm{dS}} = \frac{A_{\mathrm{dS}}}{4G} = \frac{3\pi}{G\Lambda}$$

closes the same branch globally by $\Lambda = 3\pi/(GN_{\mathrm{scr}})$, the same D6 closure fixes the de Sitter static-patch package $(S_{\mathrm{dS}}, r_{\mathrm{dS}}, t_{\Lambda})$, and the observed cosmic age remains a downstream FLRW benchmark rather than an additional theorem output.

5. Compact gauge reconstruction and the realized Standard Model quotient remain tied to the transportability, symmetric-braiding, and fiber-functor premises stated in the canonical technical-premise list.

6. The particle branch is structurally rich and sector-split: structural carriers and the realized Standard Model skeleton are fixed, electroweak closure is in hand, the Higgs/top stage closes on its source-only split theorem surface, the charged-lepton lane carries exact centered read-back plus a closed common-shift no-go, the neutrino lane closes on its weighted-cycle theorem branch, and the quark and hadron lanes sit on explicit continuation fronts.

7. The string branch is a continuation of the heat-kernel edge-sector theorem, not a promoted recovered-core result.

Local unification status. At the synthesis level, one local unification surface is explicit and one uniform theorem tier for c, G, W, Z, and H is absent. The same declared local input P governs the electroweak/Higgs trunk and the gravity-side edge-entropy bridge, while the invariant causal speed c is structural from the Lorentz branch and gains its SI label only on the local readout package.

Quantity	Best value on the paper surface	Common paper-surface chain	Caveat
c	299792458 m/s	structural Lorentz/BW branch $\to$ invariant causal speed $\to$ local SI readout	structural rather than P-fitted; the SI presentation sits on the stated BW/Lorentz support package
G	$6.674299995910528 \times 10^{-11}\ \mathrm{m^3\,kg^{-1}\,s^{-2}}$ on the stated local extension surface	$P \to \bar{\ell}_{\mathrm{SU}(2)} + \bar{\ell}_{\mathrm{SU}(3)} = \bar{\ell}_{\mathrm{shared}} \to G_{\mathrm{SI}} = c^3 a_{\mathrm{cell}}/(\hbar P)$	stated relative to the declared a_{cell} datum; on that surface the D10 pixel law fixes $\bar{\ell}_{\mathrm{shared}} = P/4$, and the strict classical-regime clause remains explicit
M_W	public target-free row 80.37700001539531 GeV; exact sidecar 80.377 GeV	$P \to \alpha_U(P) \to (t_U(P), t_{\mathrm{tr}}(P)) \to v(P) \to$ target-free electroweak repair chart $\to M_W$	exact promotion requires one target-free electroweak chart-identity theorem
M_Z	public target-free row 91.18797807794321 GeV; exact sidecar 91.18797809193725 GeV	same target-free electroweak trunk	same electroweak chart-identity caveat
M_H	exact D11 split-theorem pair $M_H = 125.1995304097179$ GeV, $m_t = 172.3523553288312$ GeV	same P-driven electroweak core $\to$ D10 repair chart $\to$ ρ_{HT} and source-only split selectors $\to$ D11 Jacobian	Higgs row is exact on the declared running, matching, and threshold surface; the inverse sidecar remains compare-only; the bridge to the auxiliary direct-top PDG row is open and is tracked in issue #207

The exact-release package on that surface is local and finite: the strict classical-regime clause and the target-free electroweak chart identity for W/Z. The familiar-unit display split on that same surface is

$$L_{\mathrm{loc}} = \sqrt{a_{\mathrm{cell}}}\,\widehat{L}(P), \qquad t_{\mathrm{loc}} = \frac{\sqrt{a_{\mathrm{cell}}}}{c}\,\widehat{T}(P),$$

$$E_{\mathrm{loc}} = \frac{\hbar c}{\sqrt{a_{\mathrm{cell}}}}\,\widehat{E}(P), \qquad \Theta_{\mathrm{loc}} = \frac{\hbar c}{k_B\sqrt{a_{\mathrm{cell}}}}\,\widehat{\Theta}(P),$$

with dimensionless $\widehat{L}, \widehat{T}, \widehat{E}, \widehat{\Theta}$. The gravity-side shared-edge identity and the local SI readout are therefore fixed relative to the declared microscopic datum a_{cell}, while meters, seconds, GeV, and Kelvin remain downstream display conventions built from the structural c row and the familiar constants $\hbar$, k_B.

Companion papers. Detailed derivations appear in Refs. [1, 2, 3, 4].

9 Conclusion

This paper presents the full framework on one synthesis surface. The five-axiom basis, the conditional gravity and gauge branches, the consensus formulation, the particle-spectrum branch, the regulated screen-microphysics lane, the fixed-cutoff measurement and observer package, and the controlled large-N_{edge} worldsheet effective-description branch appear on one common synthesis surface.

The corpus is uneven but concrete. It has a sharp fixed-cutoff local picture, a conditional recovered core for relativity and Standard-Model structure, real particle-side quantitative outputs, and a usable microscopic engineering lane. The main external burdens are equally explicit: the D3 geometric-cap lift under **T2**, the D4 null-bridge inputs together with the bounded-interval projective branch, the D5 fixed-cap-stationarity / local-Lorentz tensor-upgrade step together with the D6 global screen-capacity closure of the remaining branch constant and its de Sitter static-patch parameter package, the transportable-sector compact-gauge lift, the microphysics closure passes, and the matter-family and hadron closure lanes. Refs. [1, 2, 3, 4] are the depth surfaces for sector-by-sector proofs, quantitative details, and continuation-level material. The local cross-lane numerical surface is explicit enough to summarize concisely: one declared input P organizes the

bosonic mass trunk and the gravity-side shared edge-entropy identity beneath the emitted local G row, while the same familiar-unit package reads meters and seconds from the single ruler $\sqrt{a_{\mathrm{cell}}}$ together with the structural invariant speed c, and reads GeV and Kelvin from the inverse local ruler through $\hbar$ and k_B.

A Candidate Microphysics Reference Architecture

The synthesis paper keeps the observer-facing theorem package in the main text, but the concrete finite screen architecture belongs in an appendix. The current theorem-bearing fixed-cutoff source surface for the edge heat-kernel / Casimir leaf is the microphysics companion paper; the appendix records only the synthesis-level restatement imported from that surface. The reference architecture used here is a finite cellulation of the horizon screen with boundary-fixed patch algebras, boundary-fixed overlap algebras on collars, declared shared readout packets, local repair instruments, and record registers exposed on the observer-facing interface.

Theorem (Regulated patch-net embedding). On a fixed finite cellulation and finite patch cover, the regulated microphysics construction realizes an explicit finite patch-net layer: patch vertices are genuine screen patches, overlap edges are genuine collars, packet fields are actual readouts of declared overlap observables, and allowed repair branches are concrete local channels on the same finite regulator.

Proof sketch. All patch and overlap algebras are finite-dimensional at fixed cutoff, and the declared packet fields are readouts of explicit overlap observables in those algebras. The candidate repair menu is implemented by local finite-support channels on the same neighborhoods. Therefore the abstract finite patch-net syntax is embedded into an explicit regulator object rather than introduced as a separate formal layer.

Theorem (Fixed-cutoff edge heat-kernel branch). On the thermalized overlap branch of the same regulated architecture, the stationary sector law is

$$\pi_\beta(\alpha) \propto d_\alpha e^{-\beta C_2(\alpha)},$$

and along refinement ladders converging to a compact-group Peter–Weyl decomposition [30] the same weights converge cylinderwise to the heat-kernel/Casimir law used in the OPH edge-sector branch.

Proof sketch. The regulated proposal kernel is chosen so that the weighted proposal symmetry $d_\alpha q_{\alpha\beta} = d_\beta q_{\beta\alpha}$ holds. Detailed balance with the Casimir acceptance rule then fixes the stationary distribution in finite dimension, and the compact-group heat-kernel law is the corresponding refinement lift.

Extended derivation. The full fixed-cutoff architecture, validation package, and theorem stacks are developed in *Screen Microphysics, Patches, and Observer Synchronization in OPH* [3].

B Interpretive Epilogue: State-and-Law Habitat

This appendix records the OPH state-and-law habitat available inside the framework and leaves any strange-loop closure theorem explicitly open.

The OPH inputs used here are:

1. the patch net $P \mapsto A(P)$ of von Neumann algebras together with isotony and overlap restriction maps;
2. the global state on the inductive-limit algebra, whose restrictions witness a nonempty overlap-consistent local sector;
3. the finite Axiom-3 MaxEnt/refinement branch, which supplies a common finite family of gauge-invariant local constraint observables across cutoffs;
4. on the fixed-cutoff branch, the derived finite type-I presentations and compact boundary gluing groups used elsewhere in the OPH package.

The law slot is encoded by the finite expectation-value coordinates of the retained Axiom-3 constraint family. That choice furnishes a compact convex state-and-law habitat for the framework.

Theorem (Internal state-and-law habitat theorem). Fix finitely many screen patches $P_1, \dots, P_N$ in the standard OPH setup. For each i, let

$$M_i := A(P_i), \qquad M_{ij} := A(P_i \cap P_j),$$

where M_i is the patch von Neumann algebra and $M_{ij} \subset M_i, M_j$ is the overlap algebra. Let

$$S_i := S(M_i) \subset M_i^*$$

be the full state space of M_i, endowed with the weak* topology $\sigma(M_i^*, M_i)$.

Choose a finite family of self-adjoint gauge-invariant local constraint observables

$$\mathcal{O} = \{O_1, \dots, O_m\}$$

from the Axiom-3 MaxEnt branch, with each O_a supported in one of the chosen patches; write $i(a)$ for an index such that $O_a \in M_{i(a)}$.

Define the overlap-consistent state sector

$$X_{\mathrm{ov}} := \left\{ (\omega_1, \dots, \omega_N) \in \prod_{i=1}^{N} S_i : \omega_i|_{M_{ij}} = \omega_j|_{M_{ij}} \text{ for all } i, j \right\}.$$

Define the OPH law-coordinate map

$$c : X_{\mathrm{ov}} \to \mathbb{R}^m, \qquad c(\omega_1, \dots, \omega_N) := \big(\omega_{i(1)}(O_1), \dots, \omega_{i(m)}(O_m)\big).$$

Let

$$X_{\mathcal{O}} := \left\{ (\omega_1, \dots, \omega_N, \ell) \in \left(\prod_{i=1}^{N} S_i \right) \times \mathbb{R}^m : (\omega_1, \dots, \omega_N) \in X_{\mathrm{ov}},\ \ell = c(\omega_1, \dots, \omega_N) \right\}.$$

Then:

1. the ambient product
$$E := \left(\prod_{i=1}^{N} M_i^*\right) \times \mathbb{R}^m$$
is a Banach space for any product norm;

2. $X_{\mathcal{O}}$ is nonempty and convex;

3. $X_{\mathcal{O}}$ is norm-closed in E;

4. $X_{\mathcal{O}}$ is compact in the product topology
$$\tau := \left(\prod_{i=1}^{N} \sigma(M_i^*, M_i)\right) \times \text{Euclidean topology on } \mathbb{R}^m;$$

Proof. Each M_i is a von Neumann algebra in the OPH patch net, hence a Banach space, so each dual M_i^* is Banach. A finite product of Banach spaces is Banach, hence so is E.

For each i, the state space
$$S_i = \{\omega \in M_i^* : \omega \geq 0,\ \omega(1) = 1\}$$
is convex and weak* compact by Banach–Alaoglu, because it is a weak* closed subset of the dual unit ball. For every pair (i, j), restriction along $M_{ij} \subset M_i$ and $M_{ij} \subset M_j$ defines affine weak* continuous maps
$$r_{ij} : S_i \to S(M_{ij}), \qquad r_{ji} : S_j \to S(M_{ij}).$$
Hence X_{ov} is an intersection of affine equalizer sets, so it is convex and τ-closed inside $\prod_i S_i$. Since $\prod_i S_i$ is compact and convex in the product weak* topology, X_{ov} is compact and convex as well.

Nonemptiness comes from the OPH global state on the inductive-limit algebra: its restrictions
$$(\omega|_{M_1}, \ldots, \omega|_{M_N})$$
belong to X_{ov} because the restrictions agree on every overlap by construction.

Each coordinate $\omega_{i(a)} \mapsto \omega_{i(a)}(O_a)$ is affine and weak* continuous, since evaluation at a fixed algebra element is weak* continuous. Thus c is affine and τ-continuous. The graph map
$$G : X_{\text{ov}} \to E, \qquad G(x) := (x, c(x)),$$
is affine and continuous, so its image $X_{\mathcal{O}} = G(X_{\text{ov}})$ is convex and τ-compact.

To see norm-closedness in E, let $(x_n, c(x_n)) \to (x, \ell)$ in norm. Norm convergence implies weak* convergence on each coordinate, so $x \in X_{\text{ov}}$ because X_{ov} is weak* closed. Since c is weak* continuous, $c(x_n) \to c(x)$, while norm convergence in $\mathbb{R}^m$ also gives $c(x_n) \to \ell$. Hence $\ell = c(x)$, so $(x, \ell) \in X_{\mathcal{O}}$.

Finally, $X_{\mathcal{O}}$ is a compact convex subset of the locally convex topological vector space E equipped with τ. □

The additional inputs needed for a strange-loop closure theorem are:

1. a closure map T built from internal operations such as overlap repair, Axiom-3 MaxEnt reprojection, collar recoverability, and record-sector coarse-graining;

2. a proof that the stronger "observer-supporting" or "selected-world" criteria carve out a nonempty T-invariant observer-supporting subset of $X_{\mathcal{O}}$;

3. any uniqueness, contraction, or Lyapunov-type stability estimate for that closure map.

The habitat theorem supplies only the ambient Banach/compact-convex setting. It does not define a canonical internal self-map, it does not identify any nonempty invariant observer-supporting subset, and it does not prove uniqueness or stability. In particular, nonemptiness and compact-convexity of the ambient habitat do not by themselves imply the existence of a nonempty invariant observer-supporting sector. Any further strange-loop reading of this habitat as a closure story for existence is interpretive only and is not part of the recovered theorem package.

B.1 Additional Problem Closures

Using the documentary node labels defined in Section 3.2.6, Phase I is D1–D5 together with D7–D9; D6 is the input-dependent global closure of the same Einstein branch; and D12 collects phenomenological continuations. Ref. [3] proves fixed-cutoff theorem packages for the measurement/Born-rule interface and for checkpoint/restoration/backup. Table 14 records the status of the broader problem-closure claims discussed in the OPH synthesis surfaces.

Table 14: Problem-closure statements sorted by claim tier.

Problem domain	Status	OPH mechanism / scope note
Quantum gravity consistency	Phase I recovered core (conditional scaling-limit)	Conditional Lorentz branch on the BW scaling surface of the extracted prime geometric subnet, with the realized transported geometric cap-local system, its derived carried-collar schedule above the transported fixed-local-collar Markov/faithfulness datum, and the later ordered cut-pair rigidity step still explicit; the fixed-cutoff cap algebras are type-I regulators, but the scaling-limit observer algebra may be non-type-I/type III, so the theorem is automorphism-level and outer in the generic continuum case, with $K_C = 2\pi B_C$ only on the special type-I limit branch; the D4 null bridge proves exact-or-controlled strip additivity, endpoint-Lipschitz renormalized half-line control, the weak tail generator, the derived half-sided modular inclusion, and the positive null-translation generator on its Stone domain with the affine half-line modular relation before the separate bounded-interval projective branch; the conditional Jacobson-type Einstein branch then adds the derived fixed-cap generalized-entropy stationarity theorem for admissible fixed-cap MaxEnt variations on the realized cap-label-preserving MaxEnt family, the internal small-ball bridge from the geometric cap generator together with the separate bounded-interval projective branch, the locally Lorentzian $d = 4$ scaling regime, the small-ball constancy and D4 carried-remainder hypotheses, and the all-directions/all-reference-states tensor-upgrade premise.

Problem domain	Status	OPH mechanism / scope note
UV completion / microscopic uniqueness	Mixed: fixed-cutoff theorem package / conditional scaling-limit branch	The recovered package gives a finite-range interacting fixed-cutoff branch, a unique physical UV branch only modulo OPH-stable equivalence, and a fixed-cutoff higher-gauge closure of the genuinely noncentral topological branch. Open items are a unique microscopic representative, identification of which scaling-limit observer algebra/state pair the realized refinement branch emits on the geometric subnet, the continuum BW lift under T2 plus the realized transported geometric cap-local system with its derived carried-collar schedule above the transported fixed-local-collar Markov/faithfulness datum and the later ordered cut-pair rigidity step, and the refinement-limit transportable compact-gauge lift behind T1/T4–T6.
Measurement problem	Mixed: fixed-cutoff theorem package / interpretive synthesis	Ref. [3] proves a fixed-cutoff central-record / Born-Lüders package for observer-accessible events. Broader interpretive closure claims remain outside the theorem package.
Cosmological principle and horizon homogeneity	Conditional theorem package	SO(3)-invariant constraint data plus MaxEnt uniqueness force an isotropic reference state and perfect-fluid stress tensor; if the same isotropy holds for all observers, the Schur/Bianchi argument gives constant-curvature spatial slices; with the conditional Einstein branch and positive Λ from finite screen capacity, the metric is FLRW; Markov control supplies the explicit horizon-homogeneity error bound and collar benchmark. No theorem states that the bare axioms select the symmetric constraint branch in every cosmological sector.
Problem of time	Interpretive or organizational synthesis	Modular flow from the BW_{S^2} geometric branch provides a physical time parameter for each observer patch. It is an organizational principle rather than a separate recovered-core theorem.

Problem domain	Status	OPH mechanism / scope note
Black-hole information / evaporation package	Mixed: structural theorem package / phenomenological continuation	The retained structural package is narrower than a full evaporation closure: on the fixed-cutoff edge-center collar carrier in the exact-Markov or idealized exact-recoverability limit, the edge-center decomposition blocks naive inside/outside factorization and small-CMI recoverability gives the interior-encoding reading; on the geometric modular branch, Hawking/KMS normalization is fixed; and the integer edge-sector area operator yields the discrete area spectrum together with the Schwarzschild transition identity $\Delta E = k_B T_H \ln(d'/d)$ for actual sector changes. Integer-transition combs, QNM selectors, Page-type linewidth estimates, PBH/LIGO templates, and any Page-curve or island closure remain continuation-level.
Cosmological constant and de Sitter parameter package	Phase II input-dependent global closure	The D5→D6 local/global theorem stack is: local null data leave the Einstein equation fixed only up to a metric term, the external input N_{scr} together with the screen-capacity identification $N_{\mathrm{scr}} = S_{\mathrm{dS}}$ and the imported standard de Sitter entropy law $S_{\mathrm{dS}} = A_{\mathrm{dS}}/(4G) = 3\pi/(G\Lambda)$ fixes that same branch constant by $\Lambda = 3\pi/(GN_{\mathrm{scr}})$, and the same D6 closure emits the static-patch package $S_{\mathrm{dS}} = N_{\mathrm{scr}}$, $r_{\mathrm{dS}} = \sqrt{3/\Lambda}$, and $t_\Lambda = r_{\mathrm{dS}}/c$. The observed cosmic age remains a downstream FLRW benchmark, and the screen-capacity identification itself is still an external premise rather than part of the local recovered-core theorem.
Magnetic monopole expectation from simple GUT groups	Phase I recovered core	The realized product-group branch SU(3) × SU(2) × U(1)/$\mathbb{Z}_6$ removes the standard simple-GUT symmetry-breaking monopole channel.
Proton stability and proton spin fraction	Mixed: Phase I / Phase III	No gauge-mediated proton decay follows from the realized product-group branch, which removes the standard simple-GUT proton-decay channel. Proton spin requires nonperturbative QCD completion and is not part of Phase I.
Dark matter phenomenology	Phase III deferred continuation	Modular-anomaly response laws remain conjectural beyond the recovered gravity chain.
Baryon asymmetry scale	Phase III deferred continuation	Suppression-counting estimates are not a substitute for a derived out-of-equilibrium mechanism.
Three generations and downstream flavor structure	Mixed: Phase I / Phase III	$N_g = 3$ is recovered on the realized MAR-admissible branch; later flavor constructions require extra ansatze.

Problem domain	Status	OPH mechanism / scope note
String/worldsheet reorganization	Phase III deferred continuation with a theorem-level effective-description branch	The exact heat-kernel bridge fixes the edge partition function on the two-dimensional Yang–Mills surface. The large-N_{edge} lift then uses the standard Gross–Taylor large-N dictionary and requires a distinct $N_{\text{edge}} \neq N_c = 3$, a fixed-$\tau = tN_{\text{edge}}$ window, and uniform genus-remainder control as stated on the compact worldsheet branch. Any further lift to critical superstring structure remains a conjectural extension requiring extra worldsheet ingredients not derived in the recovered core.
Why anything exists / strange-loop closure	Interpretive epilogue only	Appendix B records only an OPH-internal state-and-law habitat under the stated setup. Missing inputs are an explicit closure self-map, a nonempty invariant observer-supporting sector, and uniqueness/stability control.

C Observer Continuation and Backup

This appendix uses the fixed-cutoff checkpoint/restoration/error/identity theorem package proved in Ref. [3] and summarizes its algebraic interface in observer language. It does not define a strange-loop closure map on an invariant observer-supporting sector or establish uniqueness and stability for such a map. Nor does it upgrade the fixed-cutoff theorem stack to continuation across redesigned environments: the proved package is same-interface restoration on observer-accessible event algebras with explicit error control.

C.1 Observer as Algebraic Pattern

Here, an observer is represented by

$$O = (P, \mathcal{A}(P), \rho, R),$$

where P is a screen patch, $\mathcal{A}(P)$ is its local algebra, ρ is the local state, and R is the record algebra. Records are carried exactly by central record projectors and, on practical readout surfaces, by approximately commuting projectors in overlap centers, so they are shareable without violating no-cloning constraints for generic quantum states.

Ref. [3] proves a fixed-cutoff observer-facing measurement interface: a finite central record algebra, Born probabilities for its event projectors, and Lüders conditioning on that same commuting record algebra. This appendix uses that fixed-cutoff measurement package in observer language. If a practical readout uses projectors that are δ_{rec}-close in operator norm to that central reference algebra and the accessible state is perturbed by at most ε in trace norm, each declared elementary record-event probability shifts by at most $\varepsilon + \delta_{\text{rec}}$.

C.2 Markov Collar Factorization

For a collar tripartition A-B-D with conditional mutual information $I(A : D|B) \leq \varepsilon$, the exact Markov limit gives

$$\rho_{ABD} = \bigoplus_{\alpha} p_\alpha \, \rho^{(\alpha)}_{Ab^\alpha_L} \otimes \rho^{(\alpha)}_{b^\alpha_R D}.$$

The sector label α is classical center data. This decomposition is the mathematical basis for extracting an interior observer state with a controlled boundary interface.

C.3 Checkpoint and Restoration Map

A checkpoint is the tuple

$$\mathcal{C} = (R, \alpha, \rho^{(\alpha)}_{\text{int}}),$$

with $\rho^{(\alpha)}_{\text{int}}$ the interior conditional state. Given a compatible target environment state $\sigma^{(\alpha)}_{\text{env}}$, a restored state is

$$\rho^{(\alpha)}_{\text{new}} = \rho^{(\alpha)}_{\text{int}} \otimes \sigma^{(\alpha)}_{\text{env}}.$$

For approximate Markov collars, recovery is controlled by the standard bound

$$\|\rho_{ABD} - (\text{id}_A \otimes \mathcal{R}_{B\to BD})(\rho_{AB})\|_1 \leq 2\sqrt{\ln 2\, \varepsilon}.$$

This gives quantitative trace-distance control on the recovered state under finite conditional mutual information (finite collar error). Interpreting that control as observer continuation requires additional modeling assumptions beyond the bound itself.

The fixed-cutoff microphysics claim is stronger than the bare recoverability estimate written above. In Ref. [3], exact checkpoint restoration preserves the full future law of observer-accessible events, while an ε-accurate restoration changes that future law by at most ε in total variation. The stronger strange-loop / substrate-transfer closure story remains open.

C.4 Physical Meaning

At fixed cutoff, Ref. [3] proves a checkpoint/restoration theorem and backup corollary for observer-accessible event algebras. The proved package stops at same-interface backup/restoration with explicit future-law error control; stronger continuation and substrate-selection claims remain outside it. This appendix states the observer-facing meaning of that theorem and its algebraic prerequisites. Open items are the stronger substrate-selection, strange-loop closure, uniqueness, and stability package.

D Cosmology, Horizons, and Modular-Anomaly Continuations

This appendix rehouses the cosmology- and continuation-facing technical statements that had been living in the former standalone technical supplement. They do not enlarge the recovered core; they make the structural continuation boundary explicit on the live synthesis surface.

D.1 Cosmological Principle

Lemma 4.1 (MaxEnt symmetry inheritance). Let G act on the screen by automorphisms α_g. If the constraint set is G-invariant and von Neumann entropy is G-invariant, then the MaxEnt optimizer ω can be taken G-invariant. If the optimizer is unique, it is automatically G-invariant.

Theorem 4.2 (Isotropy from SO(3)-invariant constraints). Under SO(3)-invariant constraints and MaxEnt uniqueness, the reference state satisfies

$$\omega \circ \alpha_g = \omega \qquad \forall g \in \mathrm{SO}(3).$$

The corresponding stress tensor has perfect-fluid form:

$$\langle T_{ab} \rangle = \rho\, u_a u_b + p(g_{ab} + u_a u_b).$$

Theorem 4.3 (Schur-type homogeneity). Let (Σ, h_{ij}) be a three-dimensional Riemannian manifold. If at every point $p \in \Sigma$, the curvature tensor is SO(3)-invariant, then

$$R_{ijkl}(p) = K(p)(h_{ik}h_{jl} - h_{il}h_{jk}),$$

and the second Bianchi identity forces $\nabla_m K = 0$, so K is constant.

Corollary 4.4. If OPH supplies isotropy for all observers, spatial geometry is a constant-curvature space form: S^3, $\mathbb{R}^3$, or H^3.

Theorem 4.5 (FLRW emergence). With the semiclassical Einstein equation from recoverable generalized entropy / entanglement equilibrium, a perfect-fluid stress tensor from SO(3)-invariant MaxEnt, and positive Λ from finite screen capacity, the emergent geometry is FLRW:

$$ds^2 = -dt^2 + a(t)^2 \left(\frac{dr^2}{1 - kr^2} + r^2 d\Omega^2 \right),$$

with Friedmann equations

$$H^2 + \frac{k}{a^2} = \frac{8\pi G}{3}\rho + \frac{\Lambda}{3}, \qquad \dot{H} - \frac{k}{a^2} = -4\pi G(\rho + p).$$

D.2 Horizon-Problem Control

Theorem 6.1 (Anisotropy bound from Markov control). For a tripartition A-B-D with collar width δ, the Markov error satisfies

$$I(A : D \mid B) \lesssim c \cdot |\partial C|_{\mathrm{UV}} \cdot e^{-\delta/\xi},$$

where ξ is the correlation length.

Corollary 6.2. For bounded observable O with $\|O\| \sim 1$,

$$|\Delta\langle O \rangle| \leq 2\sqrt{\ln 2 \cdot \varepsilon}.$$

On the CMB anisotropy target $\delta T/T \lesssim 10^{-5}$, this gives the familiar ε-scale bound used in the horizon-problem bookkeeping:

$$\varepsilon \lesssim 3.61 \times 10^{-11} \text{ bits}, \qquad \frac{\delta}{\xi} \gtrsim \ln\left(\frac{c \cdot |\partial C|_{\mathrm{UV}}}{\varepsilon_{\max}} \right) \approx 24.$$

With $\xi \approx \sqrt{a_{\mathrm{cell}}}\, \ell_P \approx 1.28\, \ell_P$, the corresponding collar benchmark is

$$\delta_{\mathrm{CMB}} \approx 31\, \ell_P.$$

Theorem 6.3 (Homogeneity as the MaxEnt default). If MaxEnt constraints are uniform across UV cells with no marked points, the MaxEnt state is invariant under triangulation automorphisms. In the continuum limit this gives SO(3) invariance.

Reading rule. The cosmological-principle and horizon-homogeneity package is a conditional theorem package on the realized symmetric MaxEnt branch. The exact theorem-level content is:

1. SO(3)-invariant constraint data plus MaxEnt uniqueness force an isotropic reference state and a perfect-fluid stress tensor.
2. If the same isotropy condition holds for all observers, the Schur/Bianchi argument forces constant-curvature spatial slices.
3. Combined with the semiclassical Einstein branch and positive Λ from finite screen capacity, the metric takes FLRW form.
4. Markov control supplies an explicit patch-overlap anisotropy bound and the collar benchmark used in the CMB homogeneity bookkeeping.

The package does not prove from the bare OPH axioms alone that the realized cosmological branch must satisfy SO(3)-invariant constraint data, that MaxEnt uniqueness holds in every cosmological sector, or that the observed universe saturates the displayed CMB benchmark.

D.3 Cosmological-Constant / Screen-Capacity Closure

Theorem 6.4 (D5–D6 local/global closure stack). On the gravity lane, assume the local Einstein branch has been recovered only modulo Λg_{ab}, the external input

$$N_{\mathrm{scr}} = \log \dim \mathcal{H}_{\mathrm{tot}},$$

the D6 screen-capacity identification

$$N_{\mathrm{scr}} = S_{\mathrm{dS}},$$

the standard de Sitter entropy relation

$$S_{\mathrm{dS}} = \frac{A_{\mathrm{dS}}}{4G} = \frac{3\pi}{G\Lambda},$$

and the standard de Sitter static-patch formulas

$$r_{\mathrm{dS}} = \sqrt{\frac{3}{\Lambda}}, \qquad t_\Lambda = \frac{r_{\mathrm{dS}}}{c}.$$

Then:

1. local null data determine the Einstein branch only modulo Λg_{ab};
2. the same branch closes globally as

$$G_{ab} + \frac{3\pi}{G N_{\mathrm{scr}}}\, g_{ab} = 8\pi G \,\langle T_{ab}\rangle;$$

3. the same D6 closure fixes

$$S_{\mathrm{dS}} = N_{\mathrm{scr}}, \qquad A_{\mathrm{dS}} = 4G N_{\mathrm{scr}}, \qquad r_{\mathrm{dS}} = \sqrt{\frac{3}{\Lambda}}, \qquad t_\Lambda = \frac{r_{\mathrm{dS}}}{c};$$

4. the observed cosmic age remains a downstream FLRW benchmark rather than an additional theorem output.

So the cosmological-constant package is one local/global theorem stack rather than a split local argument plus an unrelated global readout.

Scope boundary. The imported premises are exactly the local Einstein branch, the external input N_{scr}, the D6 identification $N_{\mathrm{scr}} = S_{\mathrm{dS}}$, the de Sitter entropy relation, and the standard static-patch formulas. What remains external is the screen-capacity identification itself: the local null-data route does not derive $N_{\mathrm{scr}} = S_{\mathrm{dS}}$ from the bare OPH axioms.

D.4 Black-Hole Structural Package and Continuation Boundary

The AMPS-style information trilemma assumes:

1. an outgoing mode B entangled with an interior partner A (smooth horizon),
2. late radiation B entangled with early radiation R (unitarity), and
3. monogamy of entanglement.

The false assumption in this setting is the naive tensor factorization

$$\mathcal{H} \stackrel{?}{=} \mathcal{H}_{\mathrm{inside}} \otimes \mathcal{H}_{\mathrm{outside}} \otimes \mathcal{H}_{\mathrm{radiation}}.$$

Theorem 7.1 (Edge-center black-hole decomposition). On the fixed-cutoff edge-center collar branch, assume the black-hole collar state lies in the exact Markov class or in the idealized exact-recoverability limit used elsewhere in the paper stack. Then the collar decomposition gives

$$\rho_{A_\delta B_\delta D_\delta} = \bigoplus_\alpha p_\alpha \left(\rho_{A_\delta b_L^\alpha} \otimes \rho_{b_R^\alpha D_\delta} \right).$$

The glue between inside and outside is the edge-sector label α in the center. Conditional on α, inside and outside factorize.

Theorem 7.2 (Hawking/KMS normalization). On the geometric modular branch where the horizon generator carries the standard 2π KMS normalization, the outside algebra is in a KMS state at inverse temperature

$$\beta = \frac{2\pi}{\kappa}$$

with respect to the horizon Killing generator.

Corollary 7.3 (Recoverability-style interior encoding). On the same fixed-cutoff collar carrier, when the collar conditional mutual information is small, the interior partner is approximately recoverable from outside plus radiation:

$$\|\rho_{ABC} - (\mathrm{id}_A \otimes \mathcal{R})(\rho_{AB})\|_1 \leq 2\sqrt{\ln 2 \cdot \varepsilon}.$$

So the interior is encoded rather than supplied by an independent tensor factor.

Corollary 7.4 (Discrete area spectrum and Schwarzschild transition identity). The same edge-center package carries a discrete area operator

$$L_C = \sum_\alpha (\log d_\alpha) P_\alpha, \qquad A_\alpha = 4G \log d_\alpha = 4\ell_P^2 \ln d_\alpha,$$

and for a Schwarzschild sector transition $d \to d'$ one has

$$\Delta E = k_B T_H \ln(d'/d).$$

Scope boundary. The retained structural theorem package is exactly Theorem 7.1, Theorem 7.2, Corollary 7.3, and Corollary 7.4. Integer-transition combs, QNM selectors, Page-type linewidth estimates, PBH burst templates, Kerr/LIGO horizon spectroscopy templates, and any Page-curve or island closure remain outside that structural core. Those items require extra premises such as an integer-transition selection rule, a semiclassical evaporation-power model, a QNM/transition identification, greybody matching, or an explicit island prescription.

D.5 Modular-Anomaly Continuation

Theorem 9.1 (Modular additivity defect). Define the collar modular additivity defect by

$$\Delta K_\delta := K_{ABD} - K_{AB} - K_{BD} + K_B.$$

Then

$$\langle \Delta K_\delta \rangle_\omega = -I(A : D \mid B)_\omega.$$

This is the precise structural identity underneath the modular-anomaly dark-sector continuation.

The corresponding continuation packages the anomalous contribution into the modified Einstein equation

$$G_{ab} + \Lambda g_{ab} = 8\pi G \left(\langle T_{ab} \rangle + \langle T_{ab}^{\text{anom}} \rangle \right),$$

with rest-frame normalization

$$\langle T_{00}^{\text{anom}} \rangle := \frac{15}{8\pi^2} \frac{\delta \langle K_C^{\text{anom}} \rangle}{\ell^4}.$$

On that continuation surface the anomaly gravitates, is dark by construction, is covariantly conserved, and is effectively classical because it is organized by central/recoverability data.

If one tentatively identifies the imported D6 static-patch radius r_{dS} as the unique IR scale available to the continuation, the natural benchmark acceleration is

$$a_0^{(\text{OPH})} := \frac{15}{8\pi^2} c^2 \sqrt{\frac{\Lambda}{3}} = \frac{15}{8\pi^2} \frac{c^2}{r_{\text{dS}}} \approx 1.03 \times 10^{-10} \text{ m/s}^2,$$

close to the empirical MOND scale. An illustrative MOND/RAR-style response would read

$$g_{\text{obs}} \approx g_b + \sqrt{a_0 g_b},$$

but that functional form is not derived from the OPH axioms.

Reading rule. The anomaly term gravitates, is dark by construction, and is covariantly conserved, but the live synthesis surface does not promote any MOND/RAR-style law, galaxy-rotation template, or cluster phenomenology to theorem status. The missing inputs are the controlled nonrelativistic limit, a derived source/response law, lensing and cluster phenomenology, cosmological abundance and structure-formation checks, and environment/stability bounds. Those remain continuation-level branches above the identity written here.

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Lane	Exact output(s)	Exact paper-surface chain	Caveat
Structural carriers	$m_\gamma = m_g = m_{\mathrm{grav}} = 0$	axioms → realized electromagnetic/color/dynamical-metric carrier skeleton → symmetry-protected zeros	theorem-grade structural exactness
Electroweak exact sidecar	exact frozen W/Z pair	axioms → gauge reconstruction → target-free electroweak quantitative-closure solve → exact frozen repair pair	exact only on the frozen authoritative repair surface; compare-only beneath the target-free theorem
Higgs/top exact sidecar	exact Higgs/top inverse pair	axioms → electroweak core → D11 Jacobian → exact inverse slice	compare-only inverse slice; the public Higgs/top pair comes from the source-only split theorem on the declared surface, while the repo-wide exact public top row is also carried by the selected-class quark theorem
Charged exact witness	exact same-family charged triple	axioms → shared excitation dictionary → ordered charged carrier → exact centered readback → closed quadratic readout theorem → same-family exact witness	same-family-only; theorem lane carries exact centered readback plus the closed common-shift no-go, then waits on $\widehat{C}_e^{\mathrm{cand}}$ and the post-promotion lift whose descended scalar is $\mu_{\mathrm{phys}}(Y_e)$, with the physical identity-mode equalizer beneath that scalar
Quark selected-class theorem	exact running quark sextet plus explicit exact forward Yukawas on f_P	axioms → shared excitation dictionary → D12 mass bridge → selected public quark frame class f_P → direct public sigma-datum descent → affine mean law → ordered three-point readout → exact forward construction	selected-class closure on the public physical quark frame class chosen by P; the sextet matches the official PDG 2025 API running-quark surface exactly, with the top coordinate taken from the PDG cross-section entry; the bridge to the auxiliary direct-top PDG row remains open and is tracked in issue #207; the same exact sextet is also realized by the same-family witness on `current_family_only` and by the restricted common-refinement transport-frame exact chain, which also carries explicit exact forward Yukawas Y_u and Y_d; the target-free mass bridge $\Delta_{ud}^{\mathrm{overlap}} = \frac{1}{6}\log(c_d/c_u)$, equivalently $\Theta_{ud}^{\mathrm{mass}} =$ `quark_d12_t1_value_law`, closes separately on the emitted D12 mass ray; the historical selector σ_{ref} is a lower negative sheet statement beneath the selected-class theorem; no global classification of all quark frame classes is claimed; a stronger class-uniform public descent theorem is separate and is not claimed
Neutrino theorem branch	emitted absolute mass triple and representative central splittings	axioms → same-label scalar certificate → weighted-cycle branch → bridge-rigidity theorem → absolute-attachment theorem	theorem-grade on the declared weighted-cycle branch; the positive-segment adapter, bridge corridor, and bridge-coordinate sidecars remain diagnostic-only